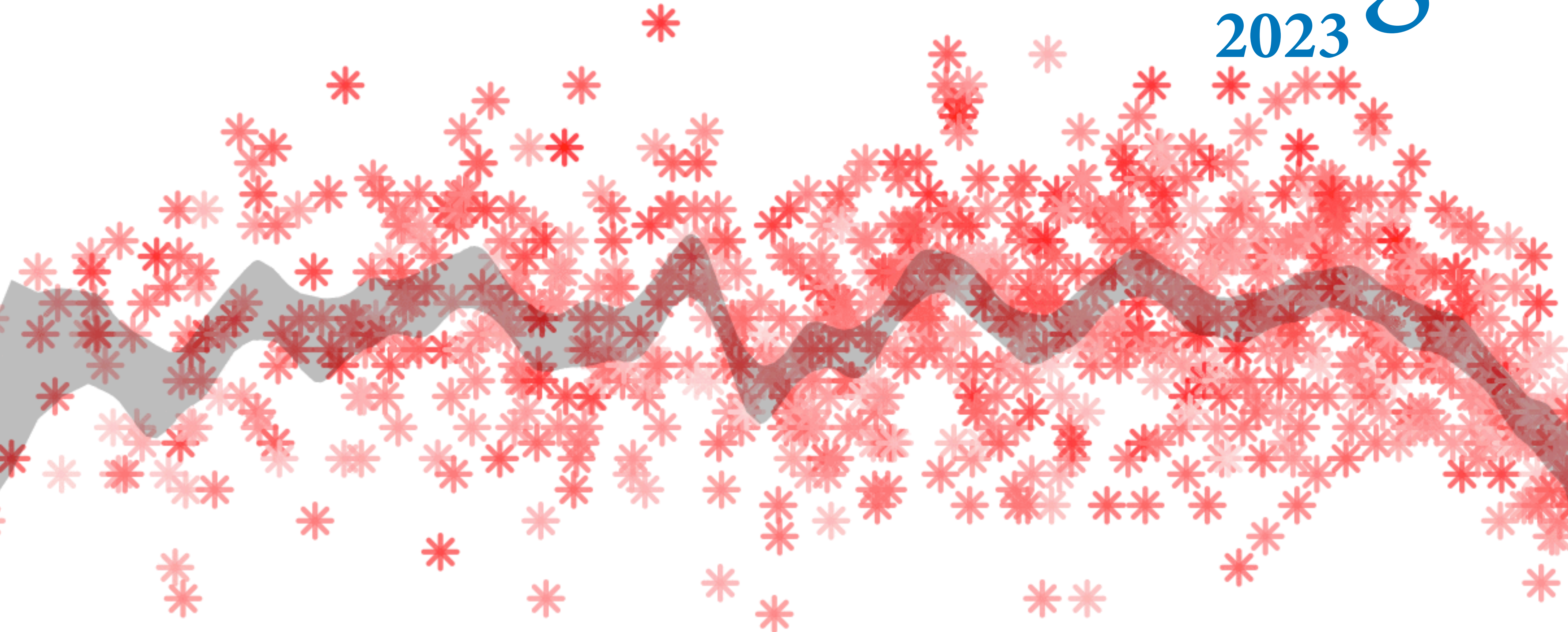


Statistical Rethinking

2023



10. Counts & Hidden Confounds

Generalized Linear Models

Linear Models: Expected value is additive (“linear”) combination of parameters

$$Y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_X X_i + \beta_Z Z_i$$

Generalized Linear Models:
Expected value is **some function** of an additive combination of parameters

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$f(p_i) = \alpha + \beta_X X_i + \beta_Z Z_i$$

Generalized Linear Models:

Expected value is **some function** of an additive combination of parameters

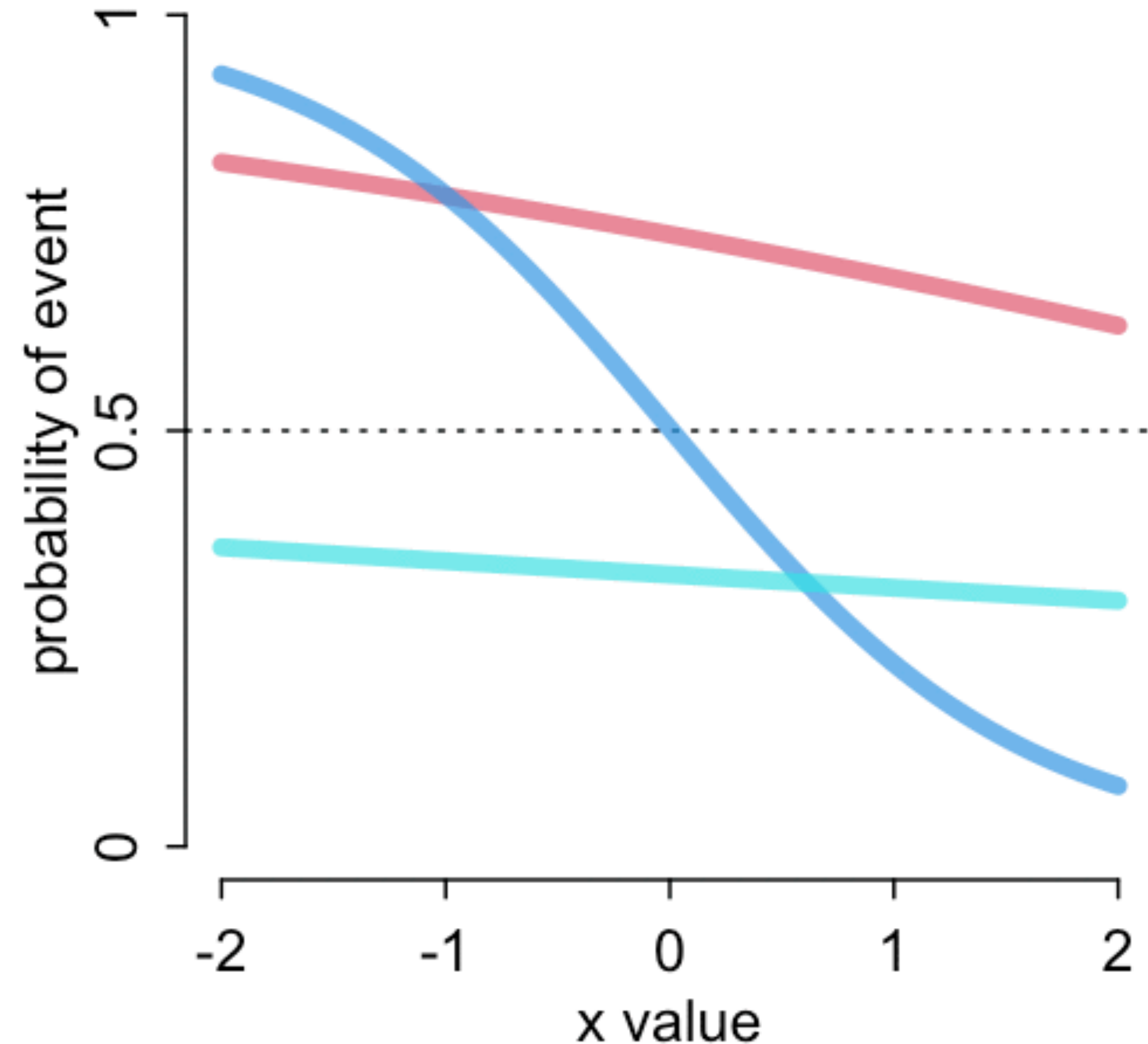
Uniform changes in predictor not uniform changes in prediction

All predictor variables interact, moderate one another

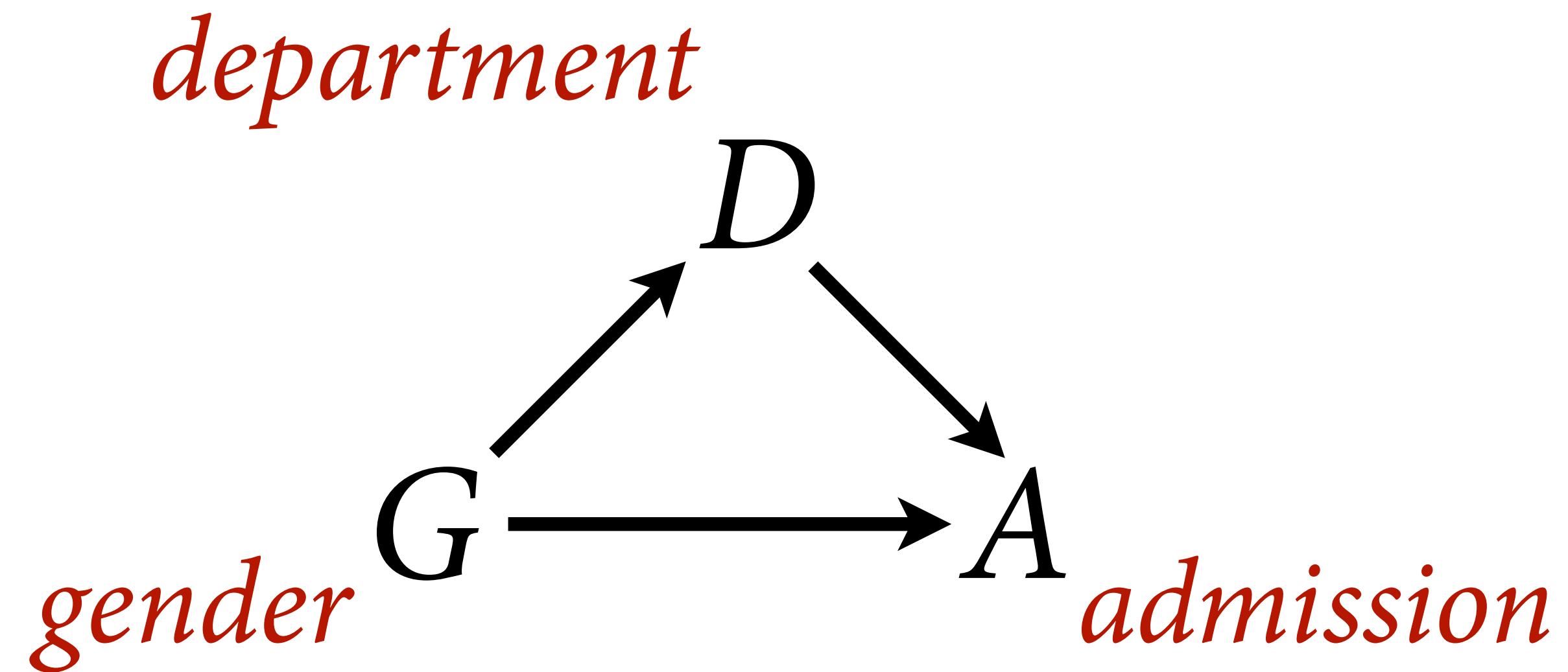
Influences predictions & uncertainty of predictions

$$\text{logit}(p_i) = \alpha + \beta x_i$$

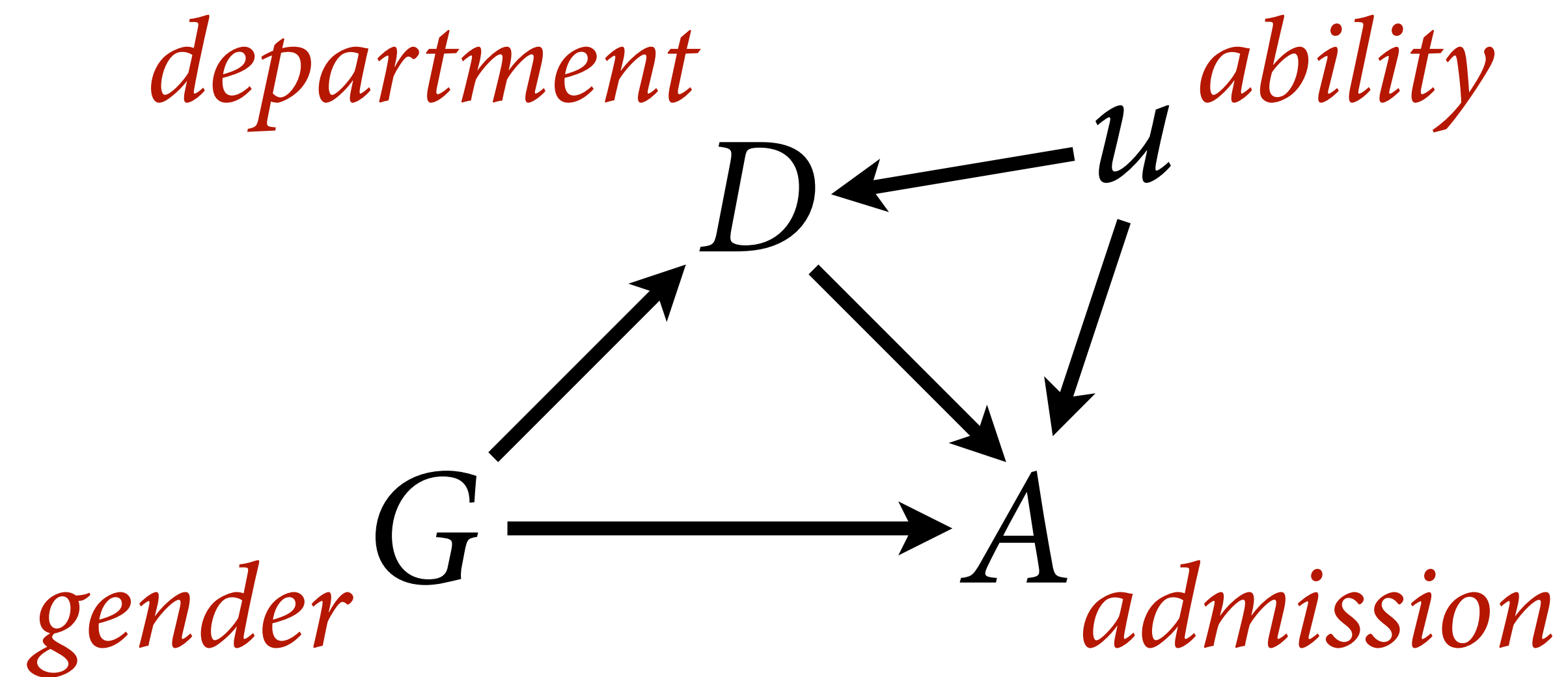
$n = 0$



Confounded Admissions



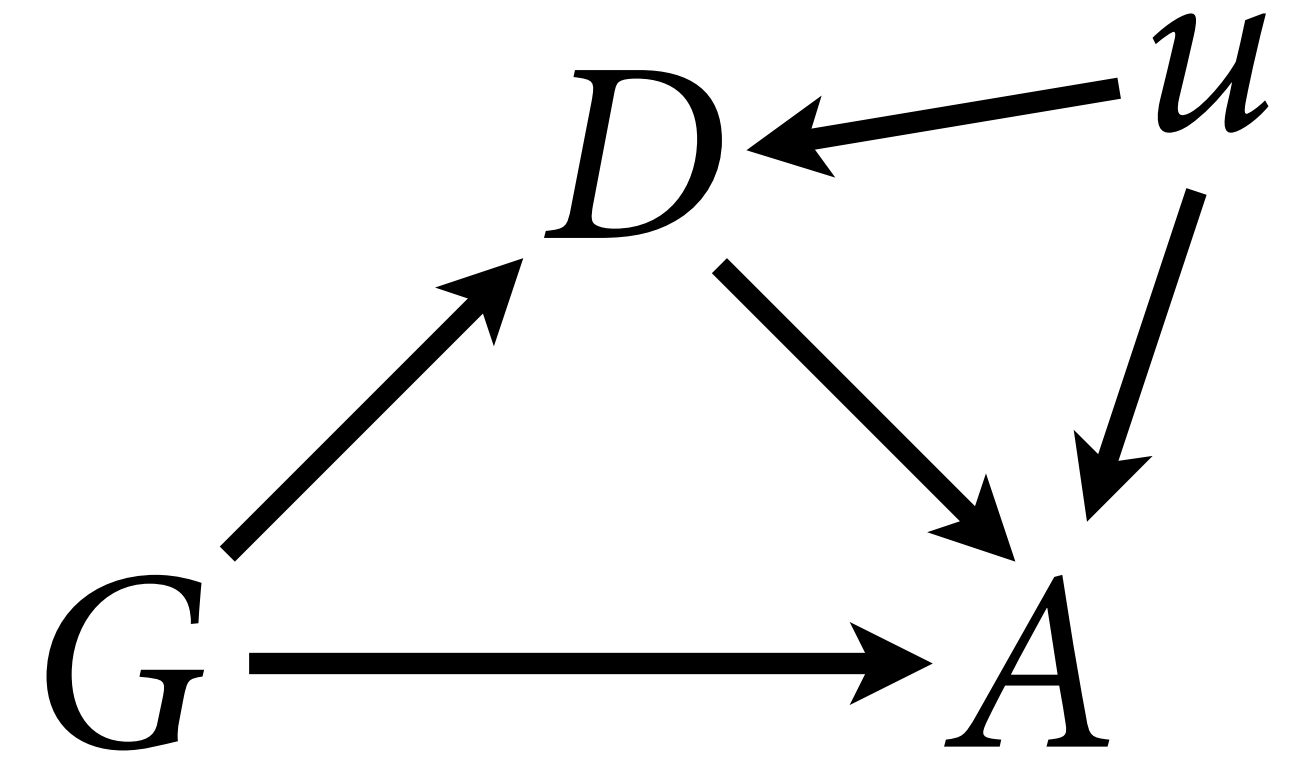
Confounded Admissions



```

set.seed(12)
N <- 2000 # number of applicants
# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# sample ability, high (1) to average (0)
u <- rbern(N,0.1)
# gender 1 tends to apply to department 1, 2 to 2
# and G=1 with greater ability tend to apply to 2 as well
D <- rbern( N , ifelse( G==1 , u*1 , 0.75 ) ) + 1
# matrix of acceptance rates [dept,gender]
p_u0 <- matrix( c(0.1,0.1,0.1,0.3) , nrow=2 )
p_u1 <- matrix( c(0.3,0.3,0.5,0.5) , nrow=2 )
p_u <- list( p_u0 , p_u1 )
# simulate acceptance
p <- sapply( 1:N , function(i) p_u[[1+u[i]]][D[i],G[i]] )
A <- rbern( N , p )

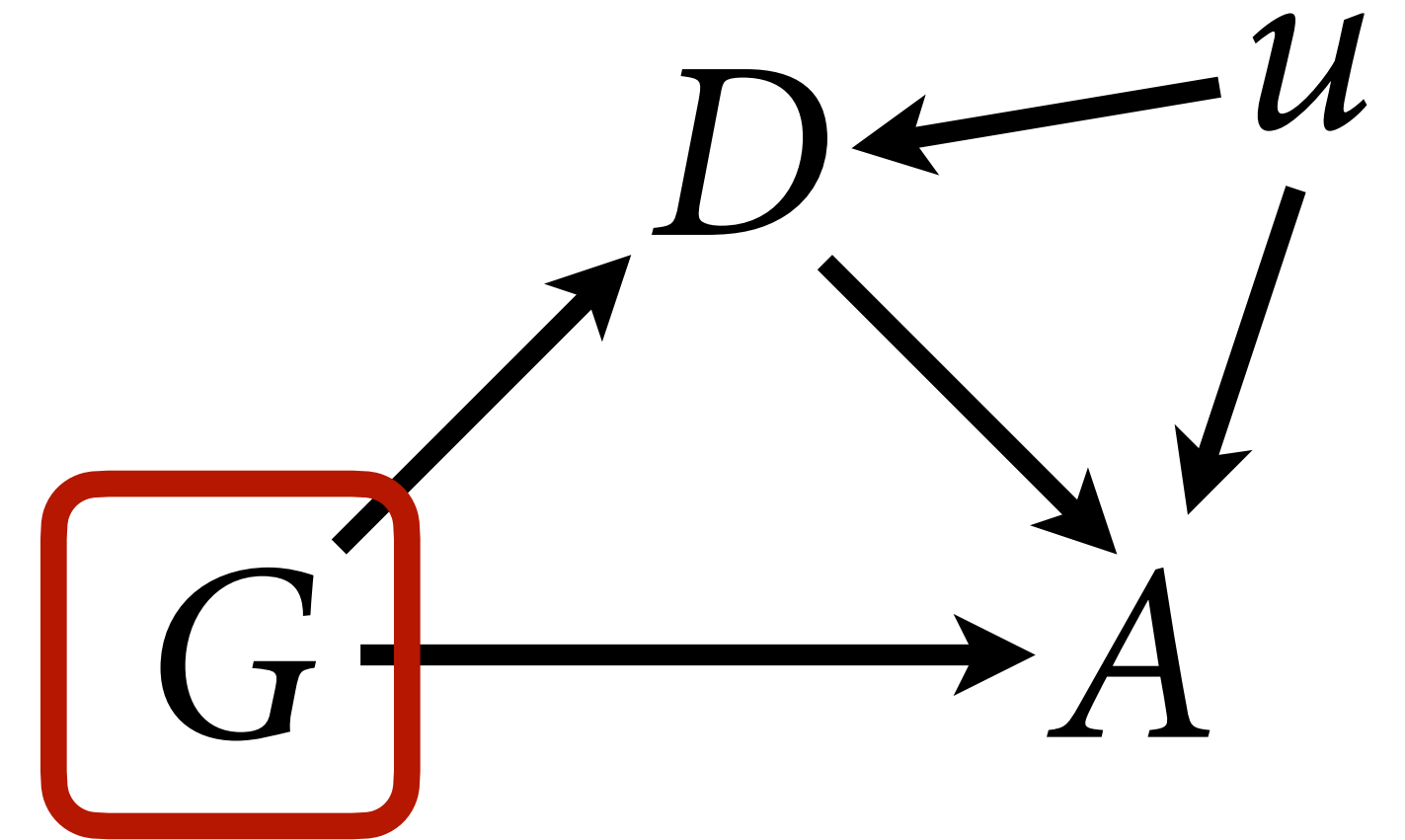
```




```

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A <- rbern( N , p )

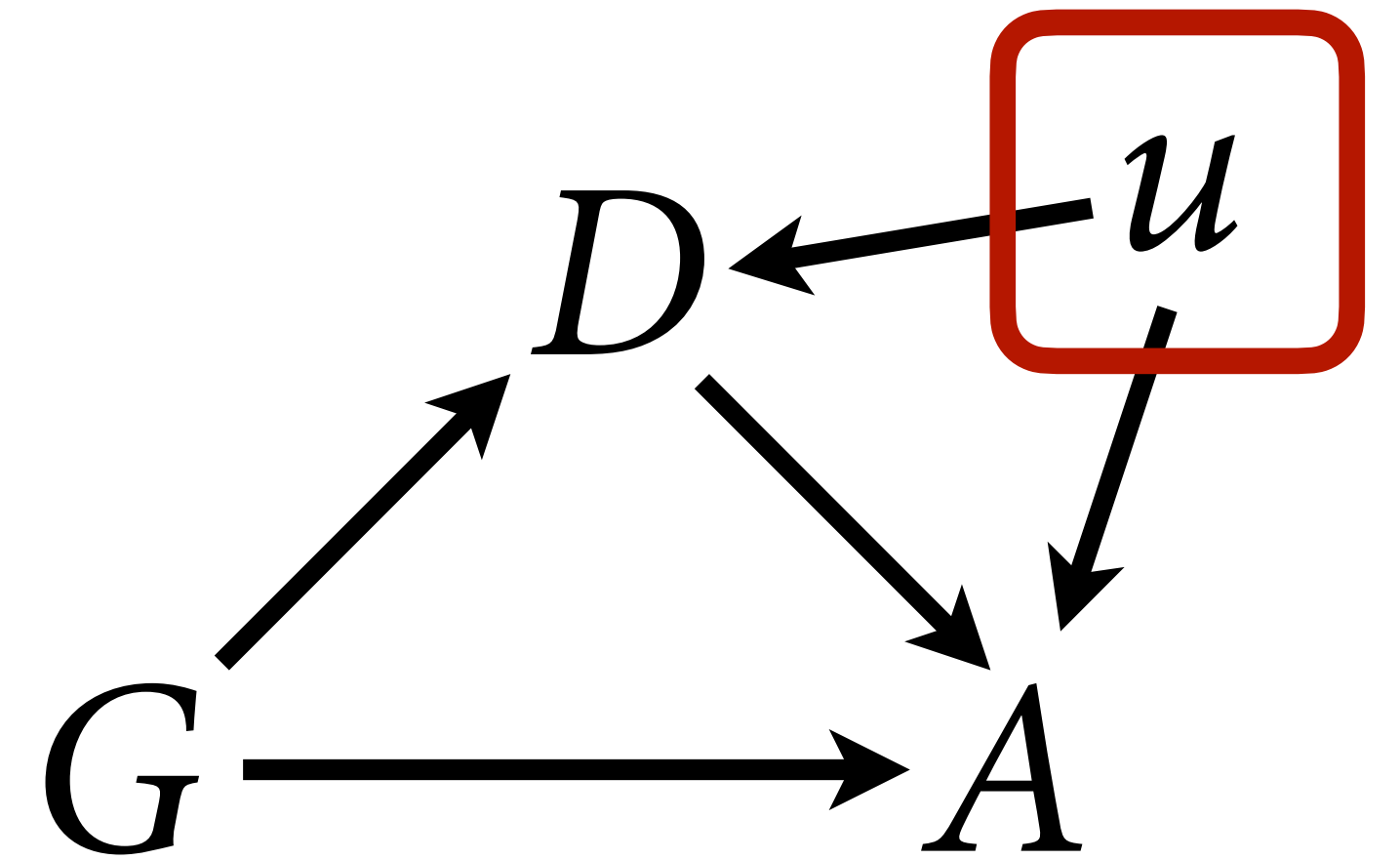
```



```

set.seed(12)
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G <- sample( 1:2 , size=N , replace=TRUE )
# sample ability, high (1) to average (0)
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p_u1 <- matrix( c(0.3,0.3,0.5,0.5) , nrow=2 )
p_u <- list( p_u0 , p_u1 )
# simulate acceptance
p <- sapply( 1:N , function(i) p_u[[1+u[i]]][D[i],G[i]] )
A <- rbern( N , p )

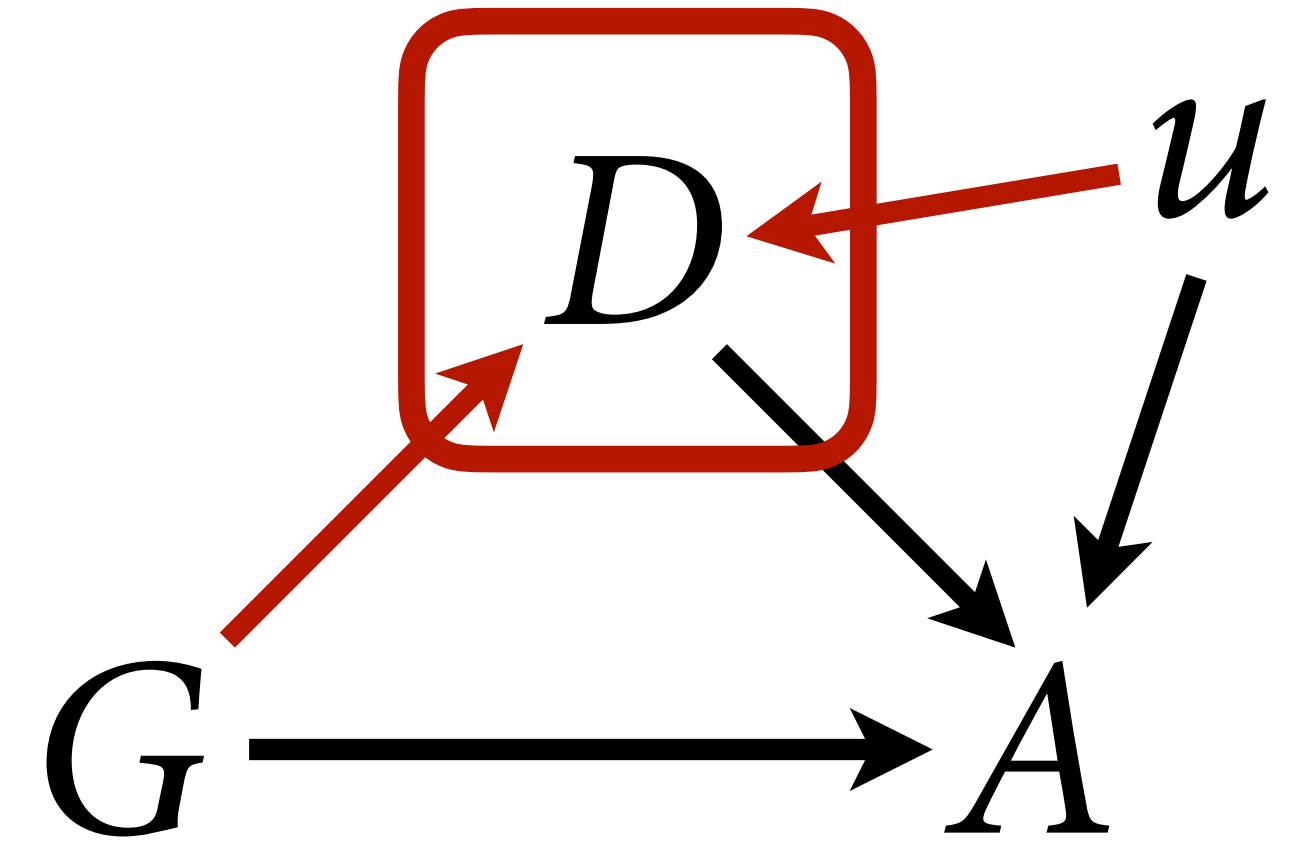
```



```

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# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# sample ability, high (1) to average (0)
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# matrix of acceptance rates [dept,gender]
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p_u1 <- matrix( c(0.3,0.3,0.5,0.5) , nrow=2 )
p_u <- list( p_u0 , p_u1 )
# simulate acceptance
p <- sapply( 1:N , function(i) p_u[[1+u[i]]][D[i],G[i]] )
A <- rbern( N , p )

```



```

> table(G,D,u)
, , u = 0
      D
G      1      2
1  908      0
2  220  645

, , u = 1
      D
G      1      2
1      0  114
2   28   85

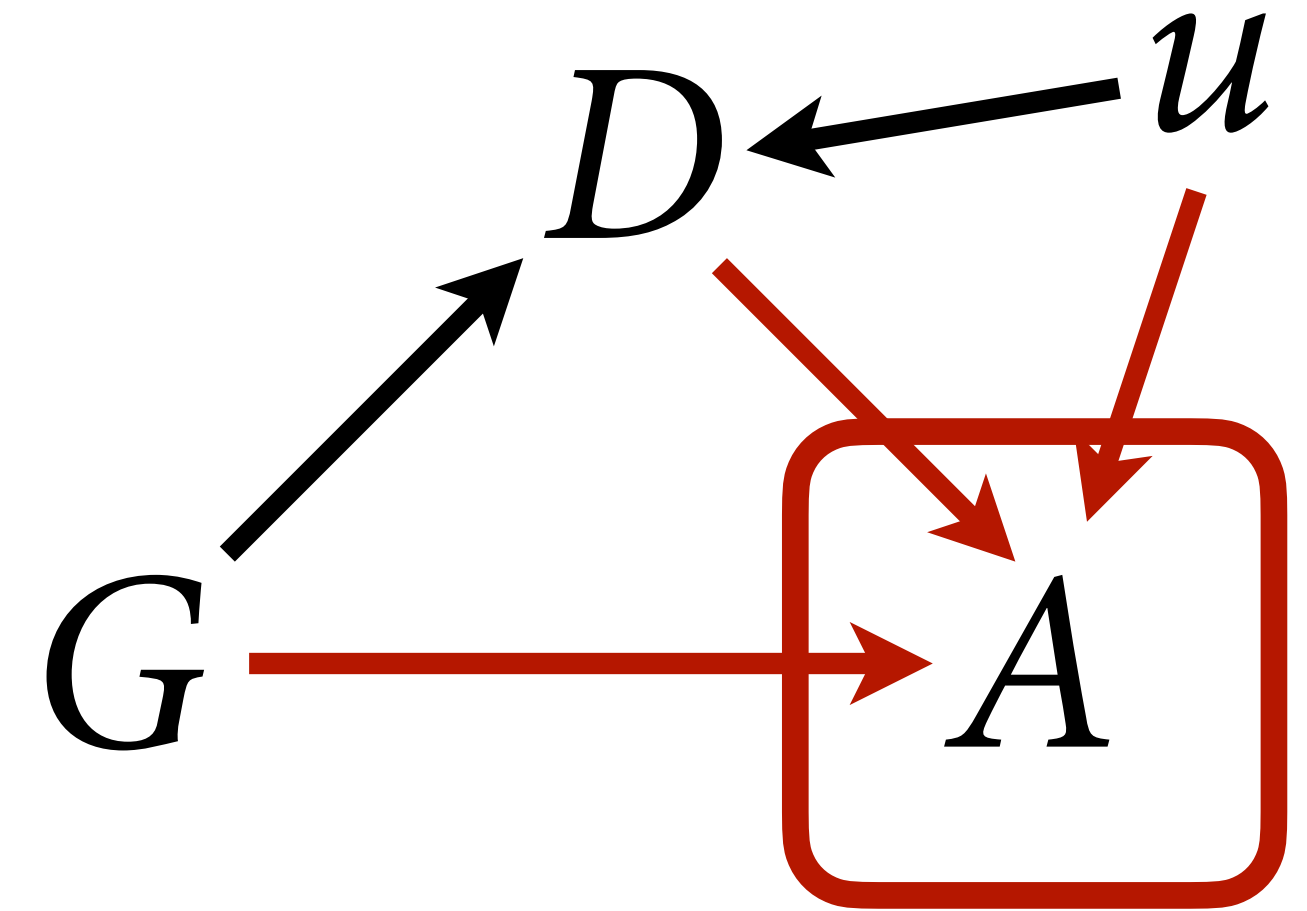
```



```

set.seed(12)
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# even gender distribution
G <- sample( 1:2 , size=N , replace=TRUE )
# sample ability, high (1) to average (0)
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p_u <- list( p_u0 , p_u1 )
# simulate acceptance
p <- sapply( 1:N , function(i) p_u[[1+u[i]]][D[i],G[i]] )
A <- rbern( N , p )

```



```

> p_u0
      [,1] [,2]
[1,]  0.1  0.1
[2,]  0.1  0.3

```

```

> p_u1
      [,1] [,2]
[1,]  0.3  0.5
[2,]  0.3  0.5

```

Department 2 discriminates against gender 1

```

dat_sim <- list( A=A , D=D , G=G )

# total effect gender
m1 <- ulam(
  alist(
    A ~ bernoulli(p),
    logit(p) <- a[G],
    a[G] ~ normal(0,1)
  ), data=dat_sim , chains=4 , cores=4 )

# direct effects - now confounded!
m2 <- ulam(
  alist(
    A ~ bernoulli(p),
    logit(p) <- a[G,D],
    matrix[G,D]:a ~ normal(0,1)
  ), data=dat_sim , chains=4 , cores=4 )

```

total effect shows disadvantage

```

> precis(m1,depth=3)
      mean    sd  5.5% 94.5% n_eff Rhat4
a[1] -1.82 0.09 -1.96 -1.67  1483    1
a[2] -0.89 0.07 -1.00 -0.78  1562    1

```

direct effect confounded

```

> precis(m2,depth=3)
      mean    sd  5.5% 94.5% n_eff Rhat4
a[1,1] -2.05 0.11 -2.23 -1.89  2482    1
a[1,2] -0.66 0.18 -0.96 -0.38  2369    1
a[2,1] -1.77 0.19 -2.06 -1.47  2278    1
a[2,2] -0.68 0.08 -0.81 -0.56  2210    1

```

```

post2 <- extract.samples(m2)
dens(inv_logit(post2$a[,1,1]),lwd=3,col=4,xli
m=c(0,0.5),xlab="probability admission")
dens(inv_logit(post2$a[,2,1]),lwd=3,col=4,lty
=2,add=TRUE)
dens(inv_logit(post2$a[,1,2]),lwd=3,col=2,add
=TRUE)
dens(inv_logit(post2$a[,2,2]),lwd=3,col=2,lty
=2,add=TRUE)

```

total effect shows disadvantage

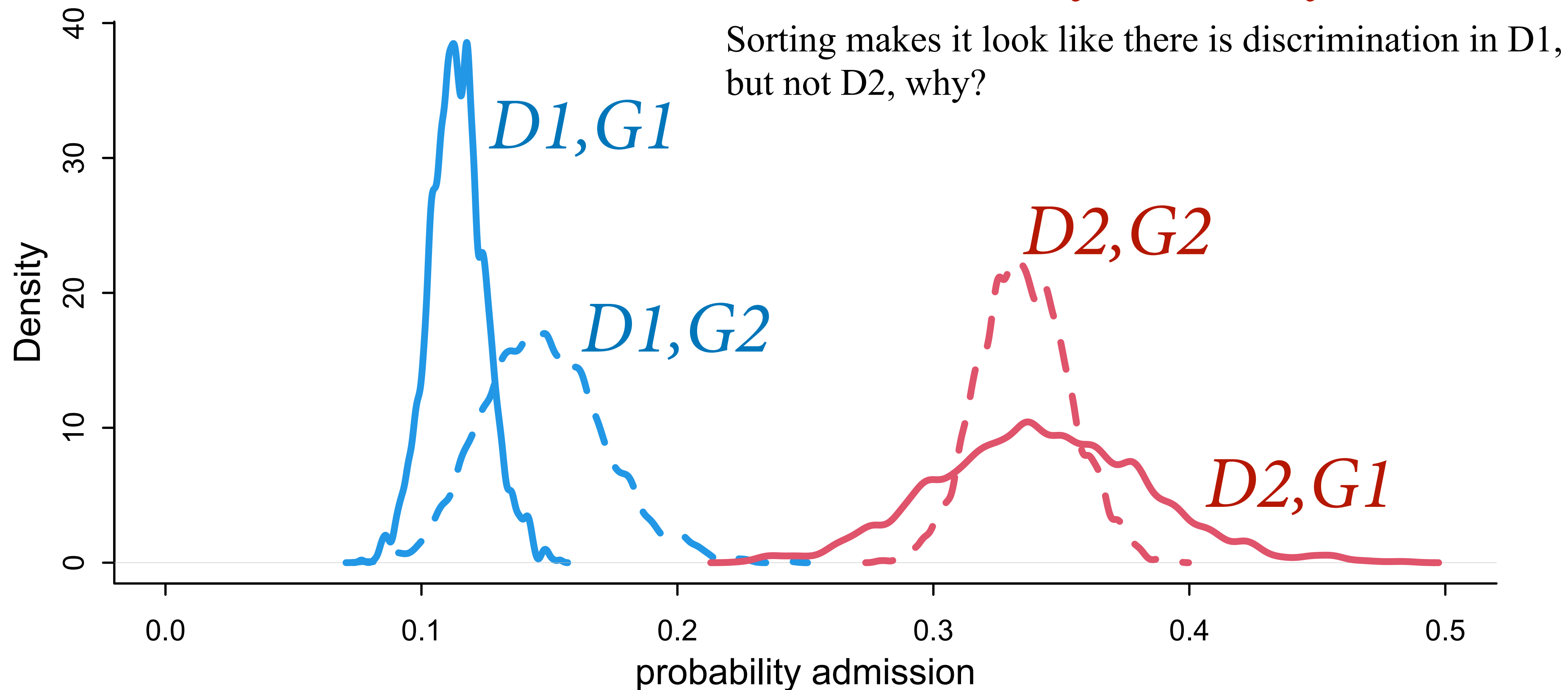
```

> precis(m1,depth=3)

```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1]	-1.82	0.09	-1.96	-1.67	1483	1
a[2]	-0.89	0.07	-1.00	-0.78	1562	1

direct effect confounded

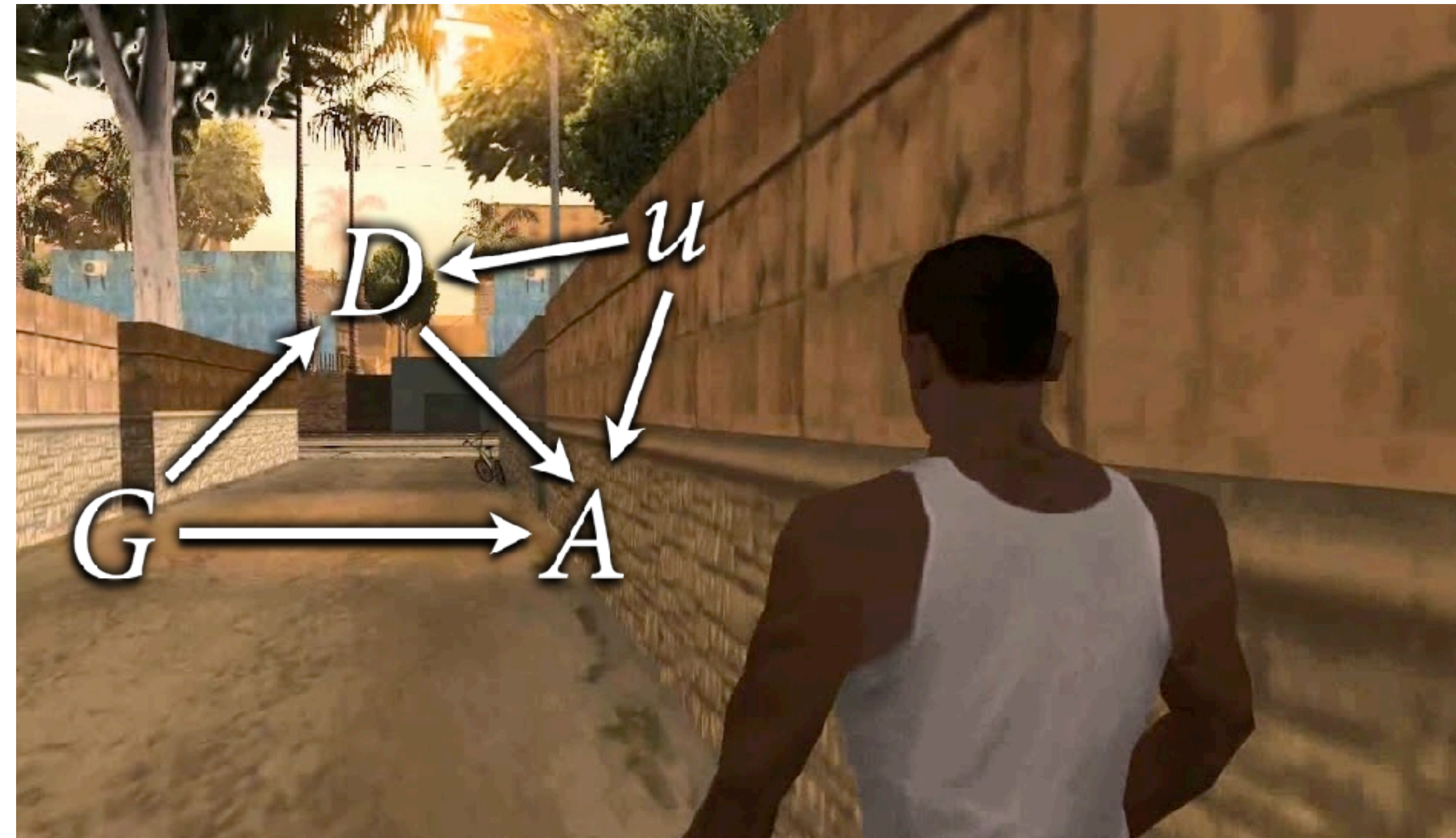


You guessed it: Collider bias

Stratifying by D opens non-causal path through u

Can estimate **total** causal effect of G , but isn't what we want

Cannot estimate **direct** effect of D or G



Worst place in the world, Collider Bias country

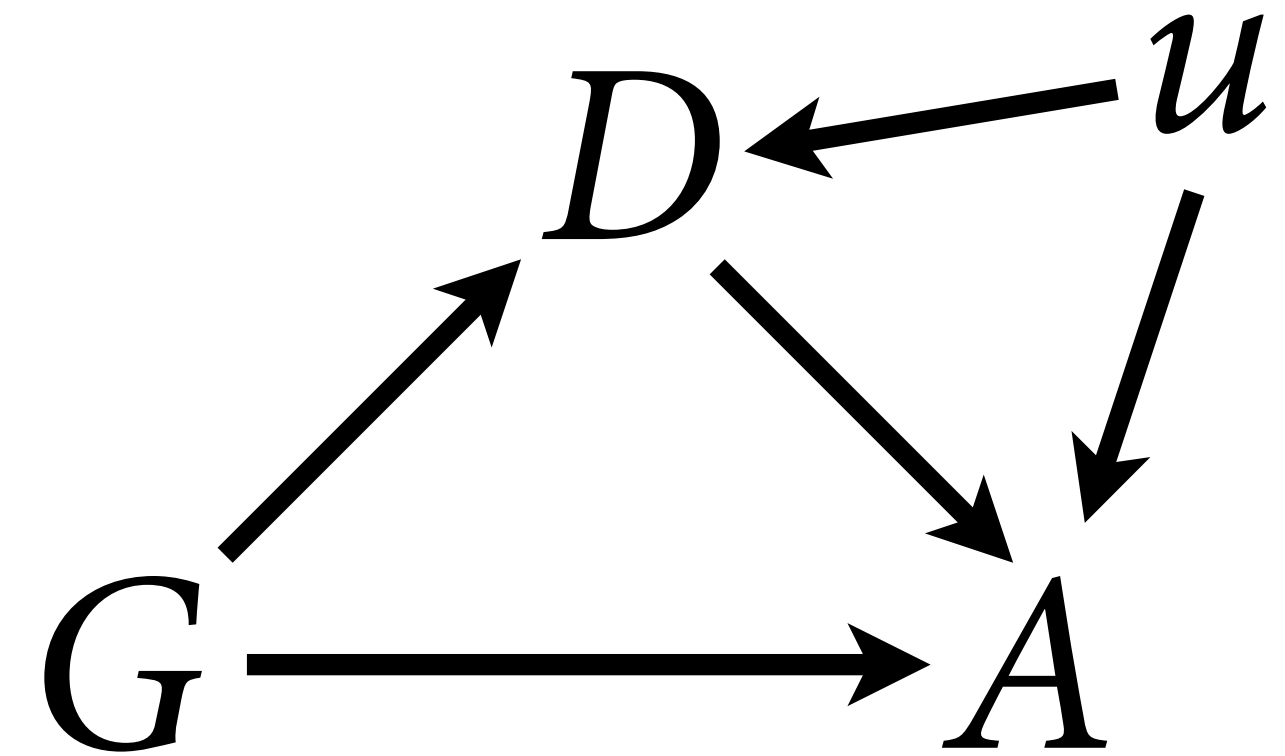
You guessed it: Collider bias

More intuitive explanation:

High ability G1s apply to
discriminatory department anyway

G1s in that department are higher
ability on average than G2s

High ability compensates for
discrimination => masks evidence





Gender Gaps at the Academies

David Card, Stefano DellaVigna, Patricia Funk & Nagore Iriberry

WORKING PAPER 30510 DOI 10.3386/w30510 ISSUE DATE September 2022

Historically, a large majority of the newly elected members of the National Academy of Science (NAS) and the American Academy of Arts and Science (AAAS) were men. Within the past two decades, however, that situation has changed, and in the last 3 years women made up about 40 percent of the new members in both academies. We build lists of active scholars from publications in the top journals in three fields – Psychology, Mathematics and Economics – and develop a series of models to compare changes in the probability of selection of women as members of the NAS and AAAS from the 1960s to today, controlling for publications and citations. In the early years of our sample, women were less likely to be selected as members than men with similar records. By the 1990s, the selection process at both academies was approximately gender-neutral, conditional on publications and citations. In the past 20 years, however, a positive preference for female members has emerged and strengthened in all three fields. Currently, **women are 3-15 times more likely** to be selected as members of the AAAS and NAS than men **with similar publication and citation records**.

PNAS

BRIEF REPORT | SOCIAL SCIENCES
COMPUTER SCIENCES

Gendered citation patterns among the scientific elite

Kristina Lerman^{a,1}, Yulin Yu^b, Fred Morstatter^a, and Jay Pujara^a

Edited by Susan Fiske, Princeton University, Princeton, NJ; received April 8, 2022; accepted August 22, 2022

Diversity in science is necessary to improve innovation and increase the capacity of the scientific workforce. Despite decades-long efforts to increase gender diversity, however, women remain a small minority in many fields, especially in senior positions. The dearth of elite women scientists, in turn, leaves fewer women to serve as mentors and role models for young women scientists. To shed light on gender disparities in science, we study prominent scholars who were elected to the National Academy of Sciences. We construct author citation networks that capture the structure of recognition among scholars' peers. **We identify gender disparities in the patterns of peer citations and show that these differences are strong enough to accurately predict the scholar's gender.** In contrast, we do not observe disparities due to prestige, with few significant differences in the structure of citations of scholars affiliated with high-ranked and low-ranked institutions. These results provide further evidence that a scholar's gender plays a role in the mechanisms of success in science.

gender | bibliometrics | science of science | gender disparities

Gender disparities persist in many fields of science. Despite long-running efforts to increase women's representation in the scientific workforce, they continue to face barriers to advancement. Women are less likely than their male peers to be mentored by eminent faculty (1) and to be hired and promoted (2, 3). Women publish in less prestigious journals (4), have fewer collaborators (5), and are underrepresented among journal reviewers and editors (6), and their papers receive fewer citations (7, 8). The multifaceted gender disparities create a "glass ceiling," an invisible barrier that fundamentally limits professional recognition for even the best women scientists (9). As a result, the share of women in higher academic positions decreases steadily (3), with relatively few becoming full professors or receiving prestigious awards. For example, among physics faculty in 4-y colleges and universities, women represent 23% of assistant professors and 18% of associate professors



Causal foundations of bias, disparity and fairness

V.A. Traag ^{a)} and L. Waltman 

Centre for Science and Technology Studies (CWTS), Leiden University, the Netherlands

(Dated: July 27, 2022)

The study of biases, such as gender or racial biases, is an important topic in the social and behavioural sciences. However, the concept of bias is not always clearly defined in the literature. Definitions of bias are often ambiguous, or definitions are not provided at all. To study biases in a precise way, it is important to have a well-defined concept of bias. We propose to define bias as a direct causal effect that is unjustified. We propose to define the closely related concept of disparity as a direct or indirect causal effect that includes a bias. Our proposed definitions can be used to study biases and disparities in a more rigorous and systematic way. We compare our definitions of bias and disparity with various definitions of fairness introduced in the artificial intelligence literature. We also illustrate our definitions in two case studies, focusing on gender bias in science and racial bias in police shootings. Our proposed definitions aim to contribute to a better appreciation of the causal intricacies of studies of biases and disparities. This will hopefully also lead to an improved understanding of the policy implications of such studies.

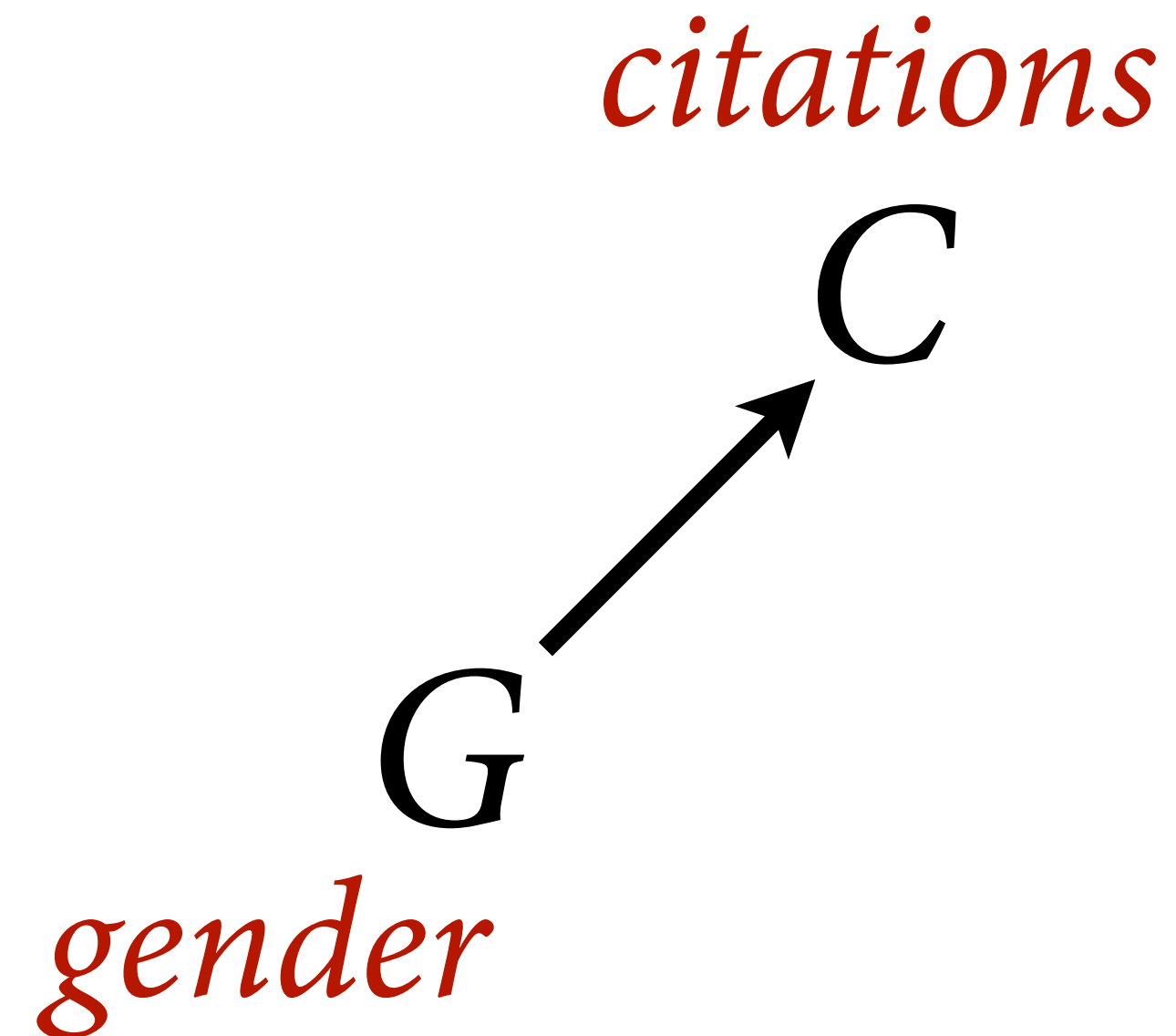
Keywords: bias; disparity; fairness; causal model; gender bias; racial bias

<https://arxiv.org/abs/2207.13665>

Citation networks

Citation networks of members of
NAS

Women associated with lower
lifetime citation rate

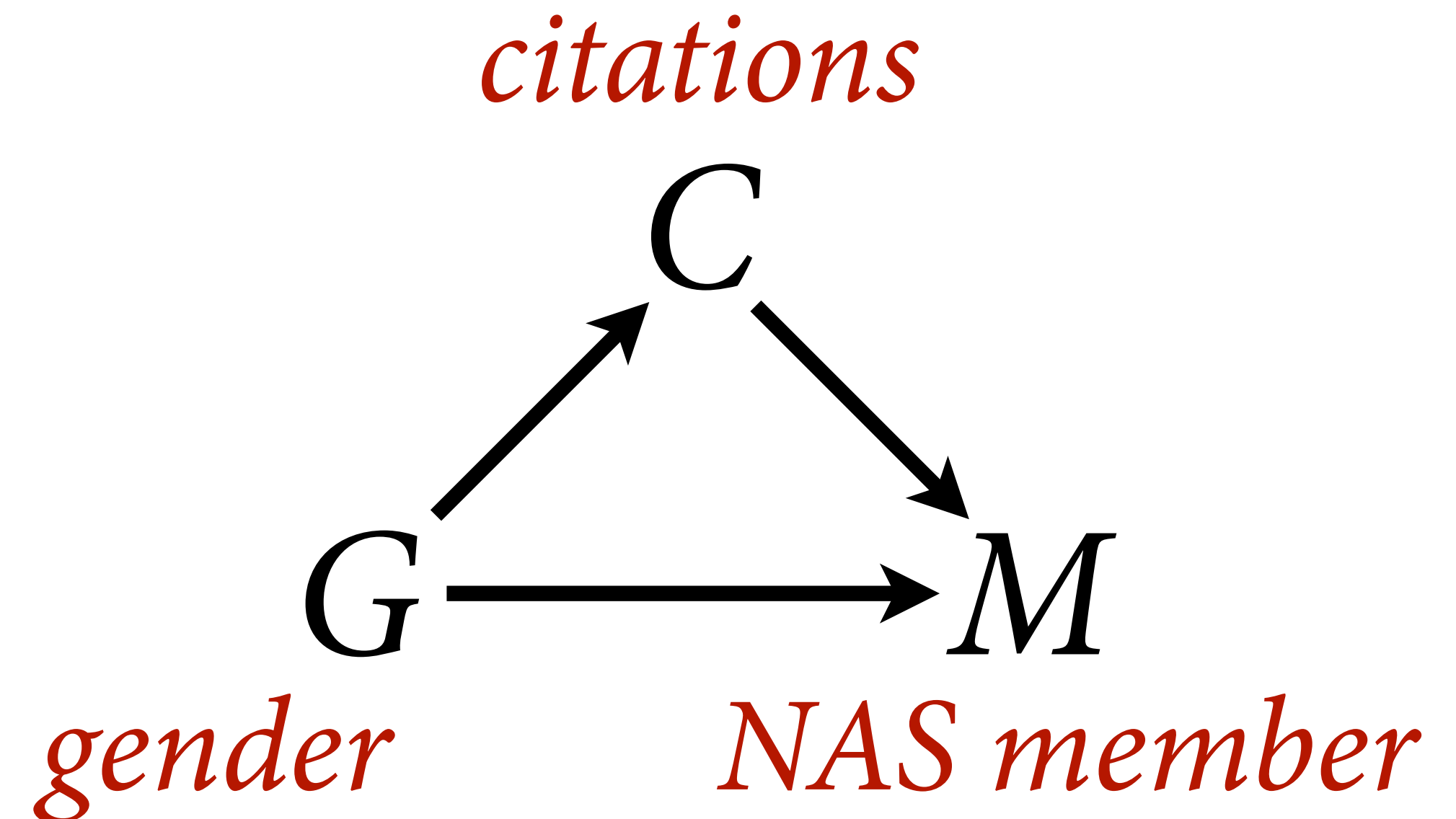


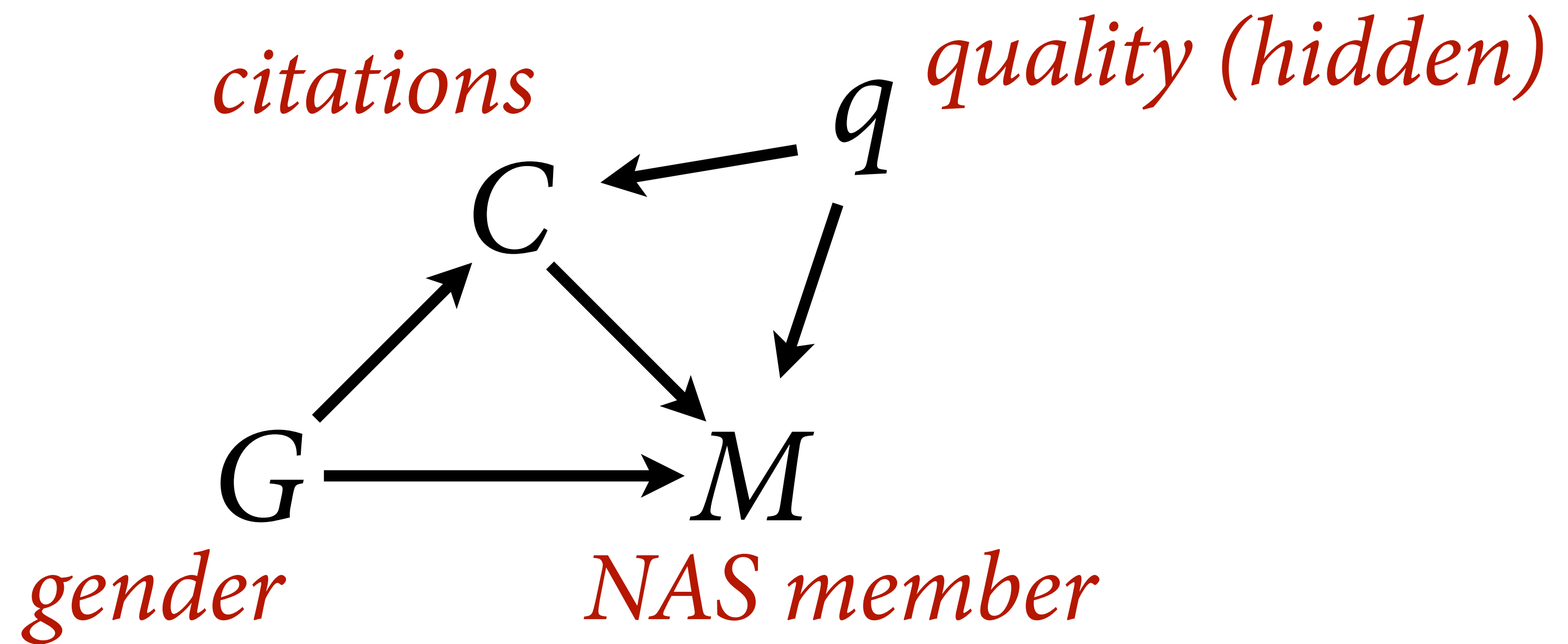
Lerman et al 2022 Gendered citation patterns among the scientific elite

Membership

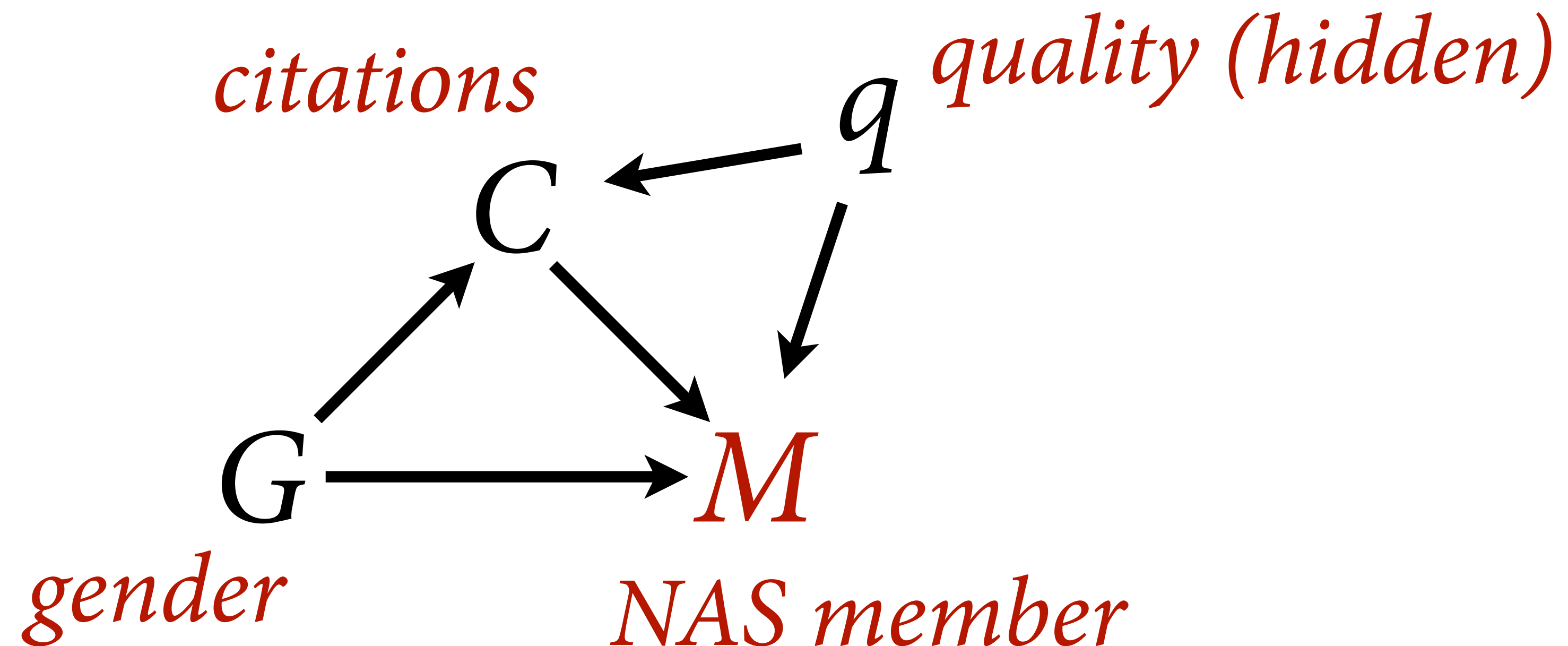
Elections to NAS

Women associated with 3–15 times
higher election rate, controlling for
citations





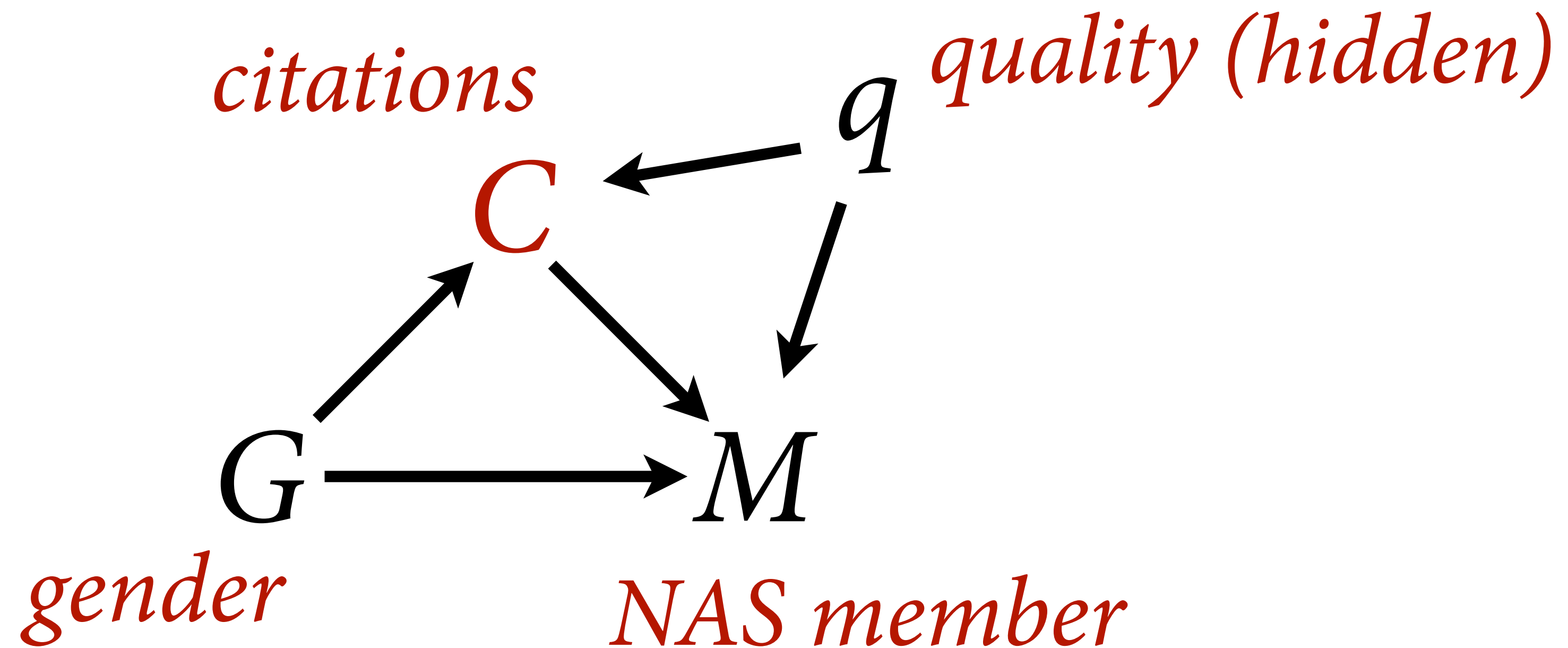
Restrict sample to NAS members, examine citations



Implication of NAS
election paper ->

If men less likely to be elected, then must
have higher q, C to compensate

Control for citations, examine elections to NAS



Implication of
citations paper
->

G is “treatment”. C is a post-treatment variable!
If women less likely to be cited (bias), then
women more likely to be elected because they
have higher q than indicated by C

No causes in; No causes out

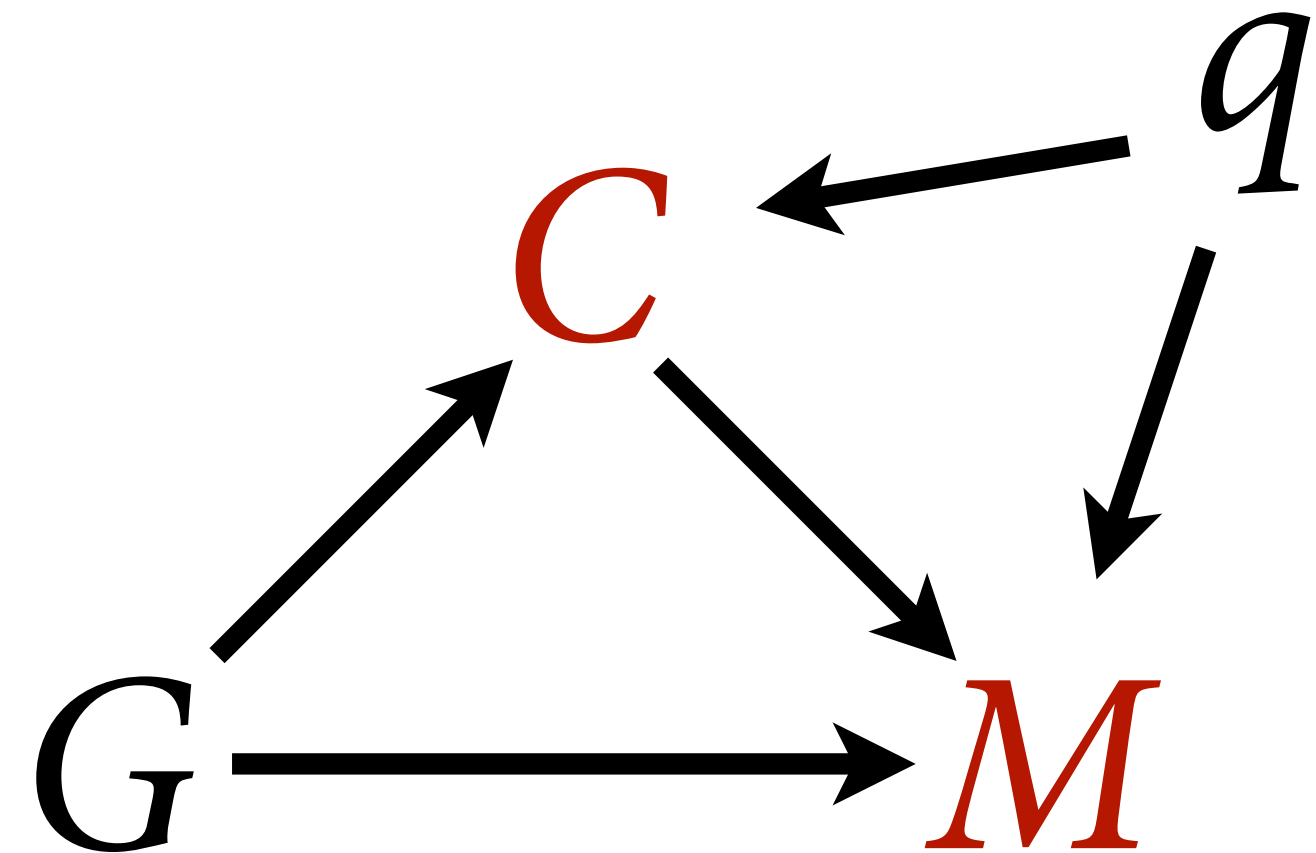
Hyped papers with vague estimands,
unwise adjustment sets

Policy design through collider bias?

We can do better

Strong assumptions required

Qualitative data useful

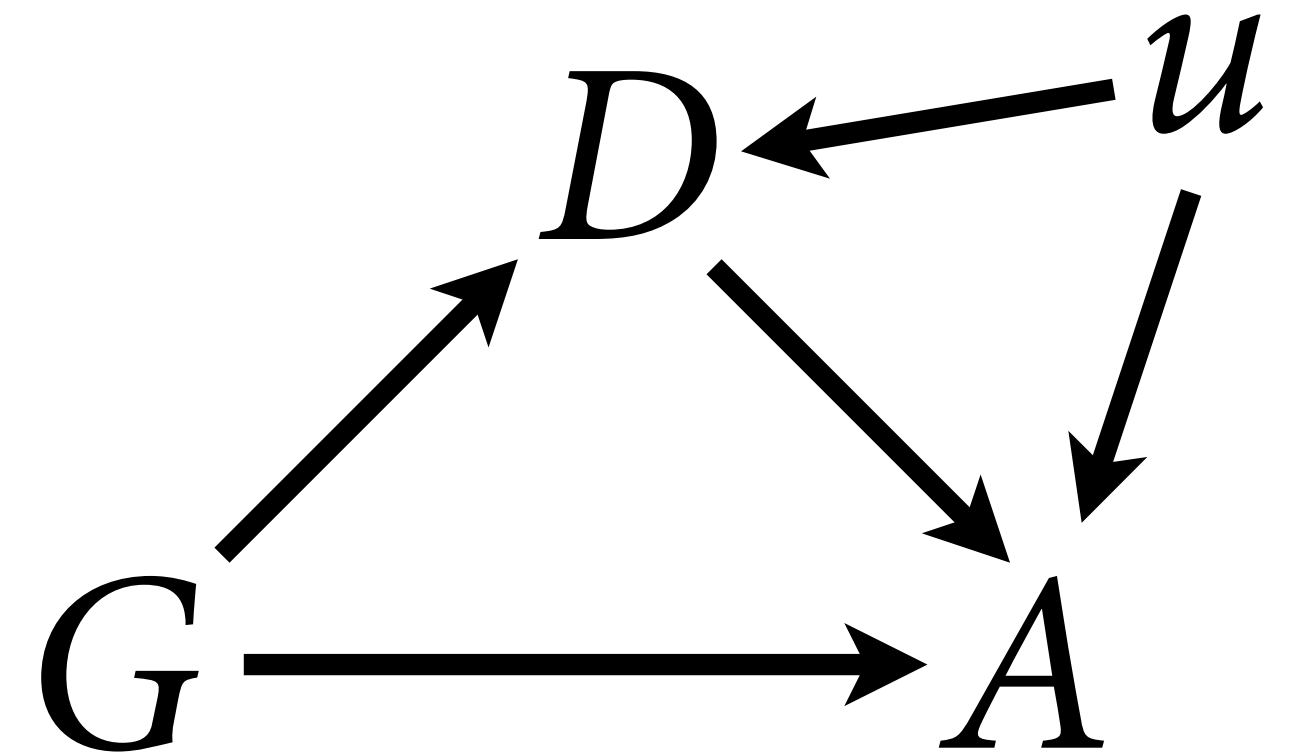


Sensitivity analysis

What are the implications of what we don't know?

Assume confound exists, model its consequences for different strengths/kinds of influence

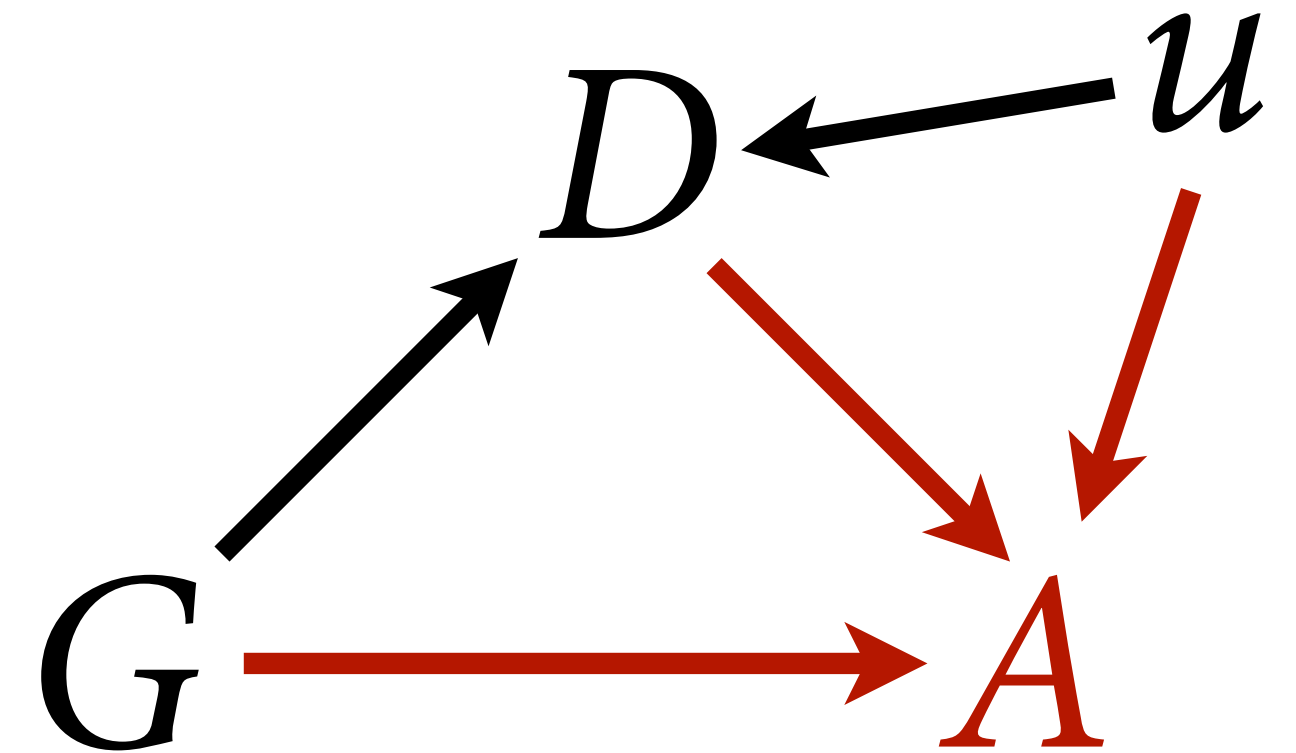
How strong must the confound be to change conclusions?



Sensitivity analysis

$$A_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha[G_i, D_i] + \beta_{G[i]}u_i$$



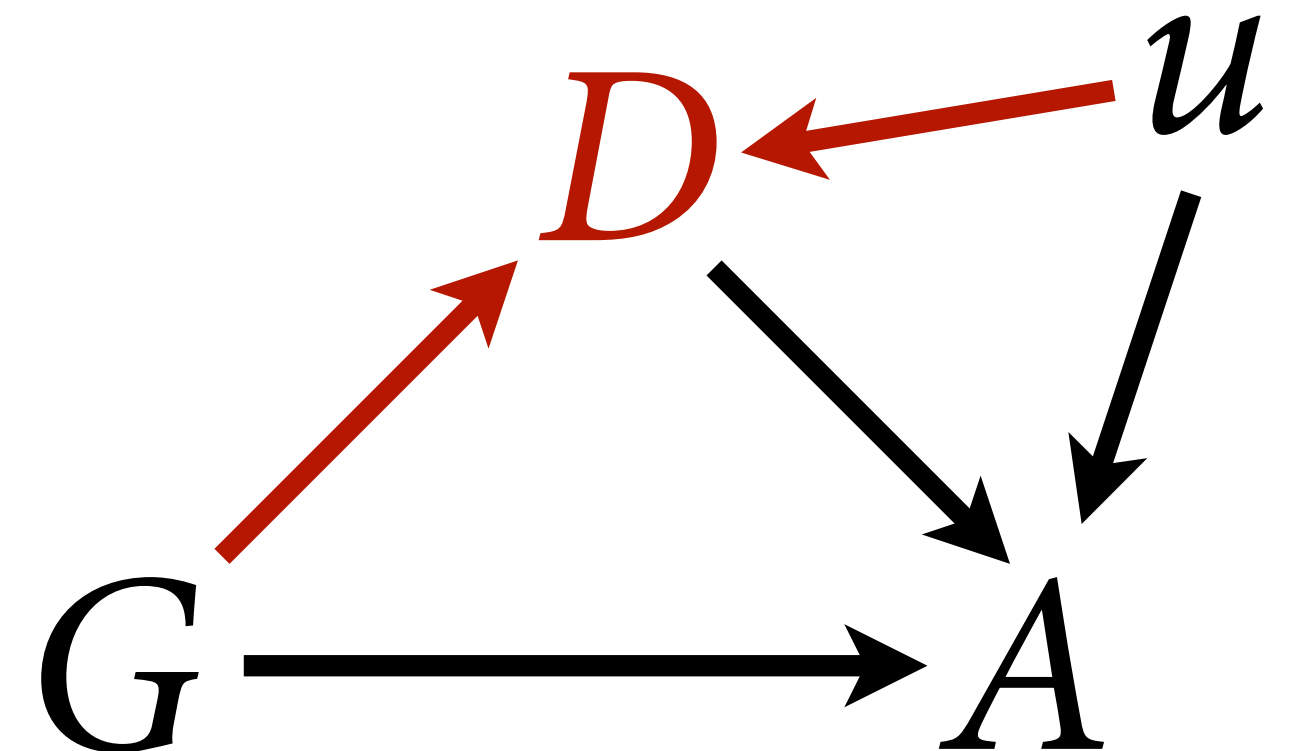
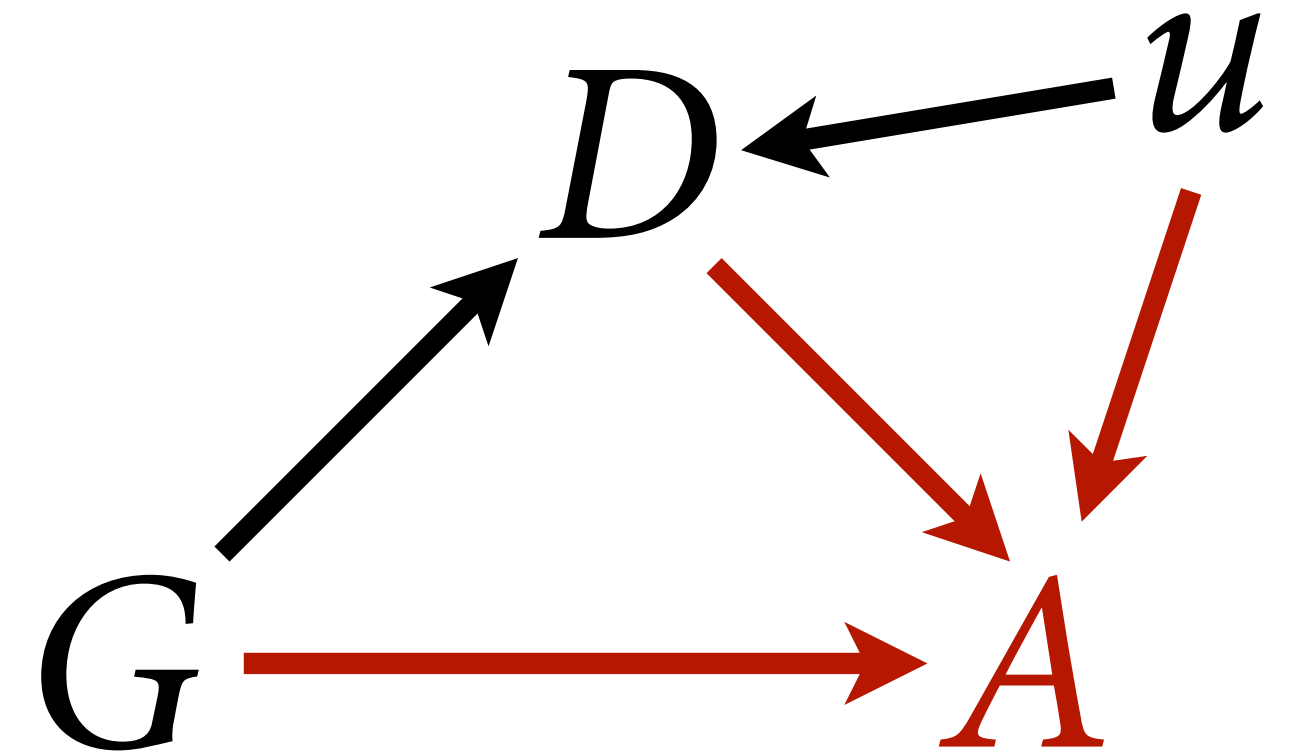
Sensitivity analysis

$$A_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha[G_i, D_i] + \beta_{G[i]}u_i$$

$$(D_i = 2) \sim \text{Bernoulli}(q_i)$$

$$\text{logit}(q_i) = \delta[G_i] + \gamma_{G[i]}u_i$$



```

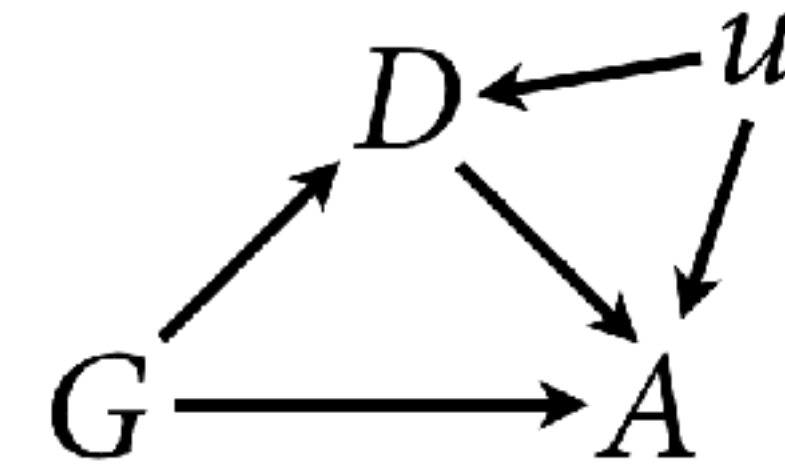
datl <- dat_sim
datl$D2 <- ifelse(datl$D==2,1,0)
datl$N <- length(datl$D)
datl$b <- c(1,1)
datl$g <- c(1,0)

mGDu <- ulam(
  alist(
    # A model
    A ~ bernoulli(p),
    logit(p) <- a[G,D] + b[G]*u[i],
    matrix[G,D]:a ~ normal(0,1),

    # D model
    D2 ~ bernoulli(q),
    logit(q) <- delta[G] + g[G]*u[i],
    delta[G] ~ normal(0,1),

    # declare unobserved u
    vector[N]:u ~ normal(0,1)
  ), data=datl , chains=4 , cores=4 )

```



$$A_i \sim \text{Bernoulli}(p_i)$$

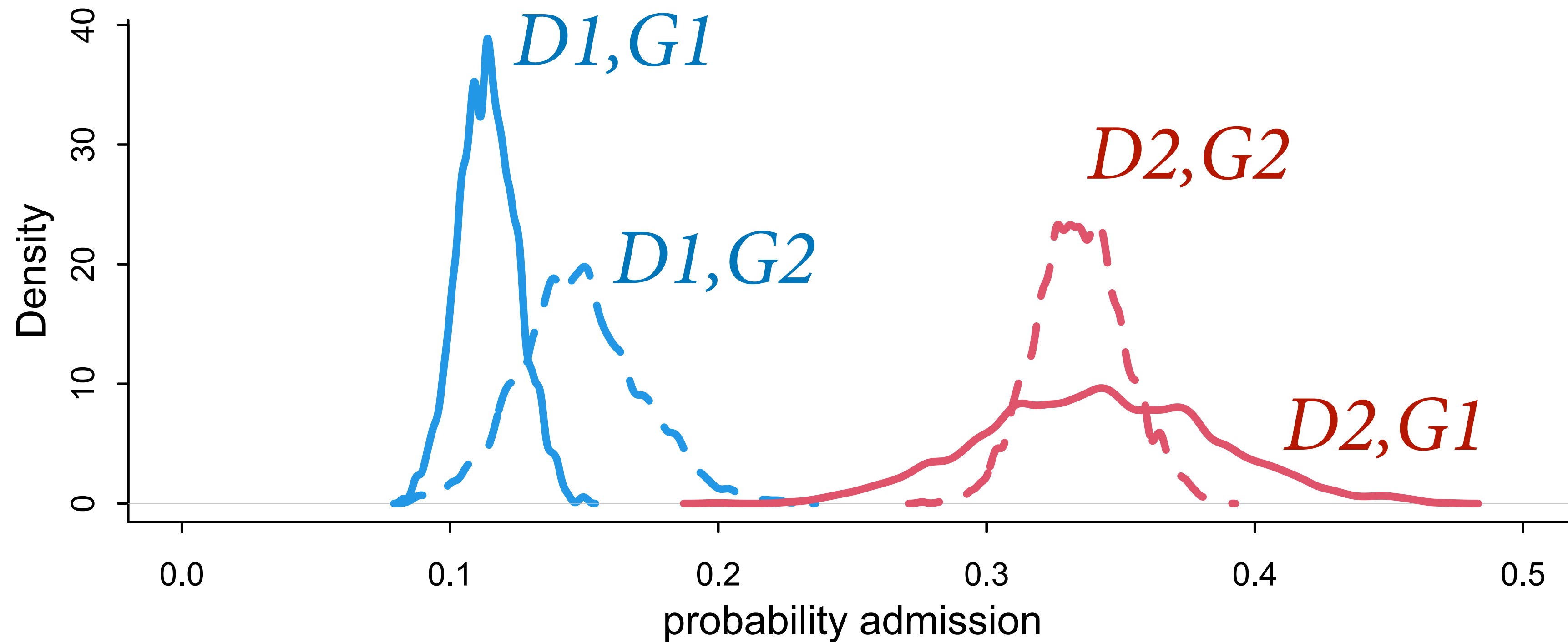
$$\text{logit}(p_i) = \alpha[G_i, D_i] + \beta_{G[i]} u_i$$

$$(D_i = 2) \sim \text{Bernoulli}(q_i)$$

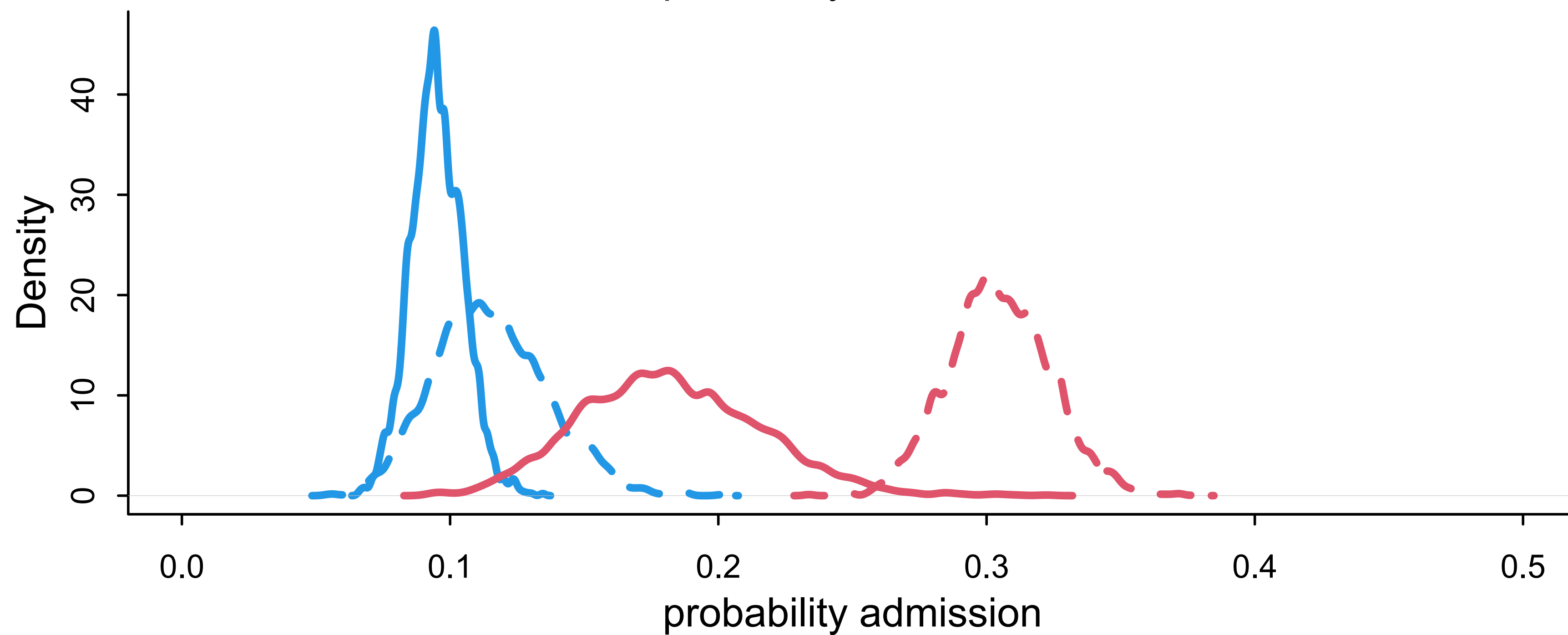
$$\text{logit}(q_i) = \delta[G_i] + \gamma_{G[i]} u_i$$

$$u_j \sim \text{Normal}(0,1)$$

Ignore
confound



Assume
confound



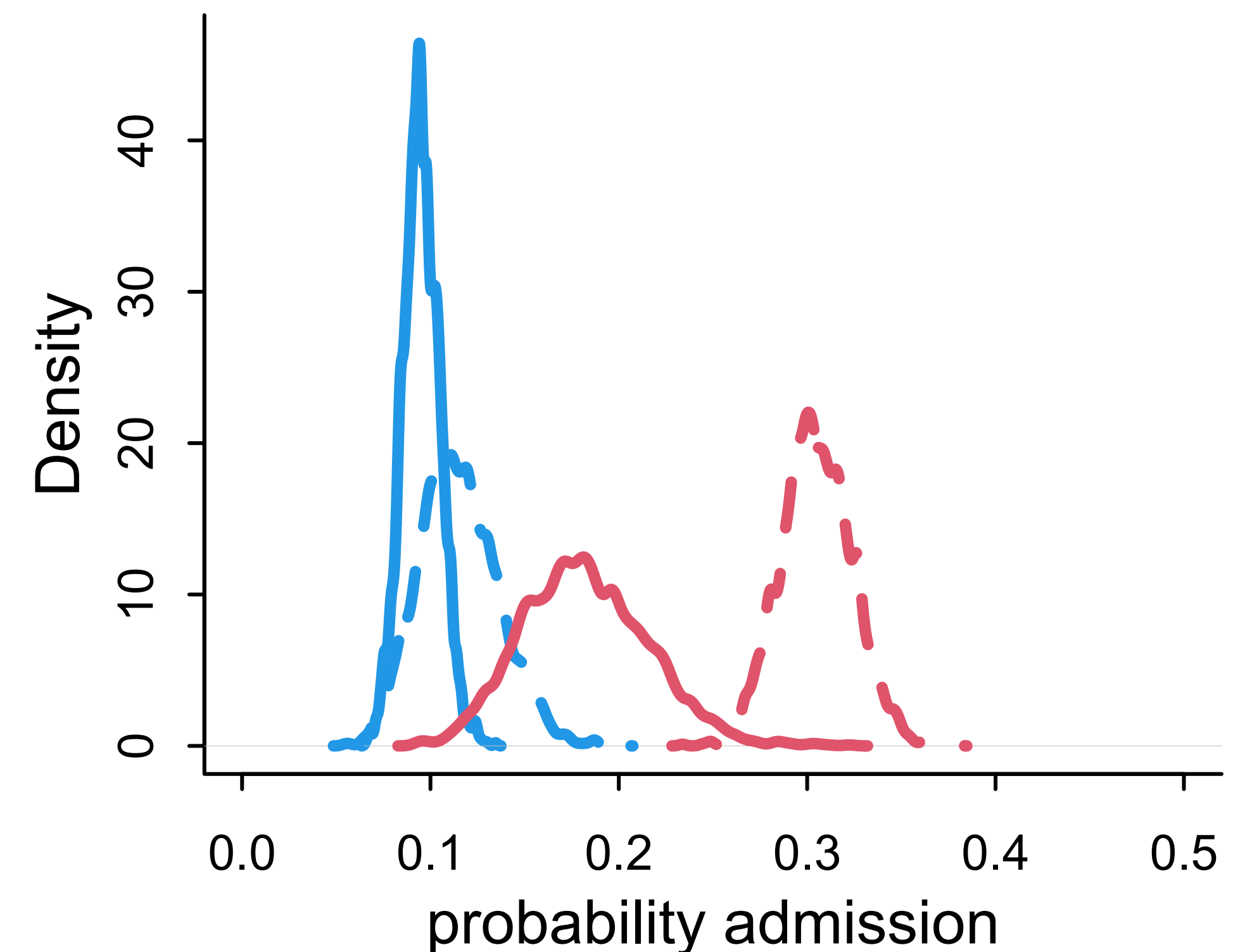
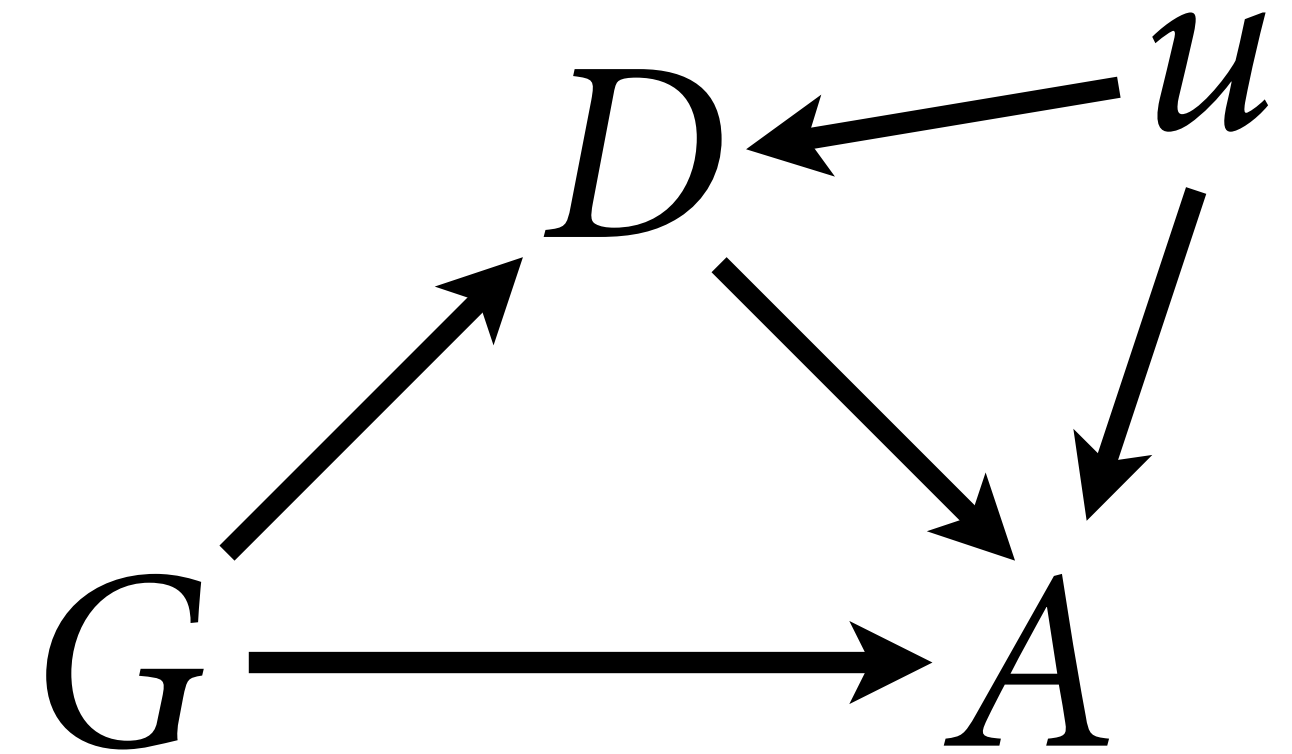
Sensitivity analysis

What are the implications of what we don't know?

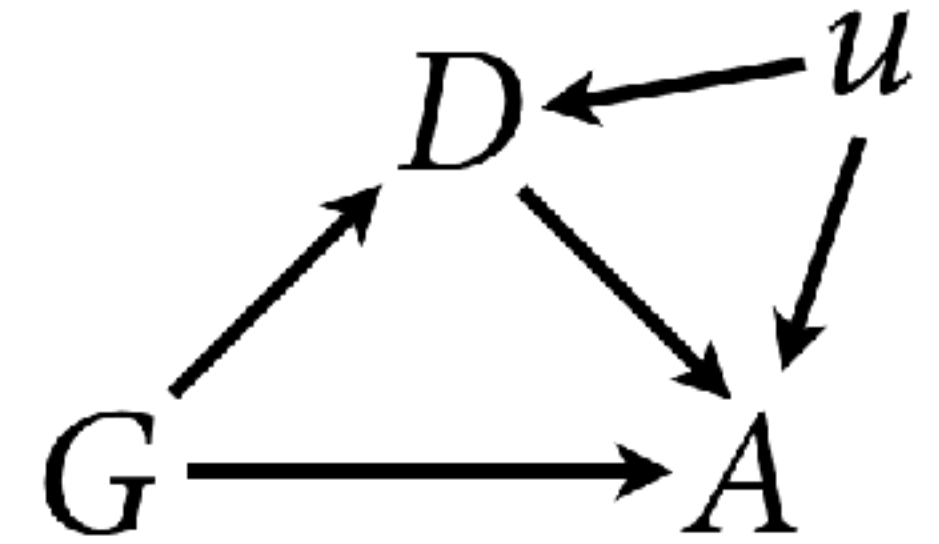
Somewhere between pure simulation and pure analysis

Vary confound strength over range and show how results change –or– vary other effects and estimate confound strength

Confounds persist — don't pretend



*More parameters (2006)
than observations (2000)!*



$$A_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \alpha[G_i, D_i] + \beta_{G[i]}u_i$$

$$(D_i = 2) \sim \text{Bernoulli}(q_i)$$

$$\text{logit}(q_i) = \delta[G_i] + \gamma_{G[i]}u_i$$

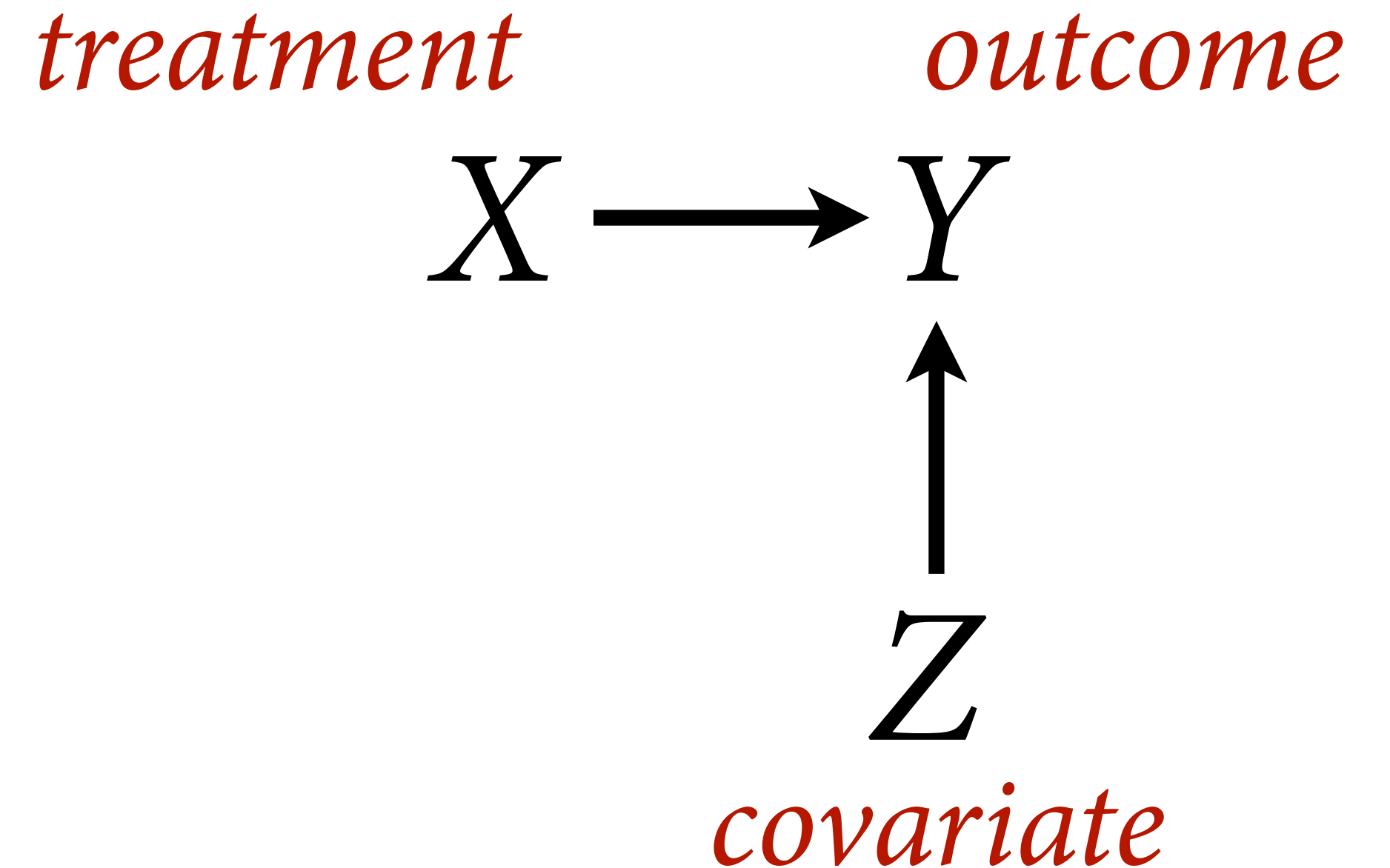
$$u_j \sim \text{Normal}(0,1)$$



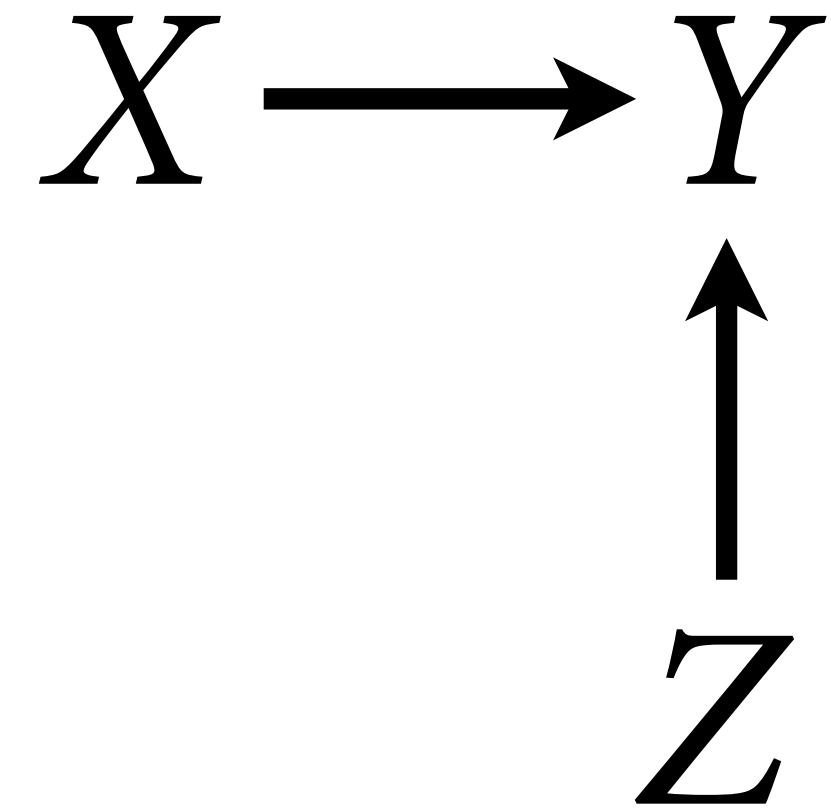
Non-linear haunting

In event models, effect reversal can arise in several ways

Example: Base rate differences



```
# base rate differences erasing effect of X
N <- 1000
X <- rnorm(N)
Z <- rbern(N)
p <- inv_logit(X + ifelse(Z==1,-1,5))
Y <- rbern(N,p)
```

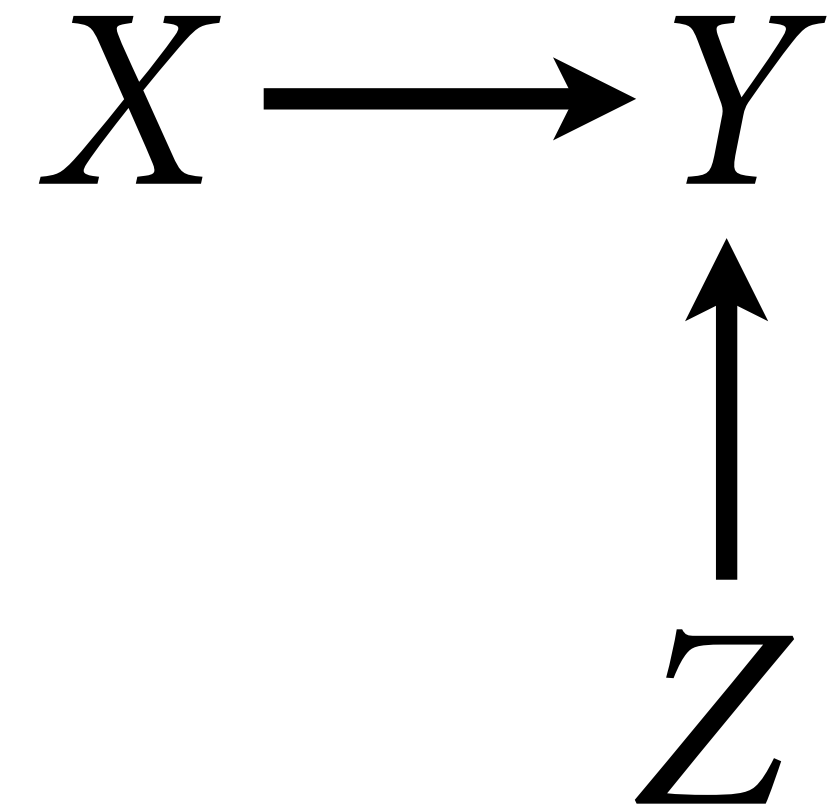



```

# base rate differences erasing effect of X
N <- 1000
X <- rnorm(N)
Z <- rbern(N)
p <- inv_logit(X + ifelse(Z==1,-1,5))
Y <- rbern(N,p)

mX <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a + b*X,
    a ~ dnorm(0,1),
    b ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X) )

```



$$\text{logit}(p_i) = \alpha + \beta X_i$$

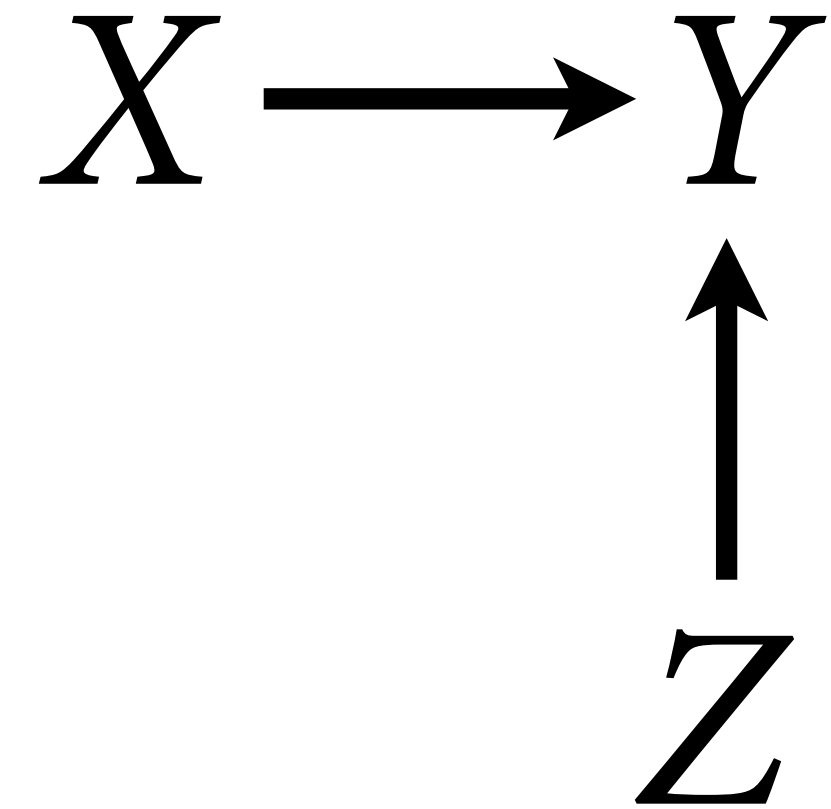
```

# base rate differences erasing effect of X
N <- 1000
X <- rnorm(N)
Z <- rbern(N)
p <- inv_logit(X + ifelse(Z==1,-1,5))
Y <- rbern(N,p)

mX <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a + b*X,
    a ~ dnorm(0,1),
    b ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X) )

mXZ <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a + b[Z]*X,
    a ~ dnorm(0,1),
    b[Z] ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X,Z=Z+1) )

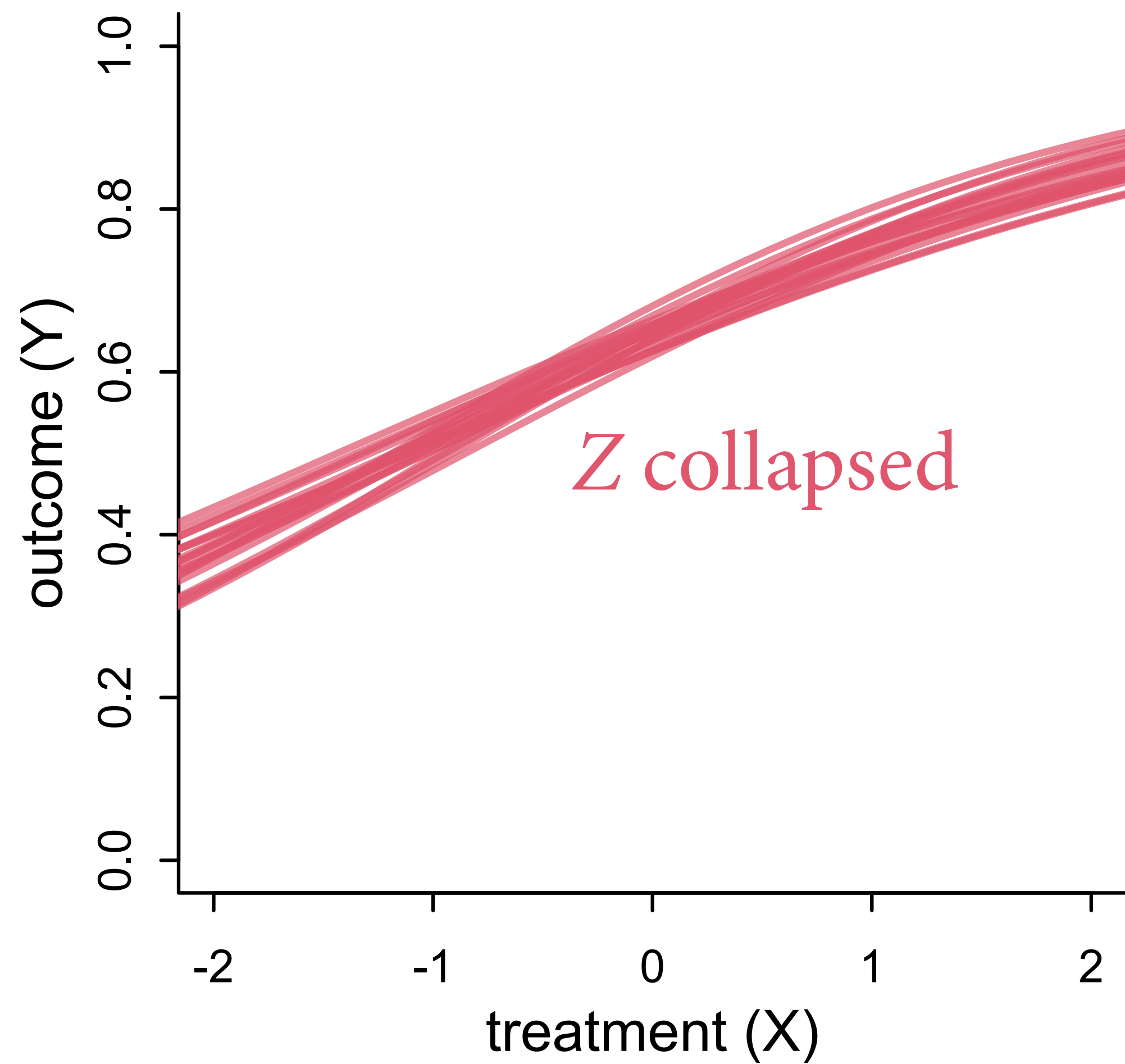
```



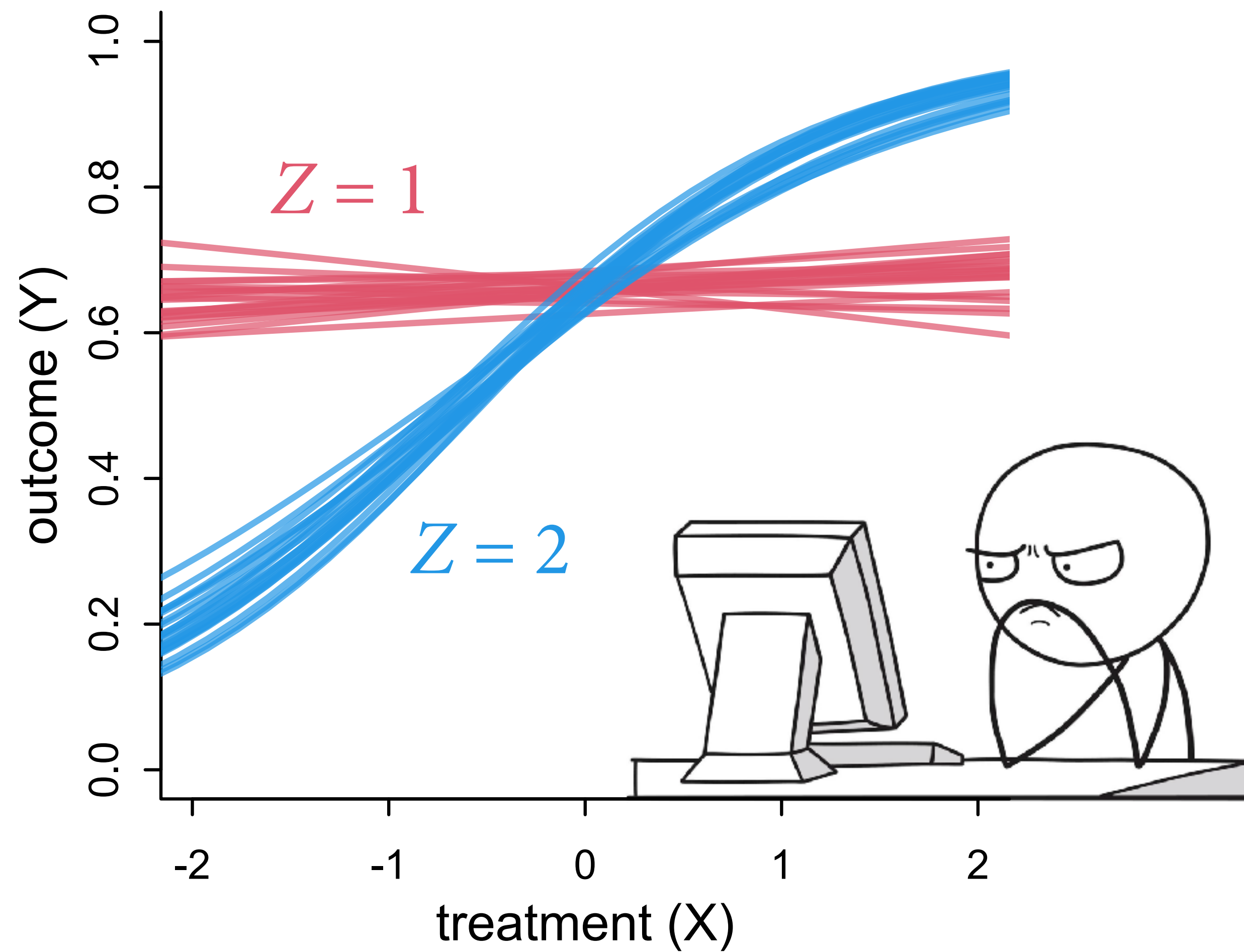
$$\text{logit}(p_i) = \alpha + \beta X_i$$

$$\text{logit}(p_i) = \alpha + \beta_{Z[i]} X_i$$

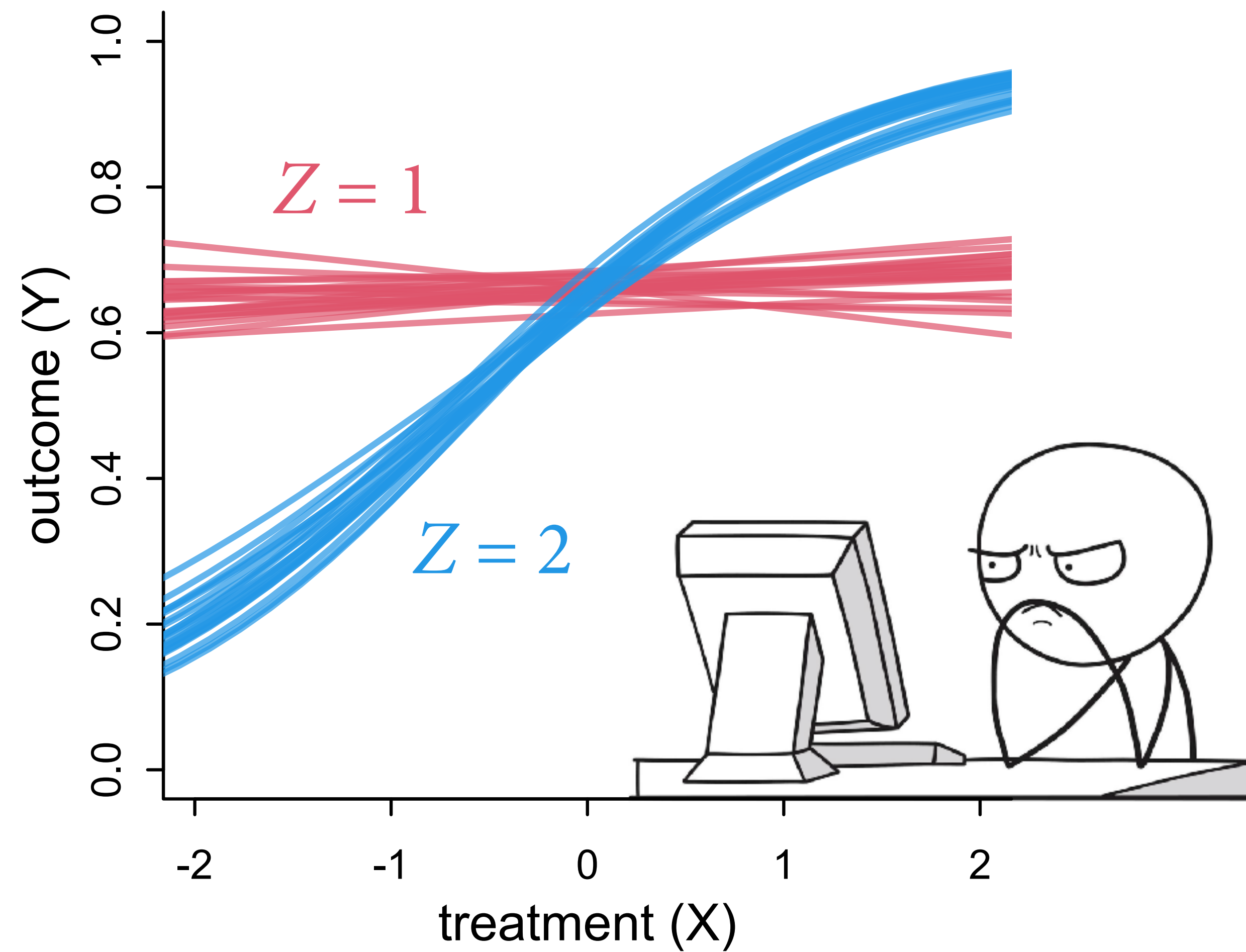
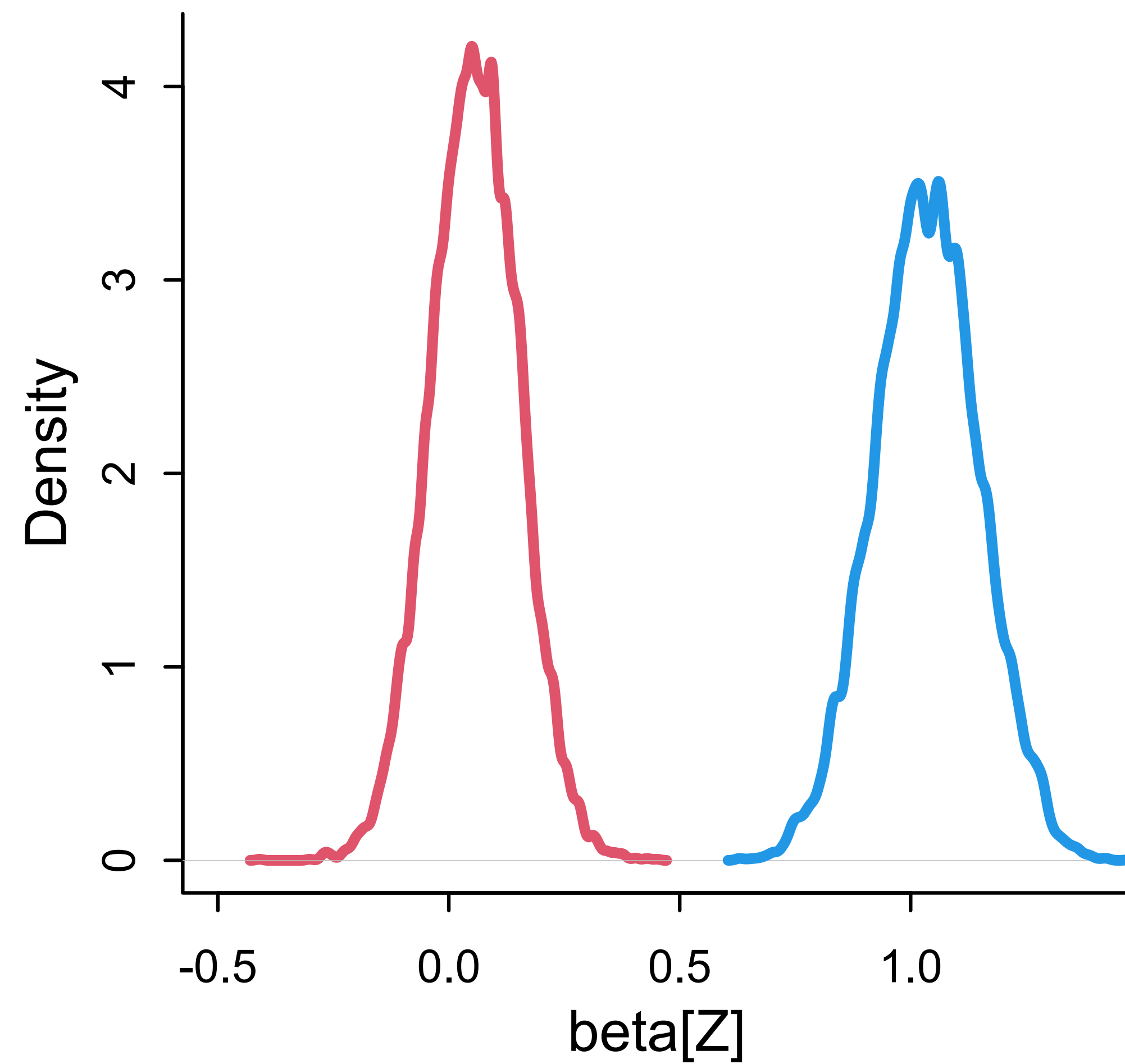
$$\text{logit}(p_i) = \alpha + \beta X_i$$



$$\text{logit}(p_i) = \alpha + \beta_{Z[i]} X_i$$



$$\text{logit}(p_i) = \alpha + \beta_{Z[i]}X_i$$




```

mX <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a + b*X,
    a ~ dnorm(0,1),
    b ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X) )

```

$$\text{logit}(p_i) = \alpha + \beta X_i$$

```

mXZ <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a + b[Z]*X,
    a ~ dnorm(0,1),
    b[Z] ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X,Z=Z+1) )

```

$$\text{logit}(p_i) = \alpha + \beta_{Z[i]} X_i$$

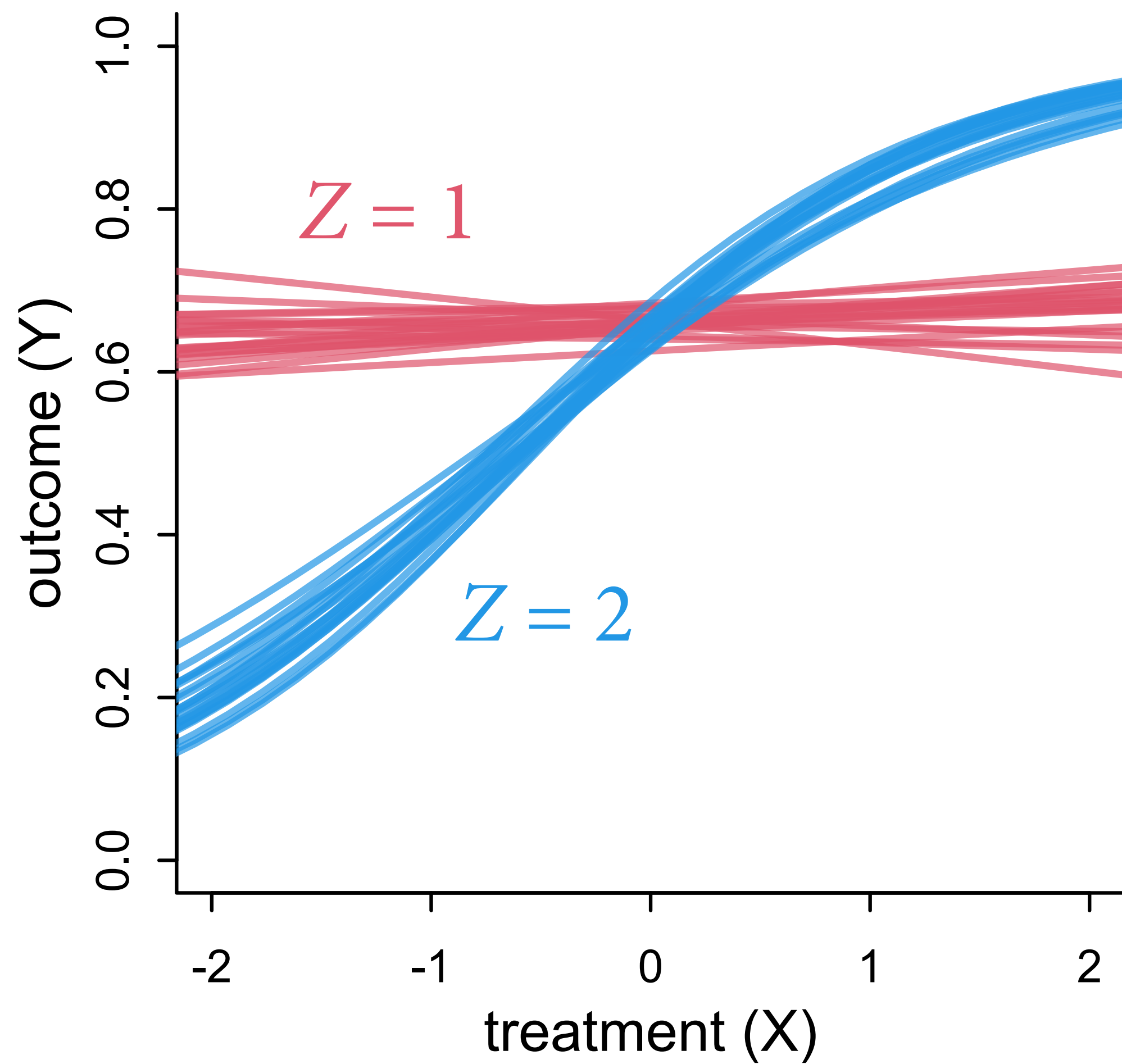
```

mXZ2 <- quap(
  alist(
    Y ~ dbern(p),
    logit(p) <- a[Z] + b[Z]*X,
    a[Z] ~ dnorm(0,1),
    b[Z] ~ dnorm(0,1)
  ) , data=list(Y=Y,X=X,Z=Z+1) )

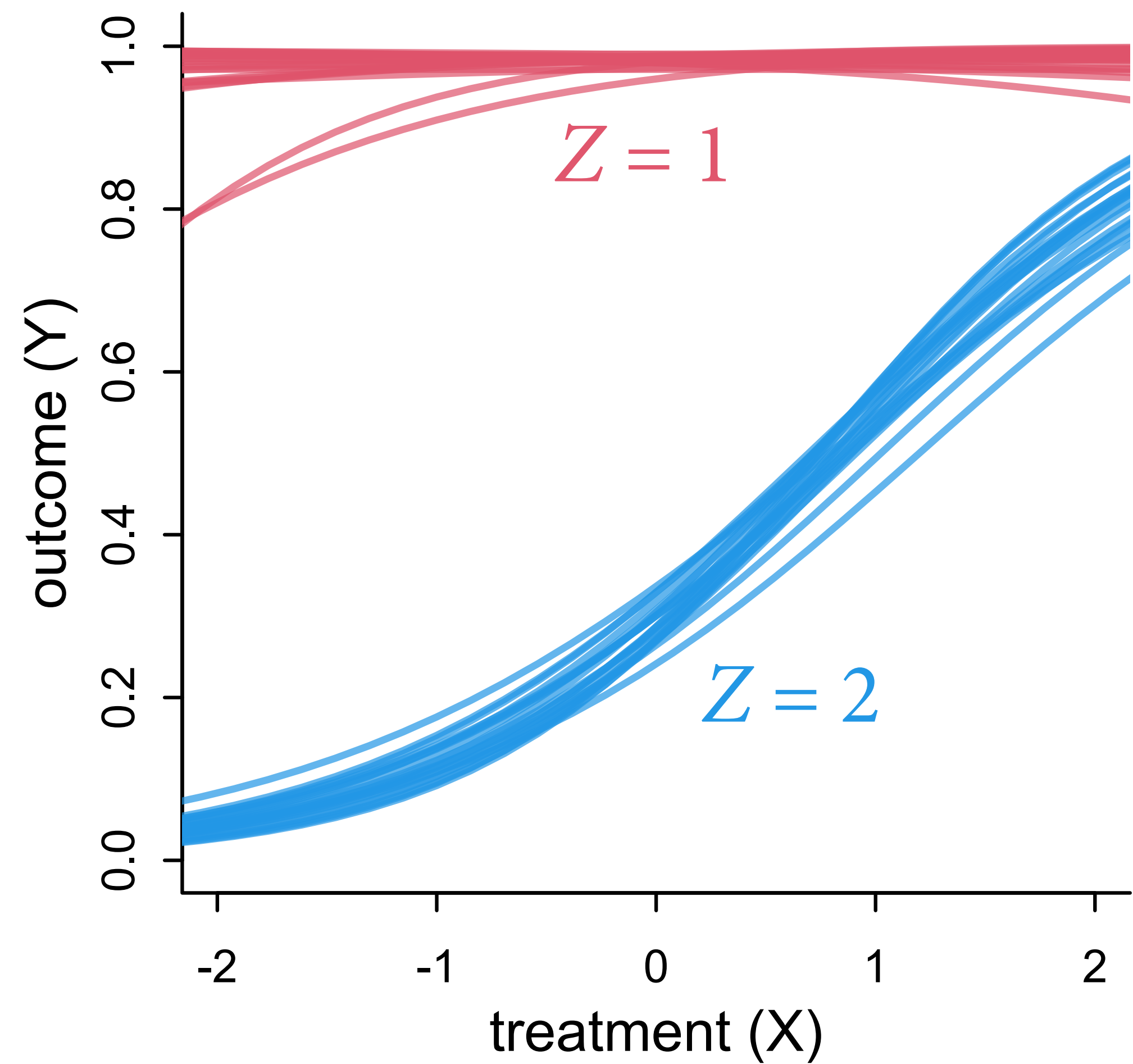
```

$$\text{logit}(p_i) = \alpha_{Z[i]} + \beta_{Z[i]} X_i$$

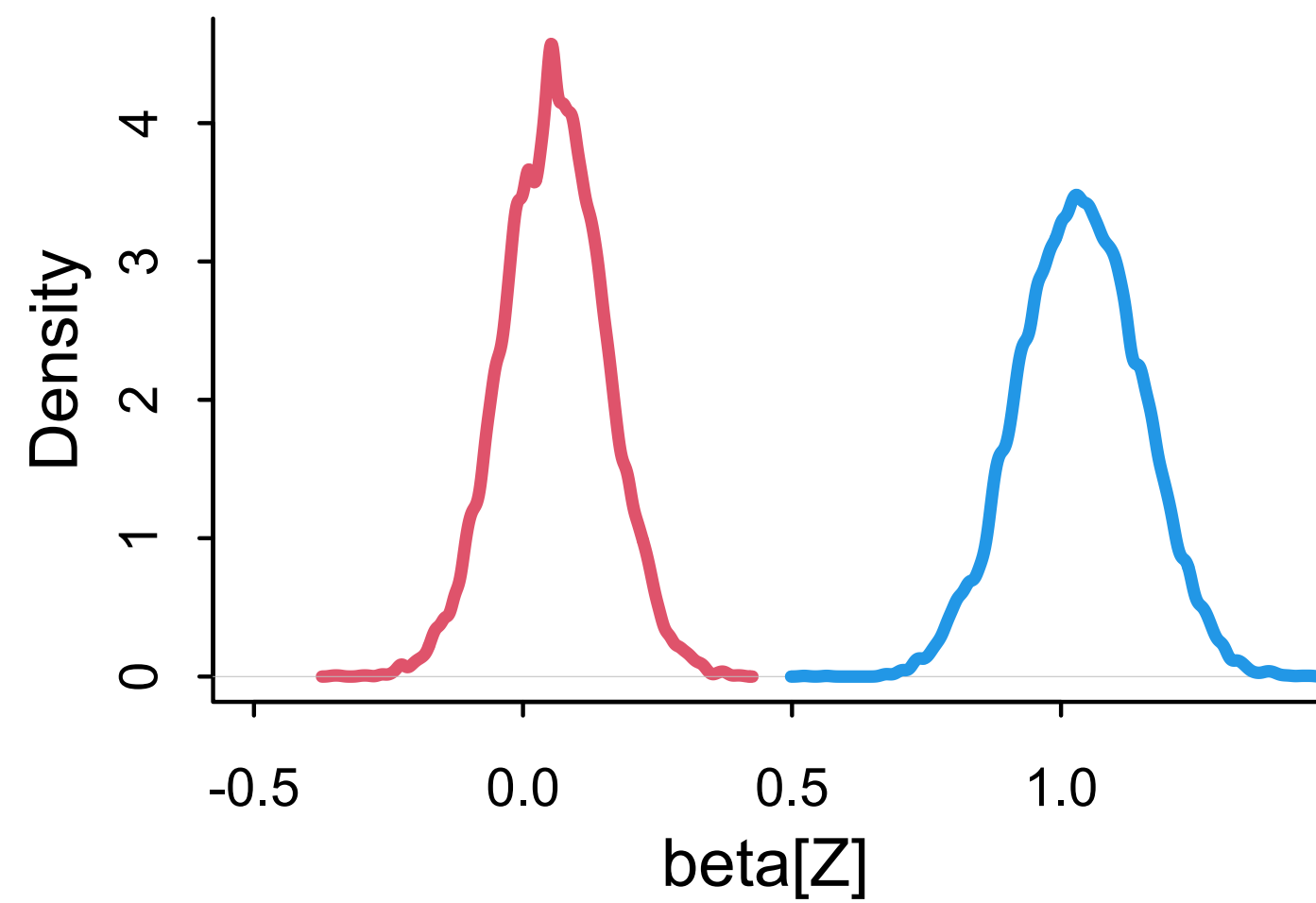
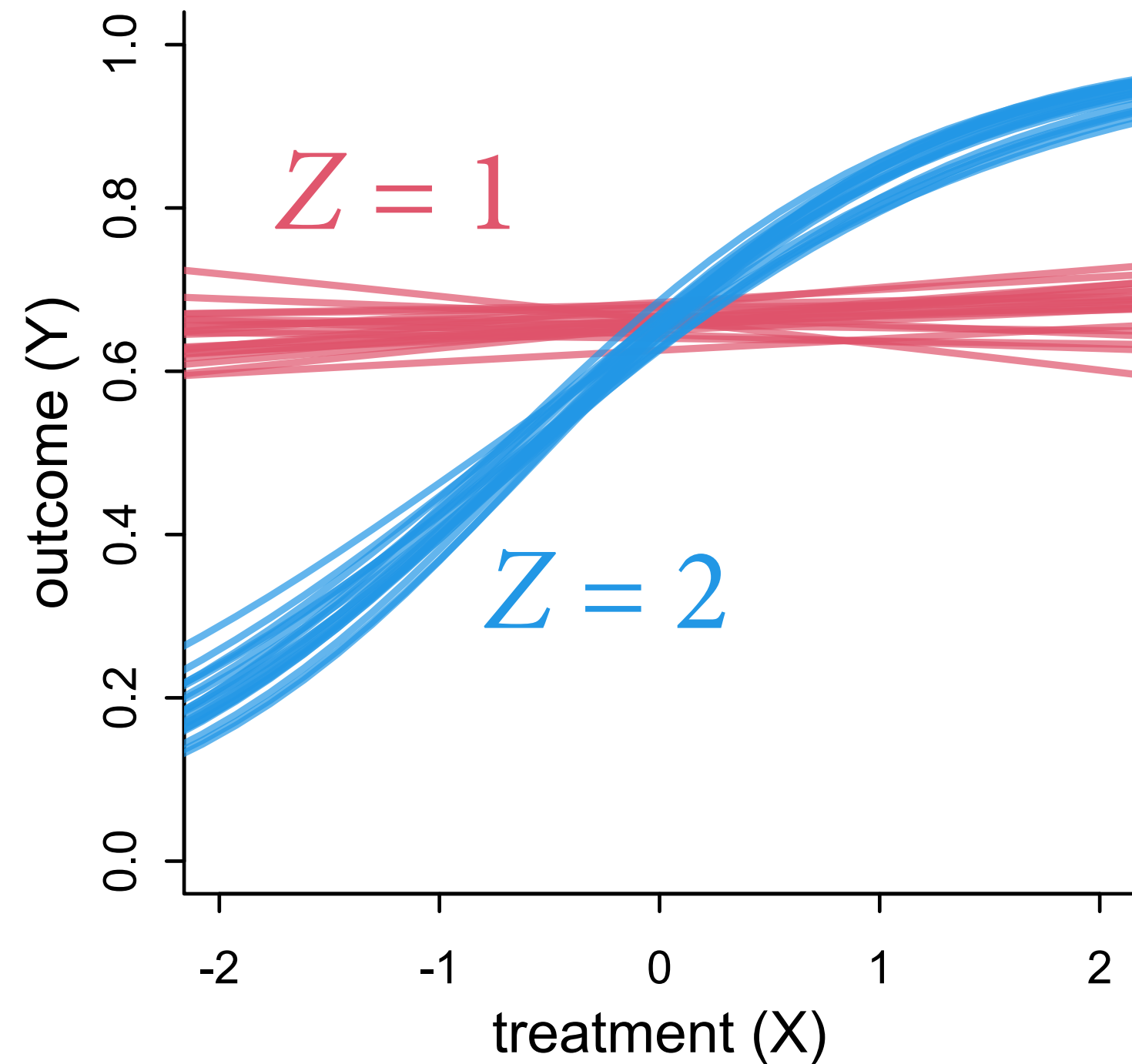
$$\text{logit}(p_i) = \alpha + \beta_{Z[i]}X_i$$



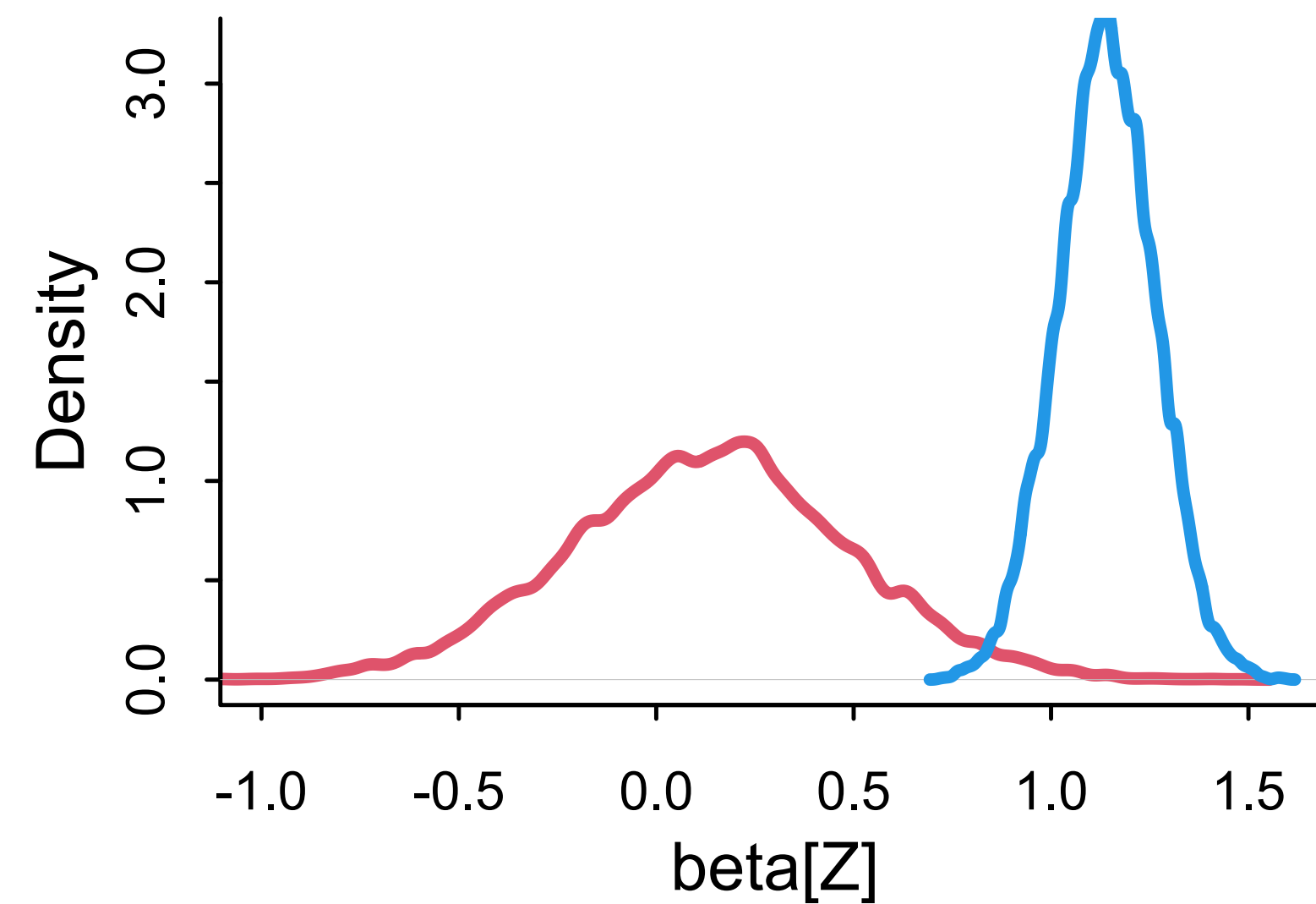
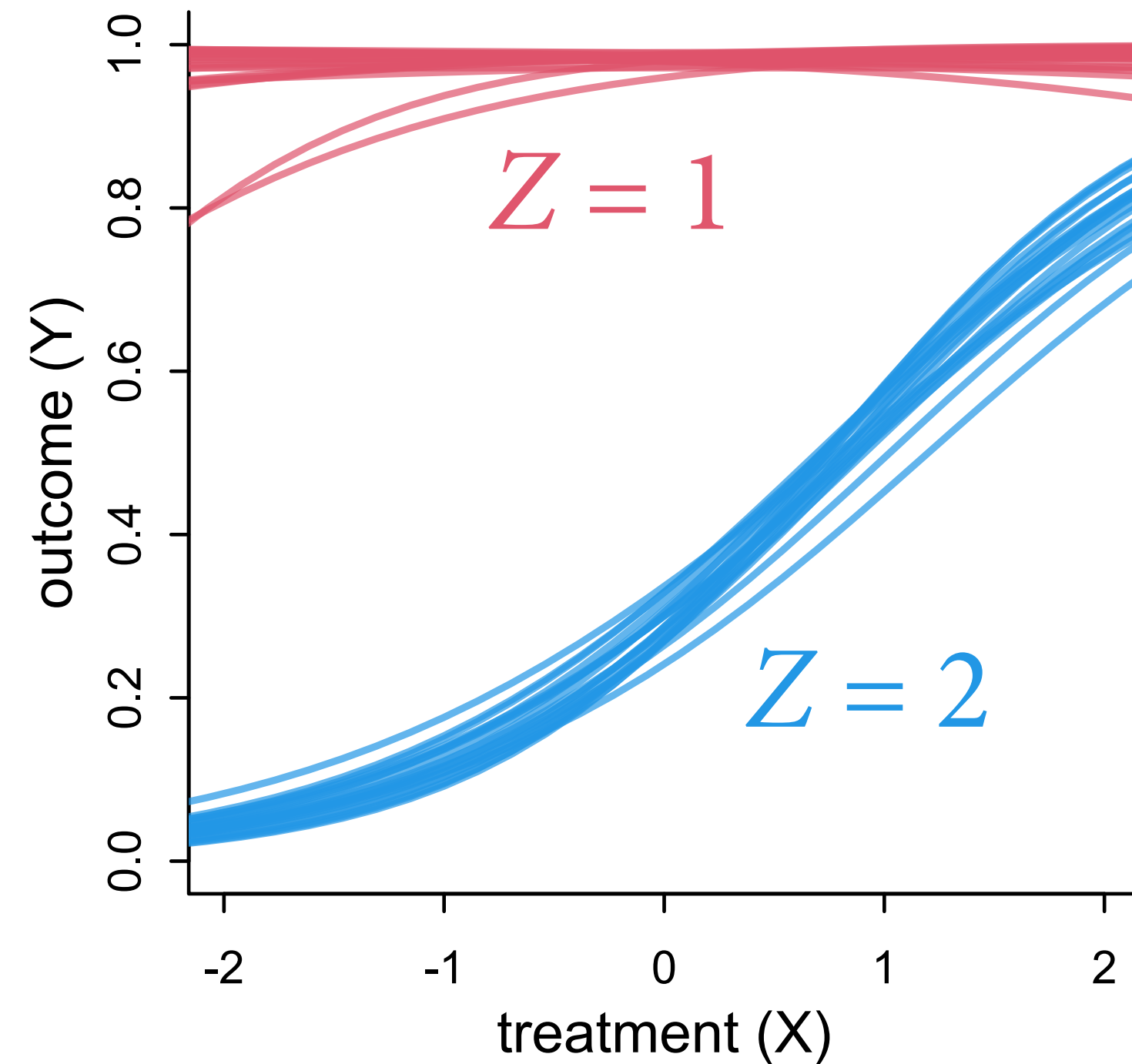
$$\text{logit}(p_i) = \alpha_{Z[i]} + \beta_{Z[i]}X_i$$



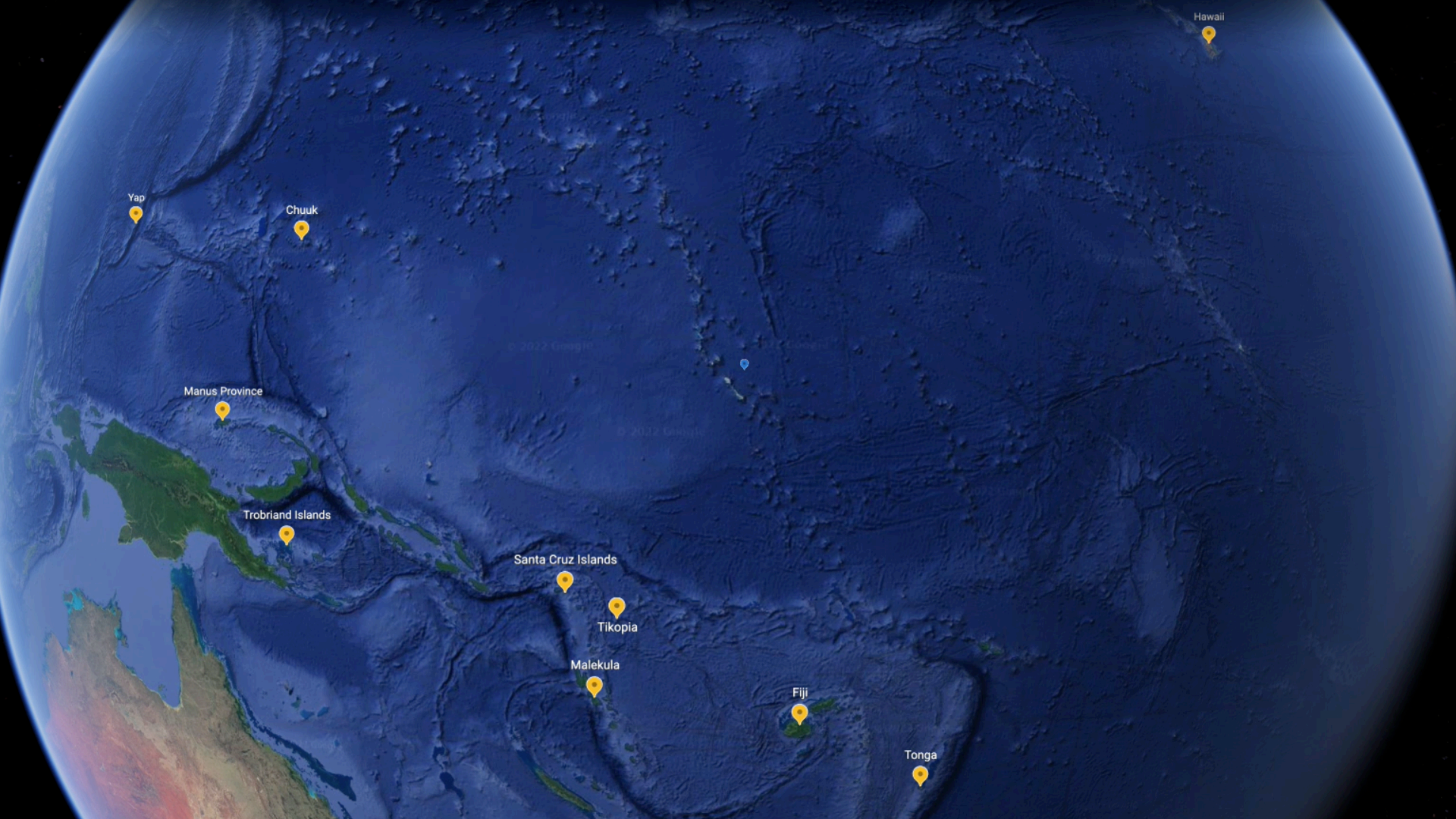
$$\text{logit}(p_i) = \alpha + \beta_{Z[i]}X_i$$



$$\text{logit}(p_i) = \alpha_{Z[i]} + \beta_{Z[i]}X_i$$



PAUSE



Hawaii

Yap

Chuuk

Manus Province

Trobriand Islands

Santa Cruz Islands

Tikopia

Malekula

Fiji

Tonga

Oceanic Technology

How is technological complexity related to population size?

To social structure?

data(Kline)

Influence of **population size** and **contact** on **total tools**



Kline & Boyd 2010 Population size predicts technological complexity in Oceania

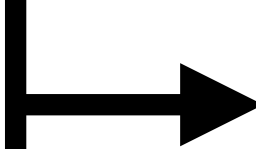
Oceanic Technology

data(Kline)

	culture	population	contact	total_tools	mean_TU
1	Malekula	1100	low	13	3.2
2	Tikopia	1500	low	22	4.7
3	Santa Cruz	3600	low	24	4.0
4	Yap	4791	high	43	5.0
5	Lau Fiji	7400	high	33	5.0
6	Trobriand	8000	high	19	4.0
7	Chuuk	9200	high	40	3.8
8	Manus	13000	low	28	6.6
9	Tonga	17500	high	55	5.4
10	Hawaii	275000	low	71	6.6

Kline & Boyd 2010 Population size predicts technological complexity in Oceania

Population



Innovations



Tools

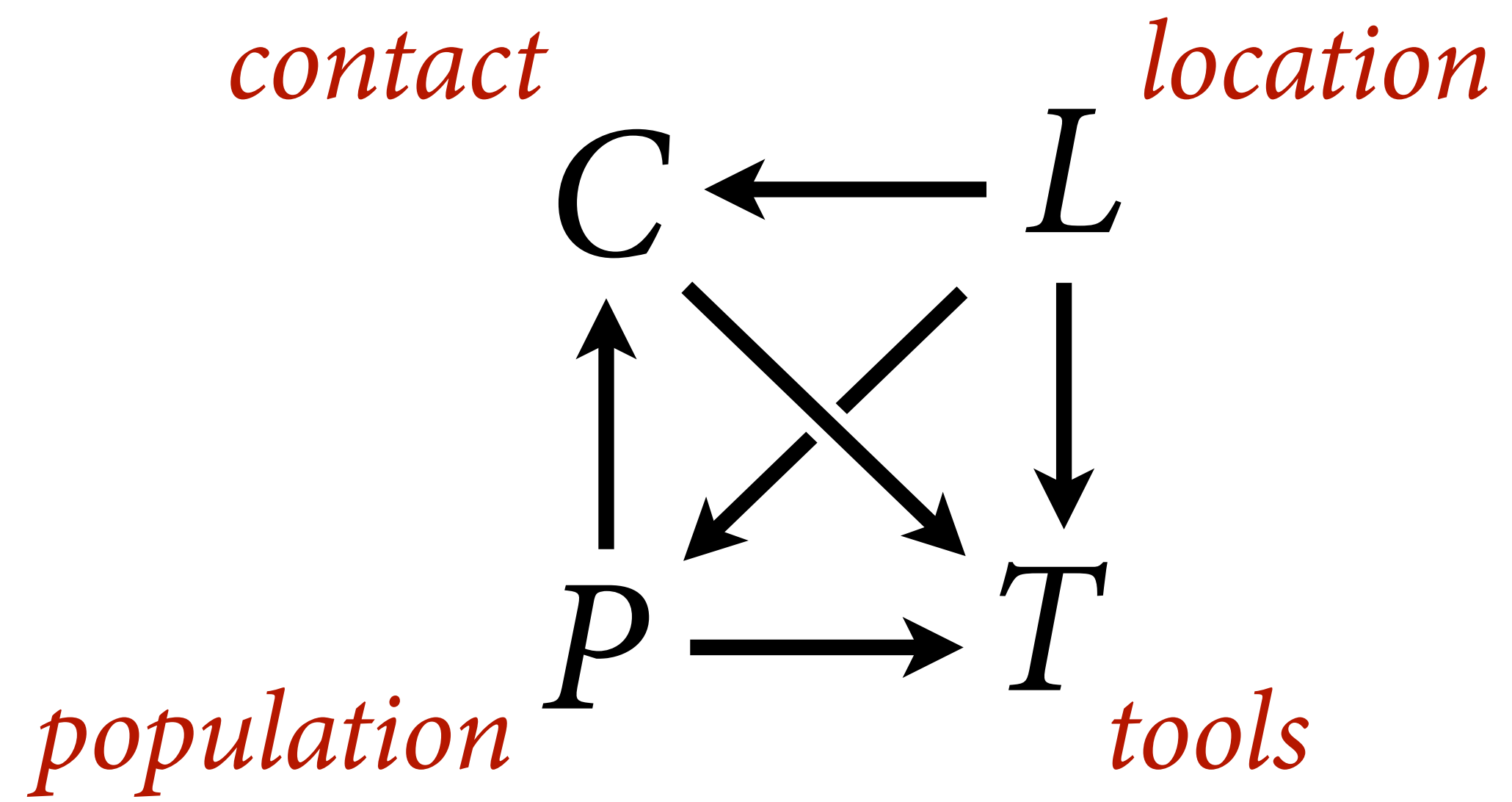


Loss

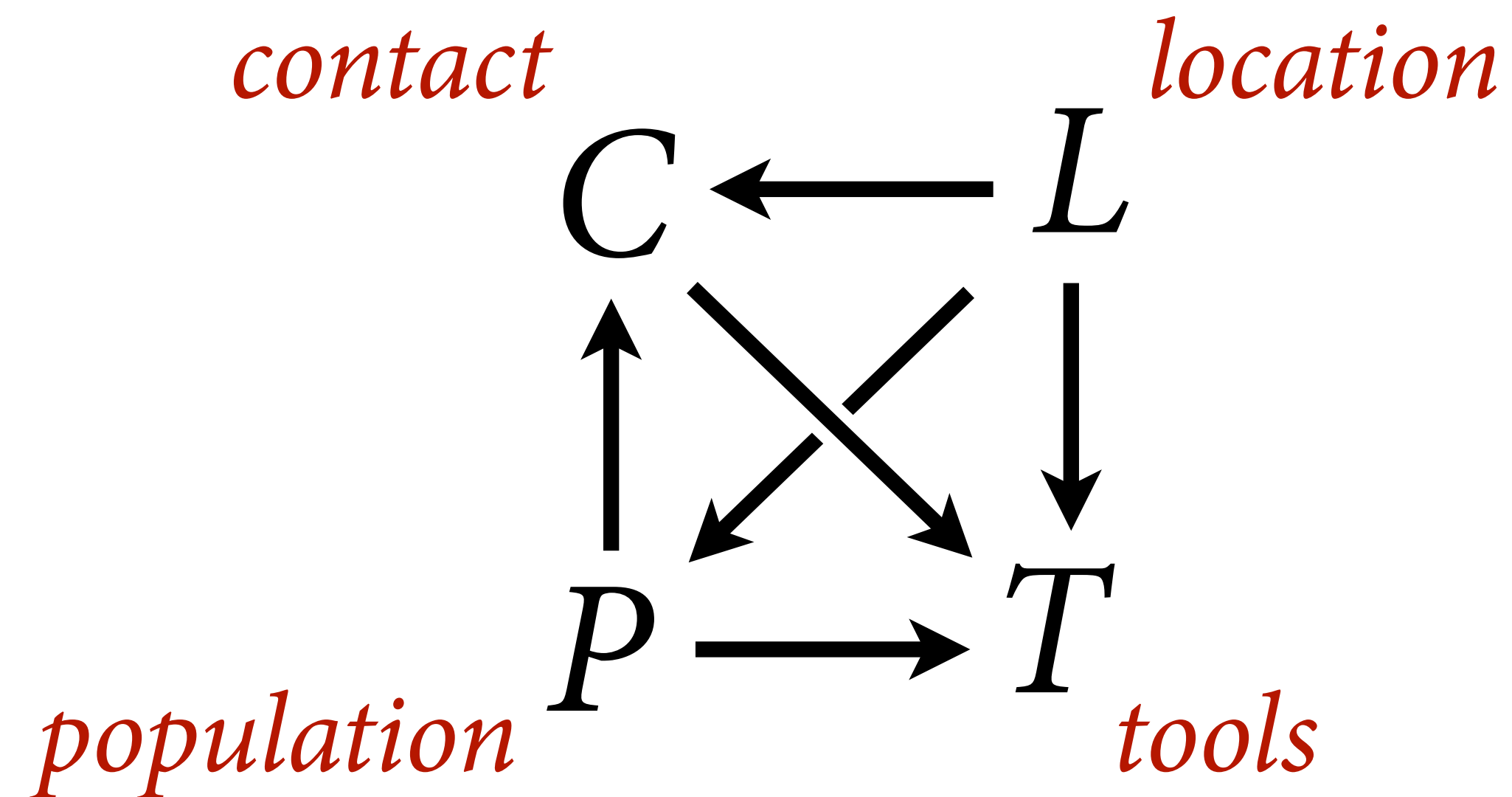


Contact





We don't include L yet,
but will show the way to model it
in a later lecture.



Adjustment set for P : L

Also want to stratify by C ,
to study interaction

Modeling tools

Tool count is not binomial: No maximum

Poisson distribution: Very high maximum and very low probability of each success

Here: Many many possible technologies, very few realized in any one place

Poisson link is log

Poisson distribution takes shape from expected value

Must be positive

Exponential scaling can be surprising!

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \alpha + \beta x_i$$

$$\lambda_i = \exp(\alpha + \beta x_i)$$

Poisson (poison) priors

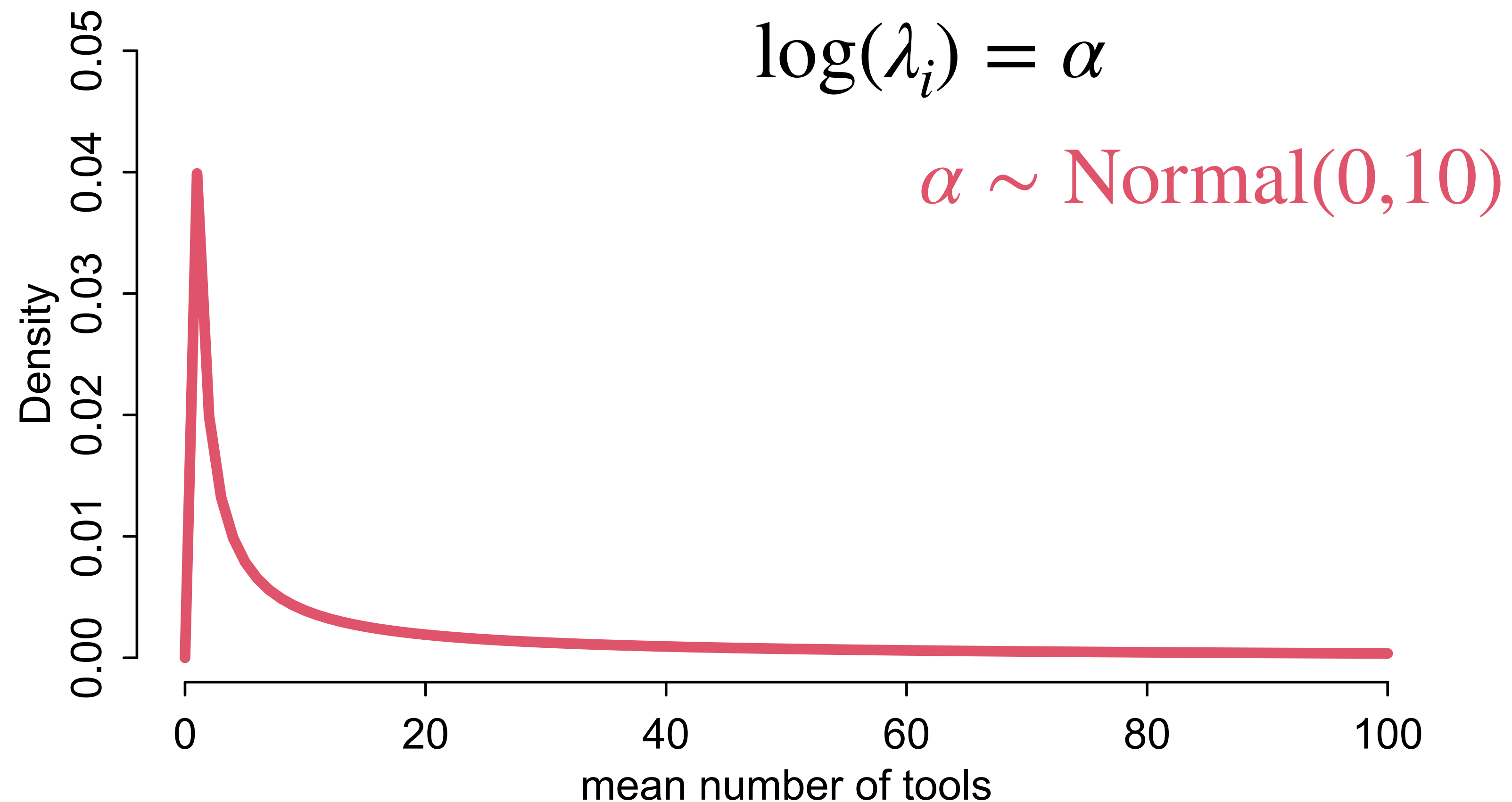
Exponential scaling can be surprising

$$\log(\lambda_i) = \alpha$$

$$\alpha \sim \text{Normal}(0, 10)$$

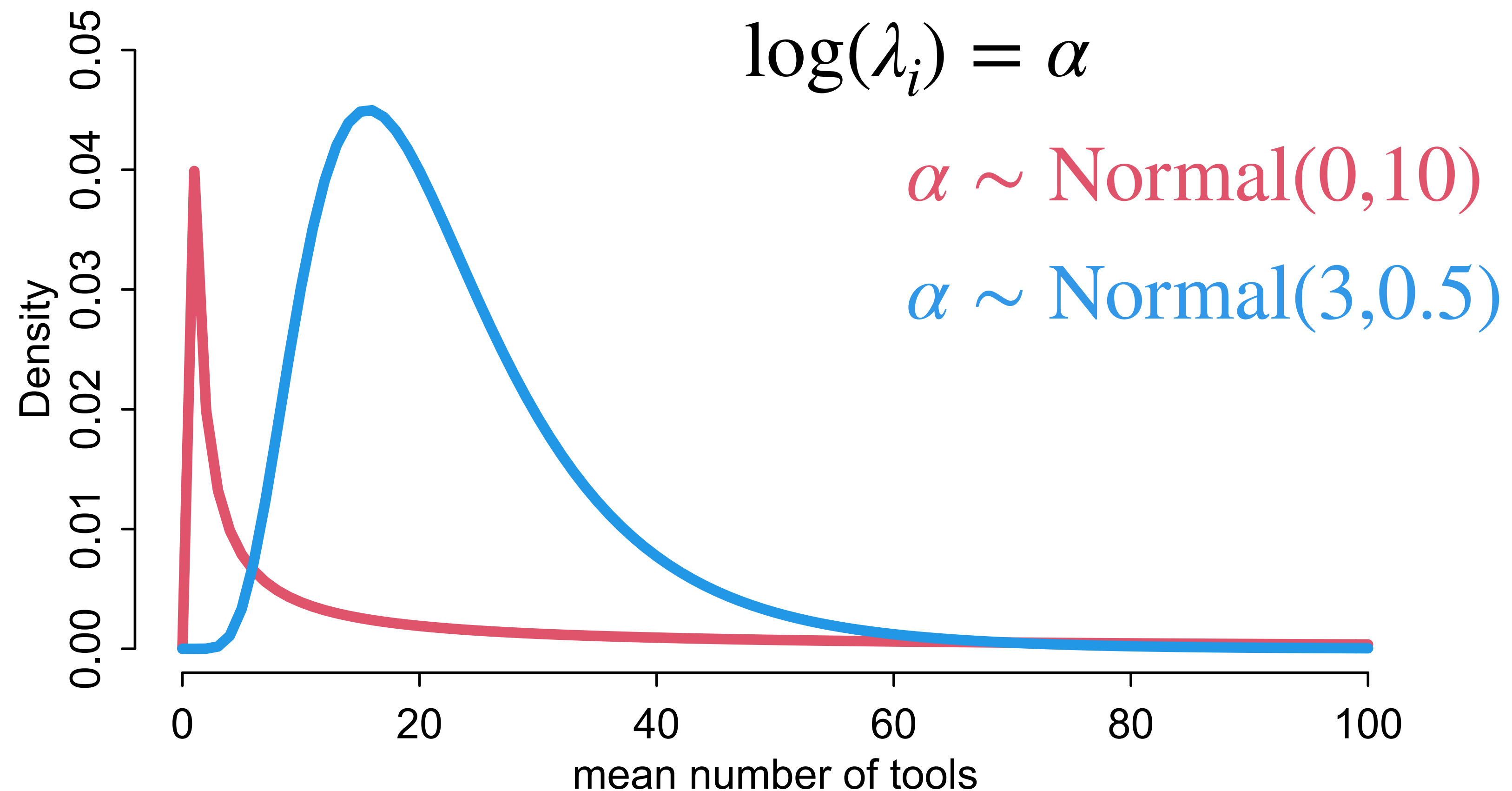
Poisson (poison) priors

Exponential scaling can be surprising



Poisson (poison) priors

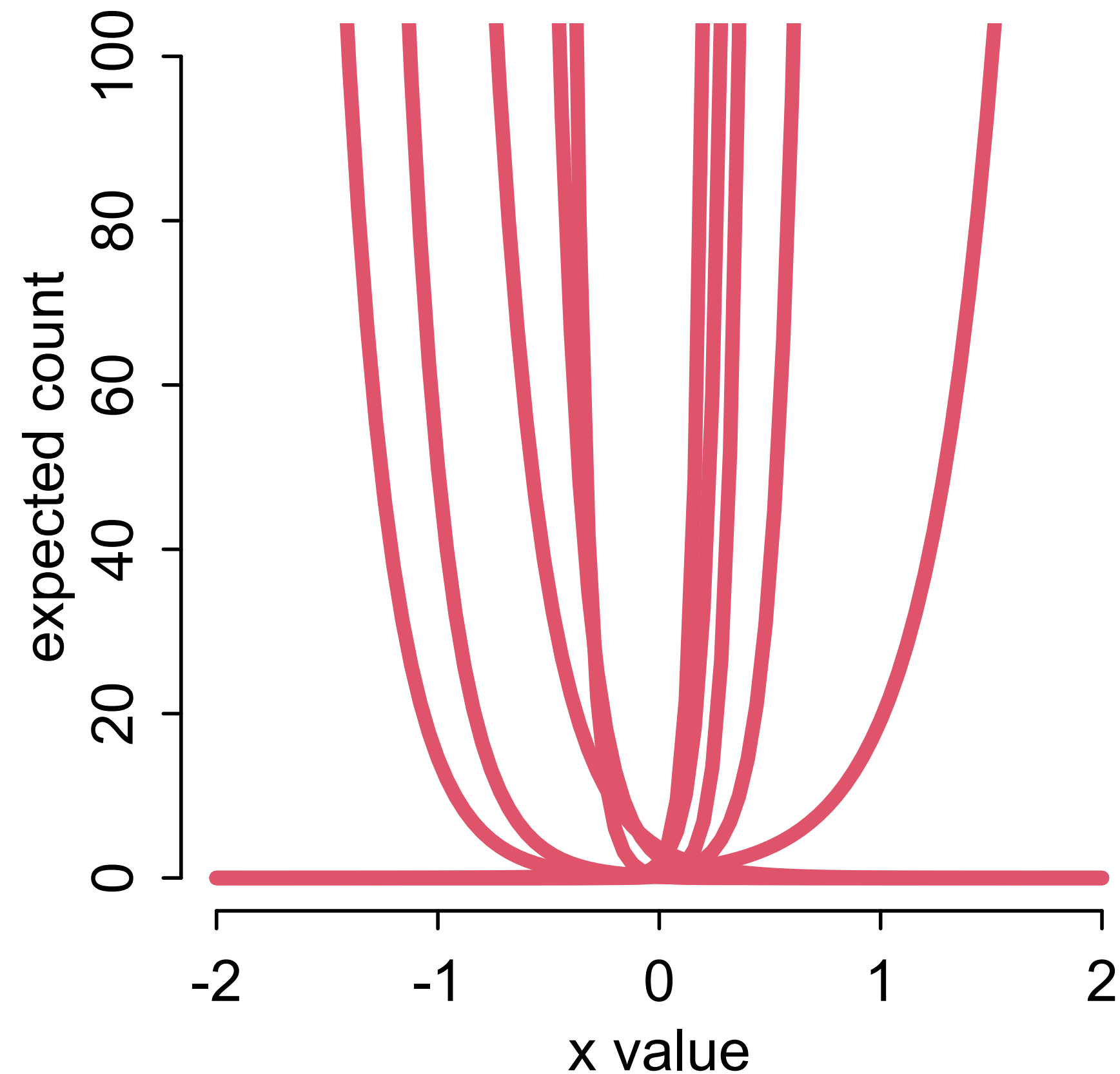
Exponential scaling can be surprising



Poisson priors

$$\alpha \sim \text{Normal}(0,1)$$

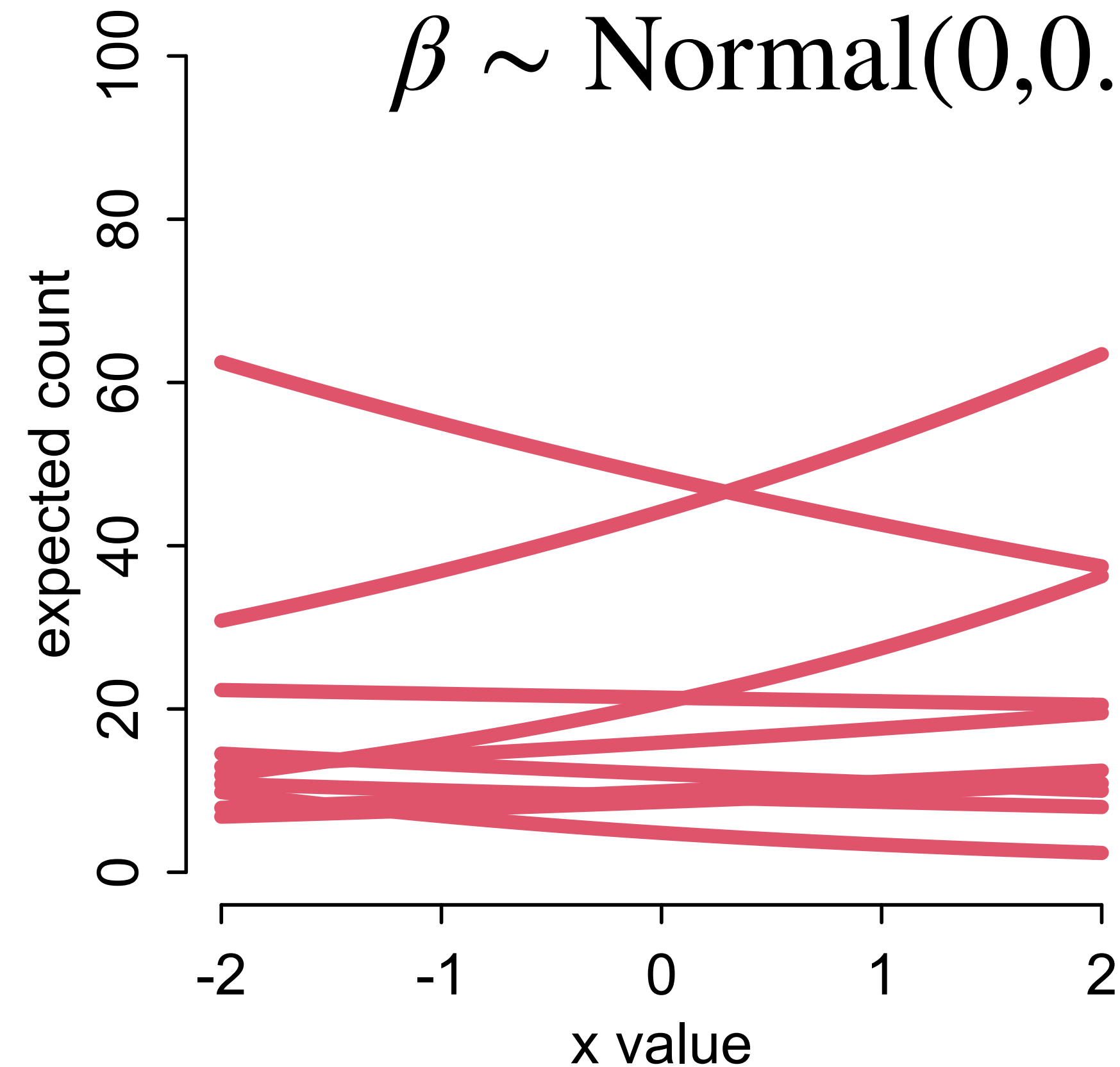
$$\beta \sim \text{Normal}(0,10)$$



$$\log(\lambda_i) = \alpha + \beta x_i$$

$$\alpha \sim \text{Normal}(3,0.5)$$

$$\beta \sim \text{Normal}(0,0.2)$$



```

data(Kline)
d <- Kline
d$P <- scale( log(d$population) )
d$contact_id <- ifelse( d$contact=="high" , 2 , 1 )
dat <- list(
  T = d$total_tools ,
  P = d$P ,
  C = d$contact_id )

# intercept only
m11.9 <- ulam(
  alist(
    T ~ dpois( lambda ),
    log(lambda) <- a,
    a ~ dnorm( 3 , 0.5 )
  ), data=dat , chains=4 , log_lik=TRUE )

# interaction model
m11.10 <- ulam(
  alist(
    T ~ dpois( lambda ),
    log(lambda) <- a[C] + b[C]*P,
    a[C] ~ dnorm( 3 , 0.5 ),
    b[C] ~ dnorm( 0 , 0.2 )
  ), data=dat , chains=4 , log_lik=TRUE )

compare( m11.9 , m11.10 , func=PSIS )

```

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \alpha_{C[i]} + \beta_{C[i]} \log(P_i)$$

$$\alpha_j \sim \text{Normal}(3, 0.5)$$

$$\beta_j \sim \text{Normal}(0, 0.2)$$

```
data(Kline)
d <- Kline
d$P <- scale( log(d$population) )
d$contact_id <- ifelse( d$contact=="high" , 2 , 1 )
dat <- list(
  T = d$total_tools ,
  P = d$P ,
  C = d$contact_id )
```

```
# intercept only
m11.9 <- ulam(
  alist(
    T ~ dpois( lambda ),
    log(lambda) <- a,
    a ~ dnorm( 3 , 0.5 )
  ), data=dat , chains=4 , log_lik=TRUE )
```

```
# interaction model
```

```
m11.10 > compare( m11.9 , m11.10 , func=PSIS )
```

Some Pareto k values are high (>0.5). Set pointwise=TRUE to inspect individual points.
Some Pareto k values are high (>0.5). Set pointwise=TRUE to inspect individual points.

	PSIS	SE	dPSIS	dSE	pPSIS	weight
m11.10	85.9	13.50	0.0	NA	7.3	1
m11.9	141.3	33.69	55.4	33.13	8.0	0

```
), data=dat , chains=4 , log_lik=TRUE )
```

```
compare( m11.9 , m11.10 , func=PSIS )
```

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \alpha_{C[i]} + \beta_{C[i]} \log(P_i)$$

$$\alpha_j \sim \text{Normal}(3, 0.5)$$

$$\beta_j \sim \text{Normal}(0, 0.2)$$


```
> compare( m11.9 , m11.10 , func=PSIS )
```

Some Pareto k values are high (>0.5). Set pointwise=TRUE to inspect individual points.
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	PSIS	SE	dPSIS	dSE	pPSIS	weight
m11.10	85.9	13.50	0.0	NA	7.3	1
m11.9	141.3	33.69	55.4	33.13	8.0	0

pPSIS is the “penalty”, the effective number of parameters.

How can a model with more parameters (m11.10) have fewer effective parameters?

m11.9 changes more when individual societies are dropped


```

k <- PSIS( m11.10 , pointwise=TRUE )$k
plot( dat$P , dat$T , xlab="log population (std)" ,
      ylab="total tools" ,
      col=ifelse( dat$C==1 , 4 , 2 ) , lwd=4+4*normalize(k) ,
      ylim=c(0,75) , cex=1+normalize(k) )

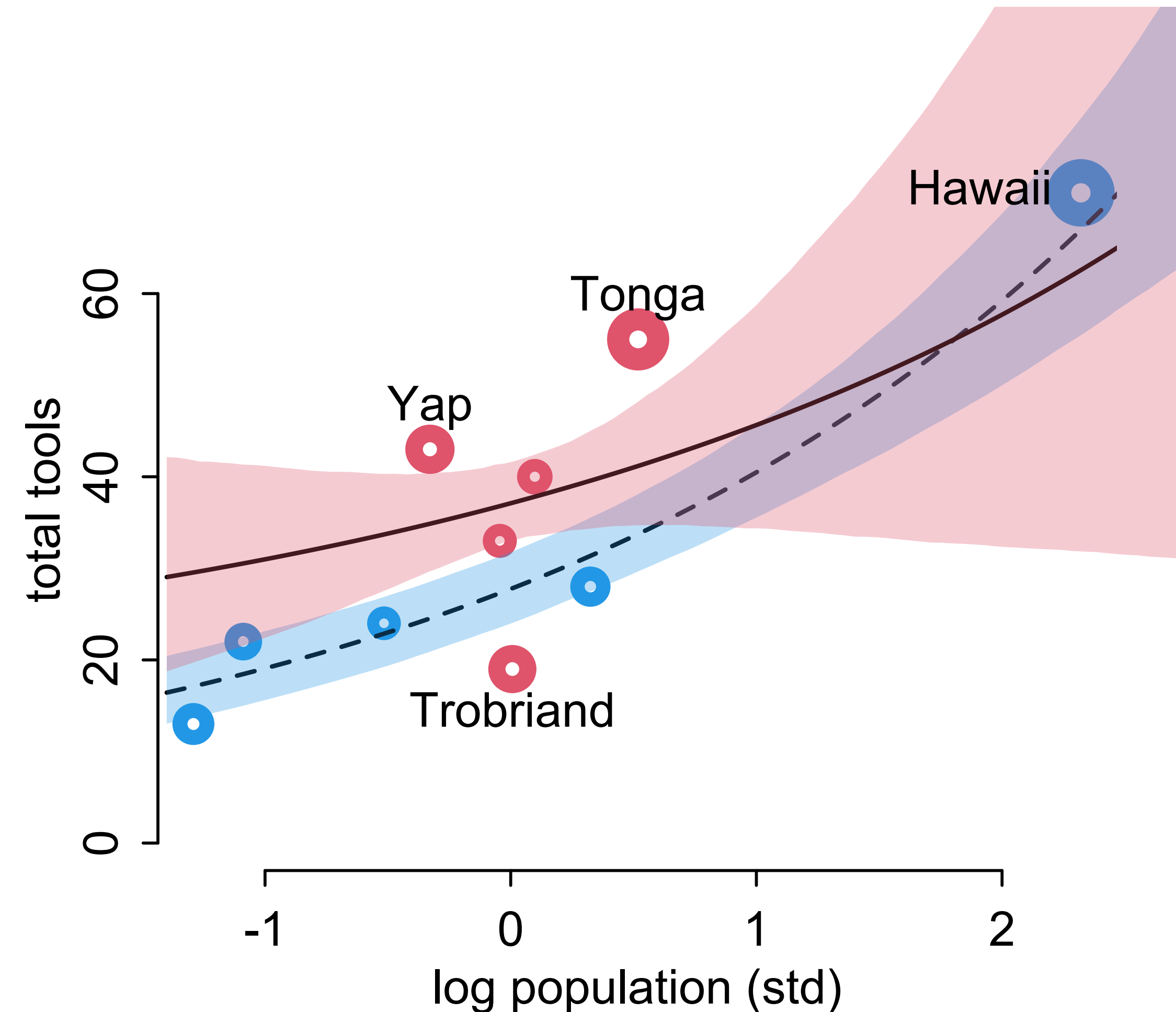
# set up the horizontal axis values to compute predictions
at
P_seq <- seq( from=-1.4 , to=3 , len=100 )

# predictions for C=1 (low contact)
lambda <- link( m11.10 , data=data.frame( P=P_seq , C=1 ) )
lmu <- apply( lambda , 2 , mean )
lci <- apply( lambda , 2 , PI )
lines( P_seq , lmu , lty=2 , lwd=1.5 )
shade( lci , P_seq , xpd=TRUE , col=col.alpha(4,0.3) )

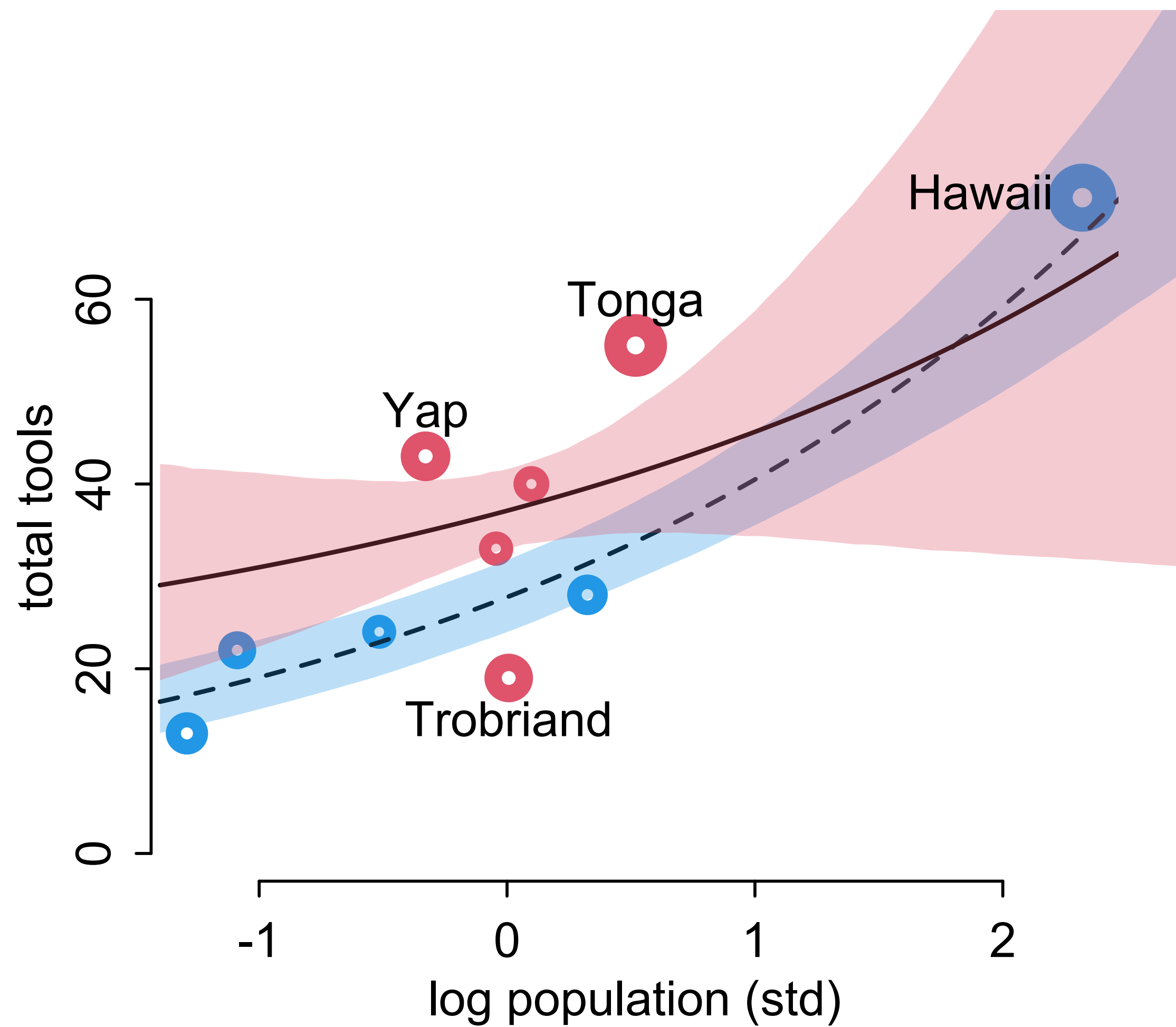
# predictions for C=2 (high contact)
lambda <- link( m11.10 , data=data.frame( P=P_seq , C=2 ) )
lmu <- apply( lambda , 2 , mean )
lci <- apply( lambda , 2 , PI )
lines( P_seq , lmu , lty=1 , lwd=1.5 )
shade( lci , P_seq , xpd=TRUE , col=col.alpha(2,0.3))

```

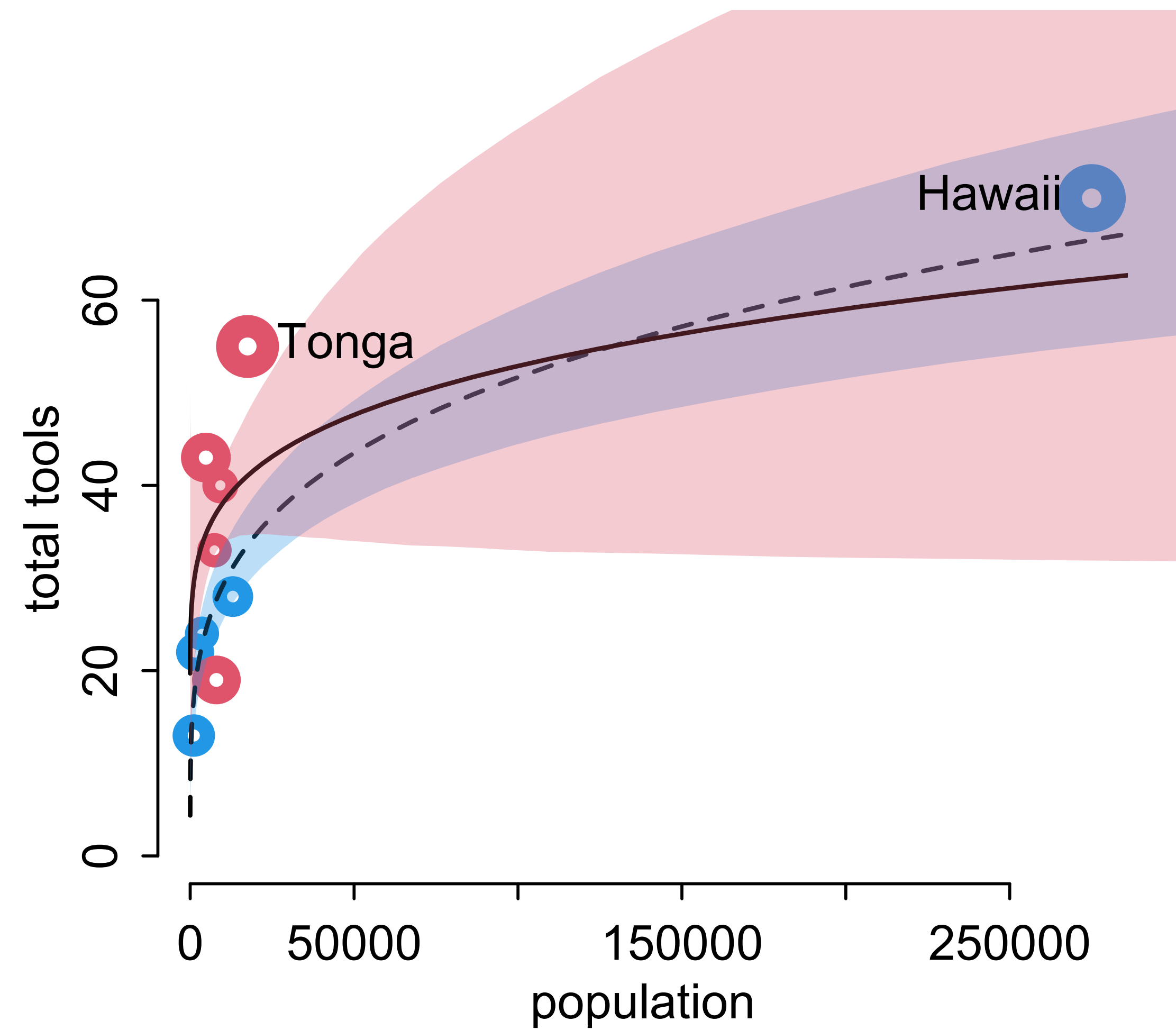
Points scaled by leverage

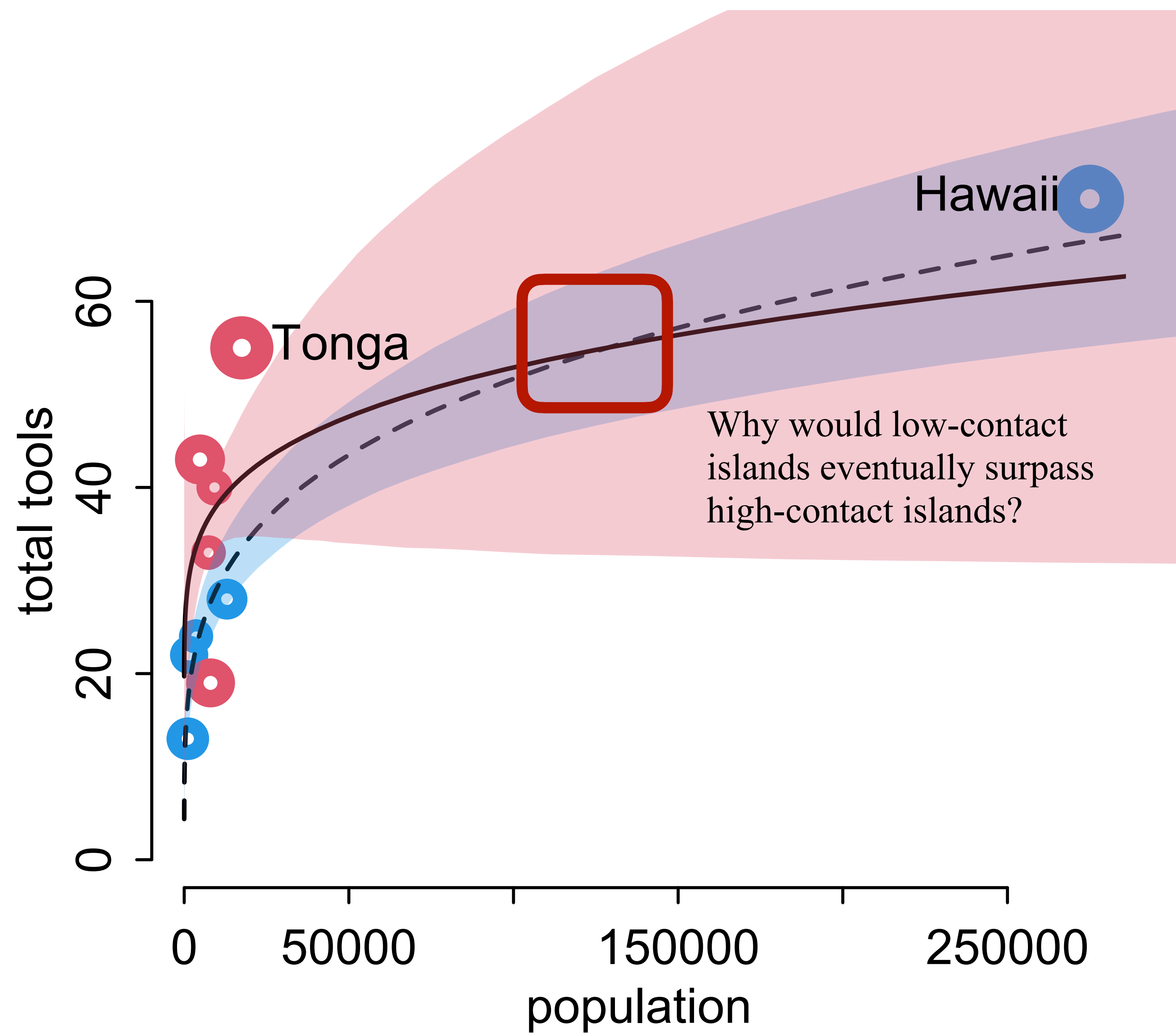


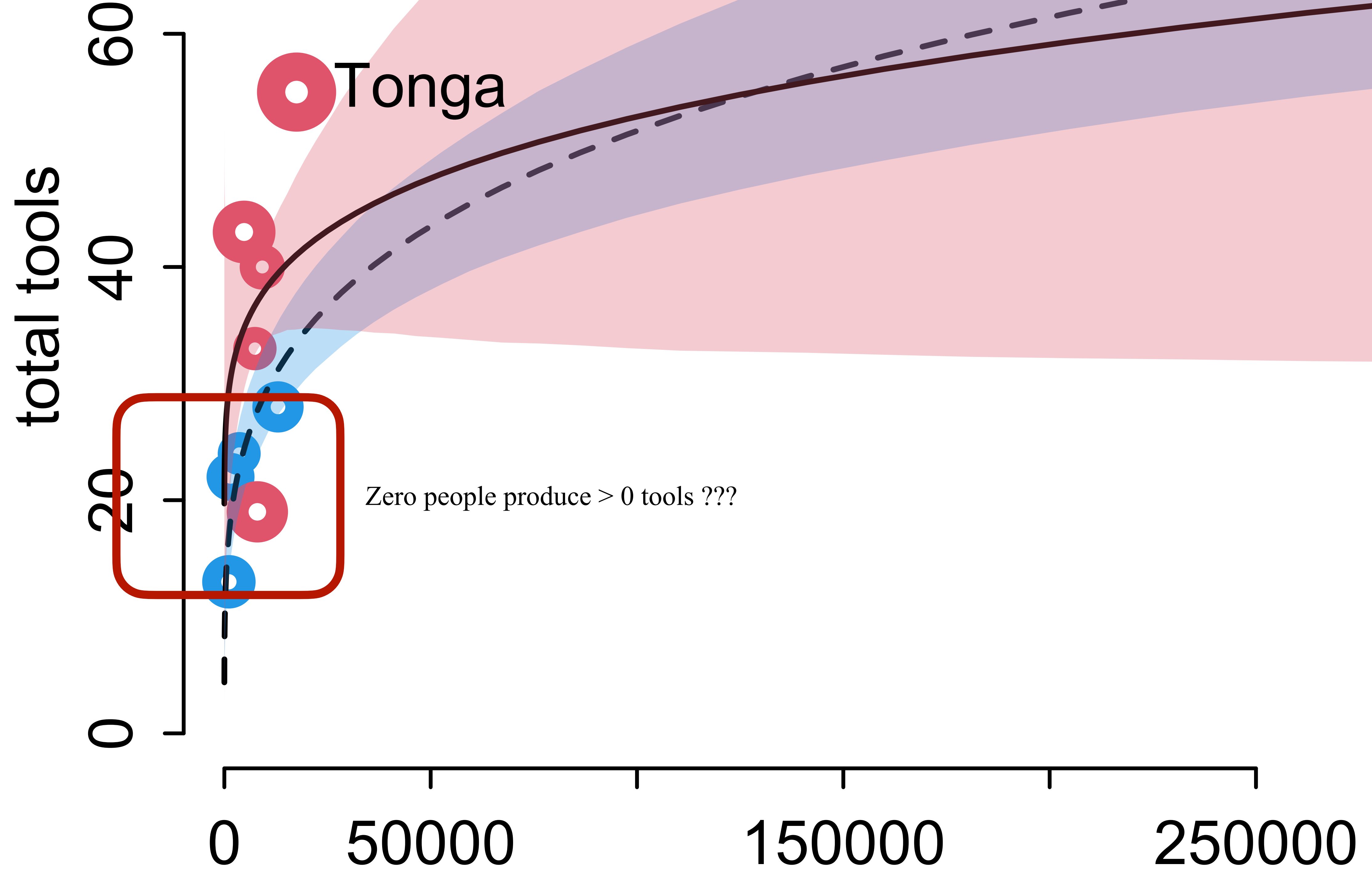
log scale



Natural scale





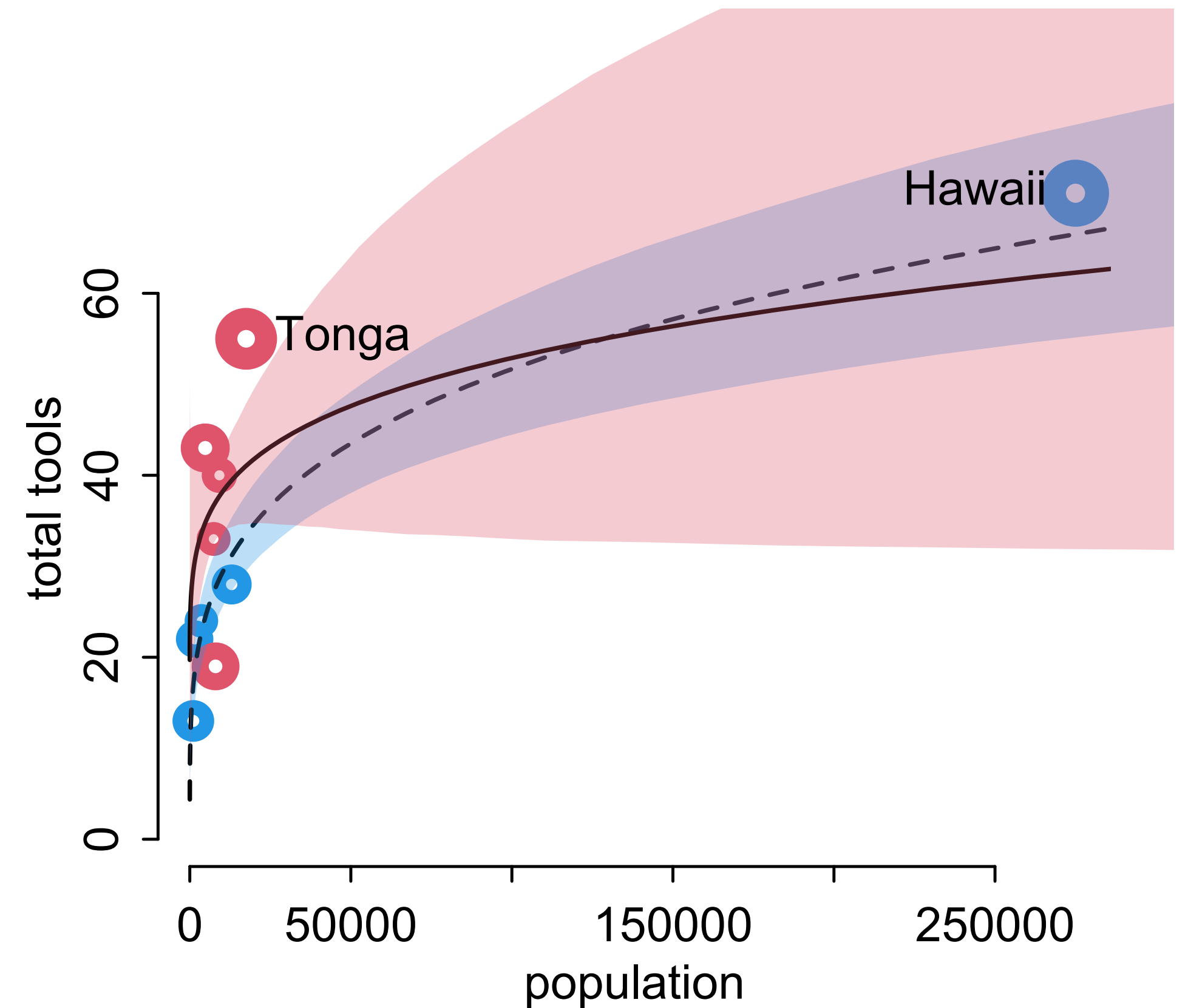


This model is wack

Two immediate ways to improve the model

(1) Use a robust model:
gamma-Poisson (neg-binomial)

(2) A principled scientific model



Population



Innovations

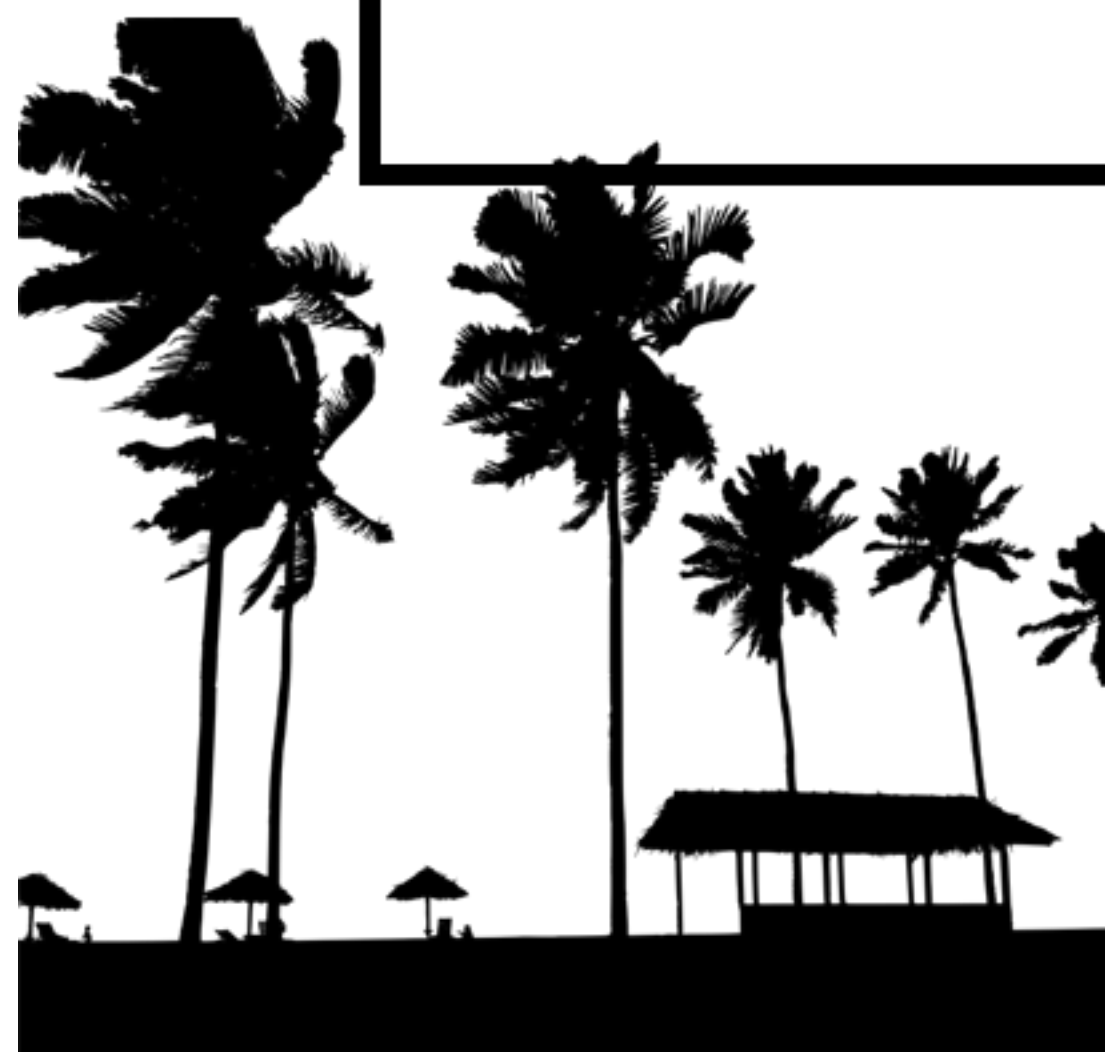


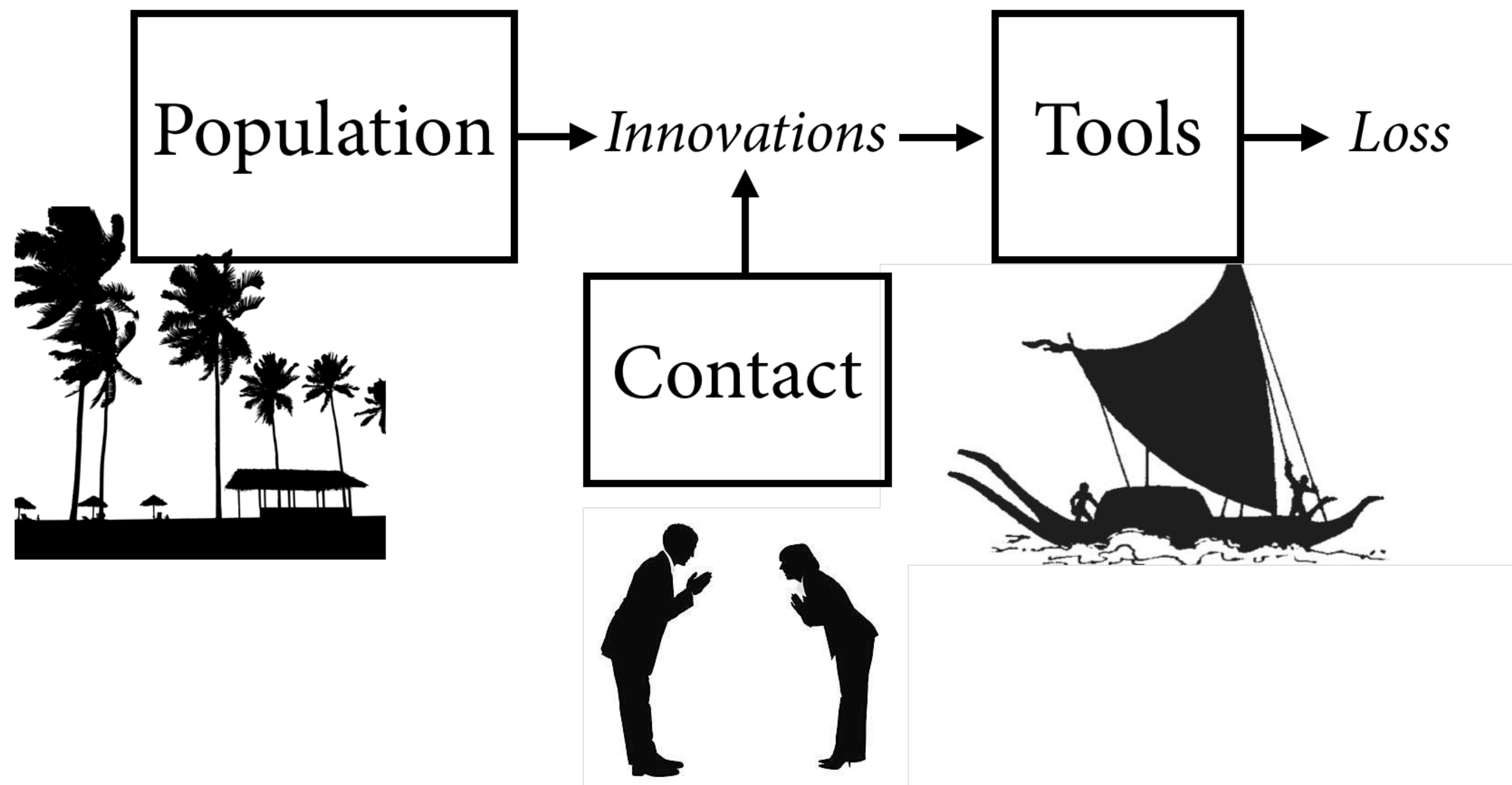
Tools



Loss

Contact





$$\Delta T = \alpha P^\beta - \gamma T$$

change in tools

$$\Delta T = \alpha P^\beta - \gamma T$$

change in tools

*innovation
rate*

$$\Delta T = \alpha P^\beta - \gamma T$$

change in tools

innovation rate

diminishing returns (elasticity)

$$\Delta T = \alpha P^\beta - \gamma T$$

change in tools

innovation rate

diminishing returns (elasticity)

rate of loss

$$\Delta T = \alpha P^\beta - \gamma T$$

*diminishing returns
depend upon contact*

$$\Delta T = \alpha_c P^{\beta_c} - \gamma T$$

*innovation depends
upon contact*

$$\Delta T = \alpha_c P^{\beta_c} - \gamma T$$

Difference equation: Says how tools change, not expected number

What is the **equilibrium**?

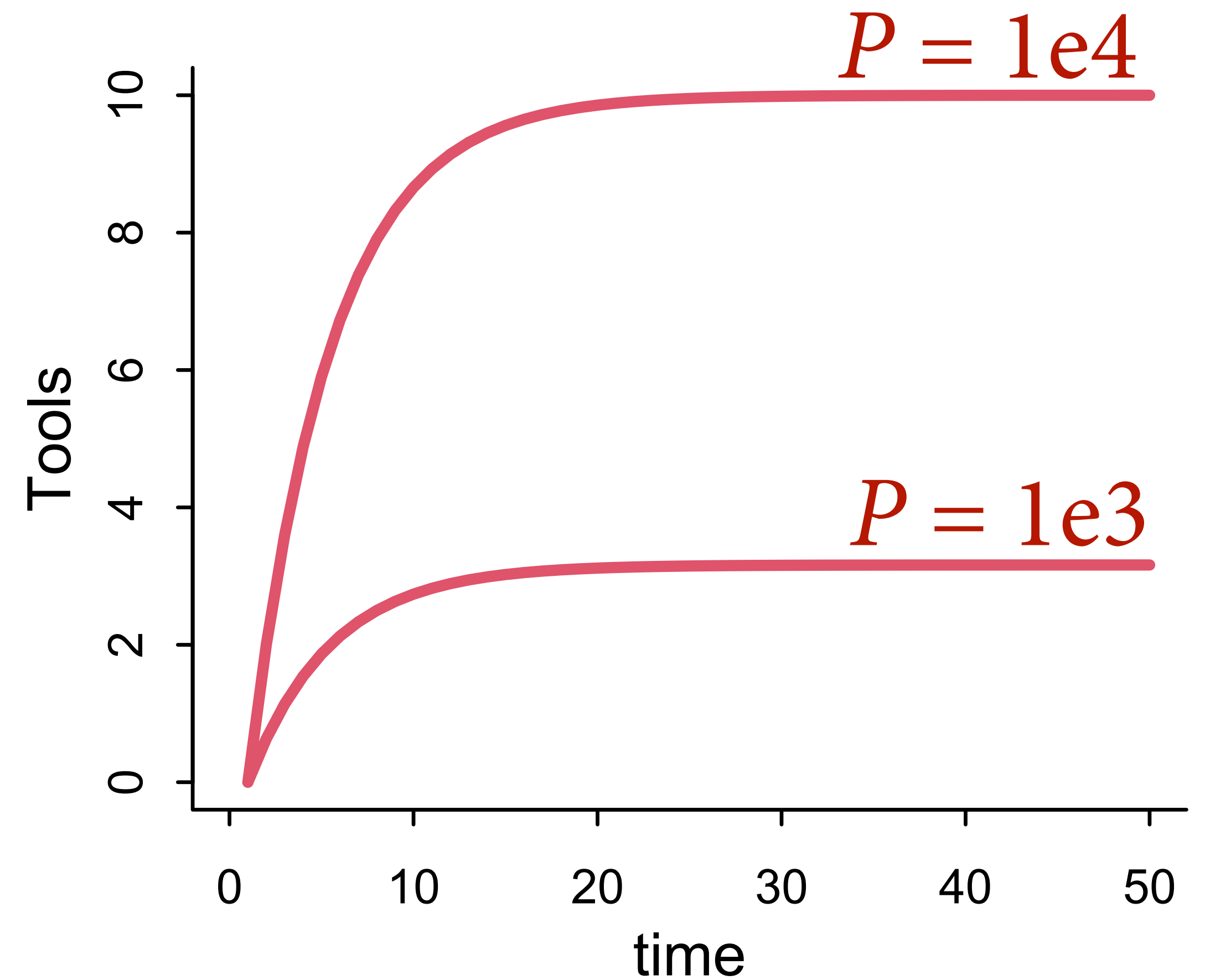
$$\Delta T = \alpha_c P^{\beta_c} - \gamma T$$

```
f <- function(a=0.02,b=0.5,g=0.2,P=1e4,t_max=50) {
  T <- rep(0,t_max)
  for ( i in 2:t_max )
    T[i] <- T[i-1] + a*P^b - g*T[i-1]
  return(T)
}

plot( NULL , xlim=c(0,50) , ylim=c(0,10) ,
      xlab="time" , ylab="Tools" )

T <- f(P=1e3)
lines( 1:50 , T , lwd=3 , col=2 )

T <- f(P=1e4)
lines( 1:50 , T , lwd=3 , col=2 )
```



$$\Delta T = \alpha_c P^{\beta_c} - \gamma T = 0$$

Solve for equilibrium T

$$\Delta T = \alpha_c P^{\beta_c} - \gamma T = 0$$

Solve for equilibrium T

$$\hat{T} = \frac{\alpha_c P^{\beta_c}}{\gamma}$$

$$\hat{T} = \frac{\alpha_c P^{\beta_c}}{\gamma}$$

$$T_i \sim \text{Poisson}(\lambda_i)$$

$$\lambda_i = \hat{T}$$

$$\hat{T} = \frac{\alpha_c P^{\beta_c}}{\gamma}$$

```
# innovation/loss model

dat2 <- list( T=d$total_tools, P=d$population,
              C=d$contact_id )

m11.11 <- ulam(
  alist(
    T ~ dpois( lambda ),
    lambda <- exp(a[C])*P^b[C]/g,
    a[C] ~ dnorm(1,1),
    b[C] ~ dexp(1),
    g ~ dexp(1)
  ), data=dat2 , chains=4 , cores=4 )
```

All parameters must be positive

Two ways to do this

(1) use exp()

- Sometimes improves calculation efficiency

(2) use appropriate prior

- More transparent

$$\hat{T} = \frac{\alpha_c P^{\beta_c}}{\gamma}$$

```
# innovation/loss model
```

```
dat2 <- list( T=d$total_tools, P=d$population,
C=d$contact_id )
```

```
m11.11 <- ulam(
```

```
  alist(
```

```
    T ~ dpois( lambda )
```

```
    lambda <- exp(a[C])
```

```
    a[C] ~ dnorm(1,1),
```

```
    b[C] ~ dexp(1),
```

```
    g ~ dexp(1)
```

```
  ), data=dat2 , chains=4
```

```
> precis(m11.11,2)
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a[1]	0.85	0.68	-0.26	1.90	698	1
a[2]	0.93	0.83	-0.39	2.31	902	1
b[1]	0.26	0.03	0.21	0.32	1149	1
b[2]	0.29	0.10	0.12	0.45	711	1
g	1.11	0.70	0.32	2.43	862	1

All parameters must be positive

Two ways to do this

(1) use exp()

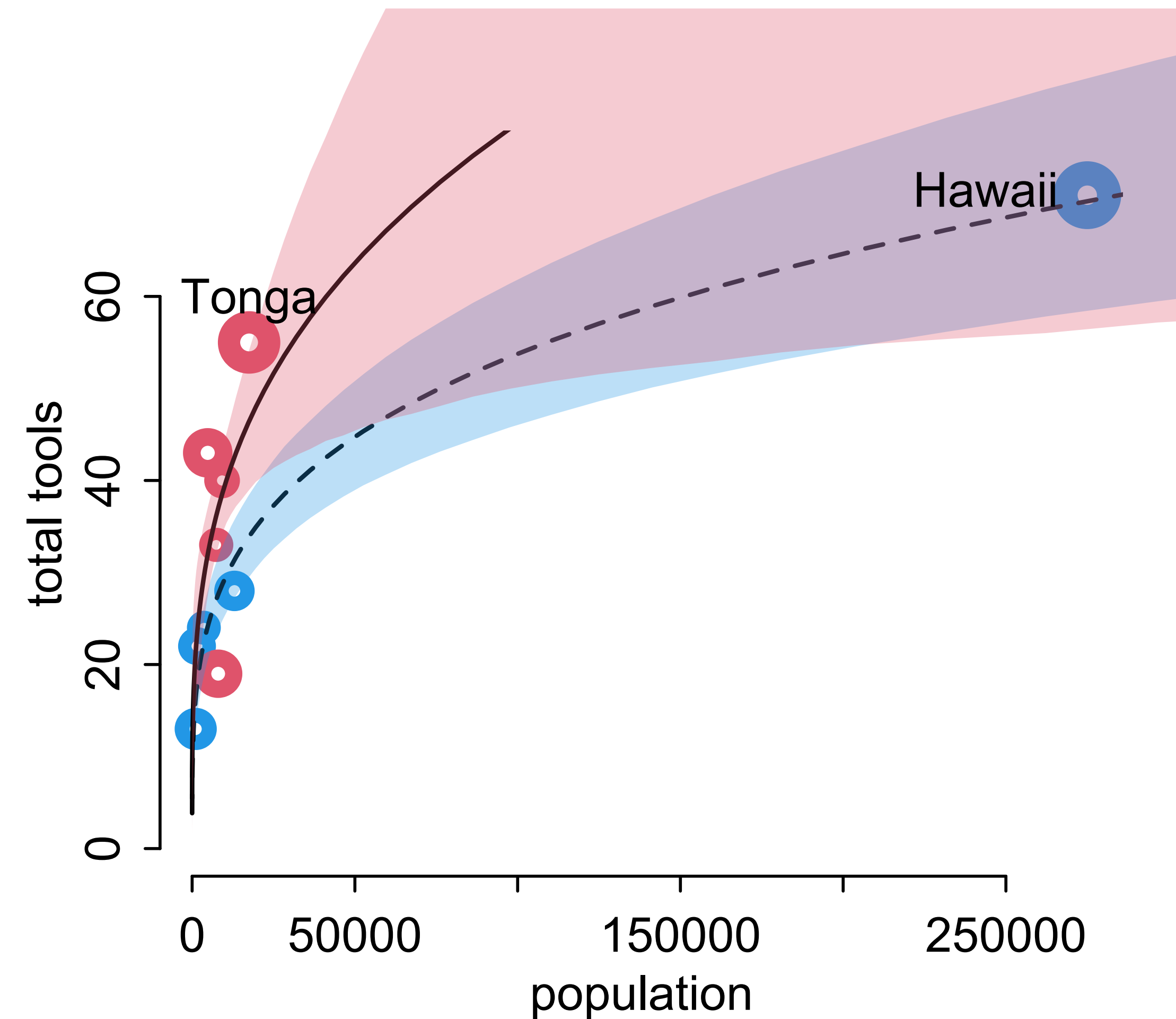
appropriate prior

$$\hat{T} = \frac{\alpha_C P^{\beta_C}}{\gamma}$$

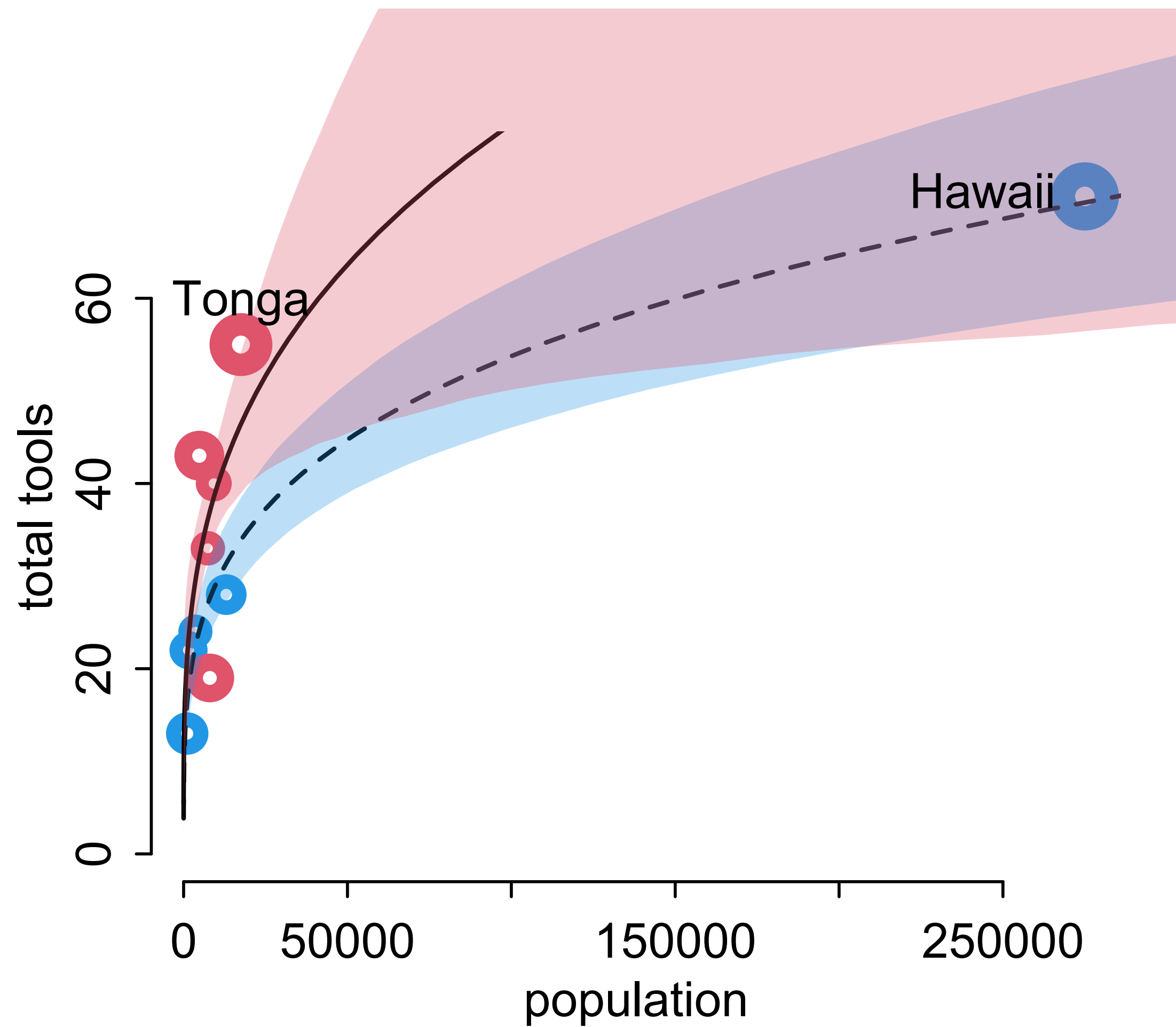
```
# innovation/loss model

dat2 <- list( T=d$total_tools, P=d$population,
              C=d$contact_id )

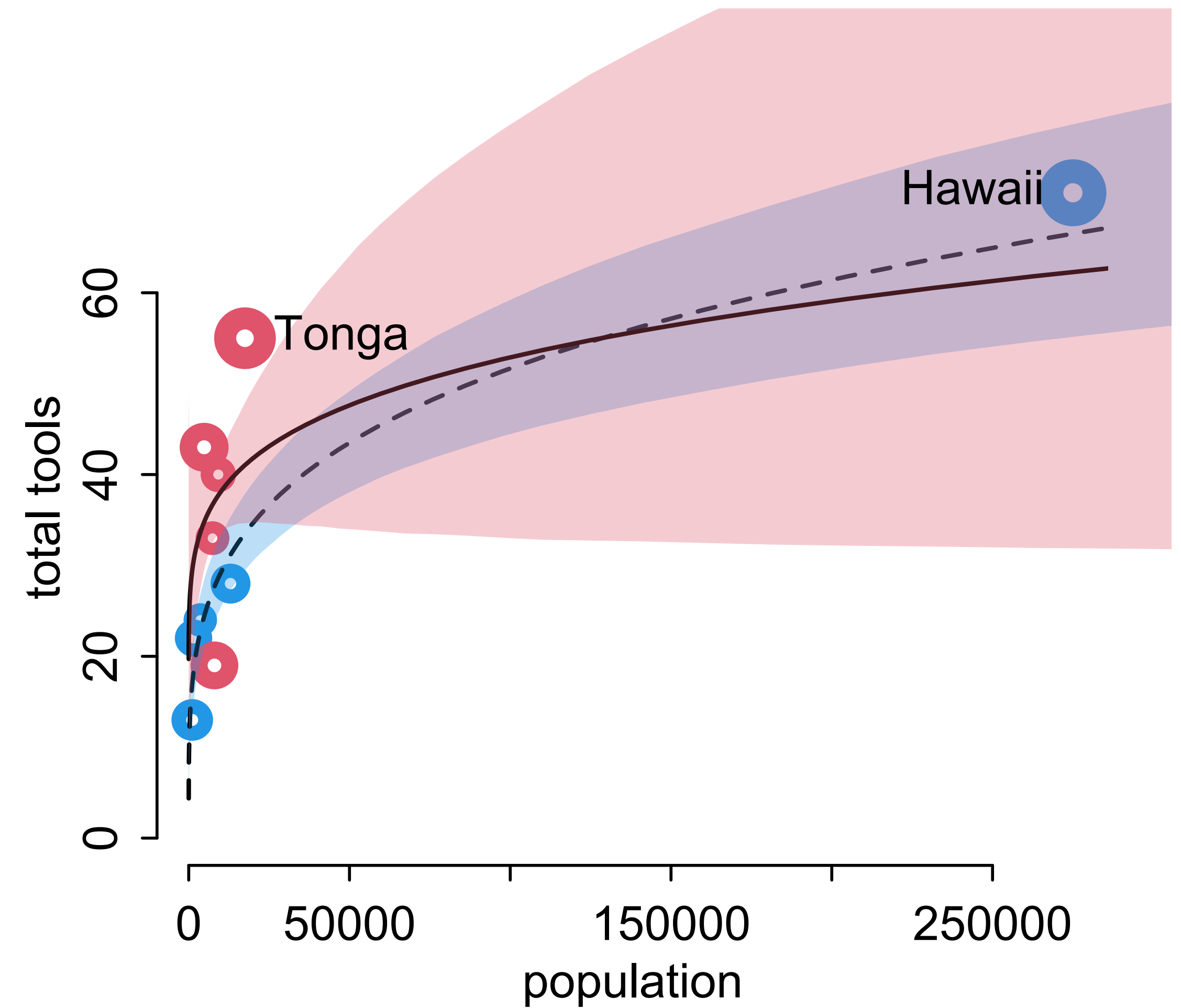
m11.11 <- ulam(
  alist(
    T ~ dpois( lambda ),
    lambda <- exp(a[C])*P^b[C]/g,
    a[C] ~ dnorm(1,1),
    b[C] ~ dexp(1),
    g ~ dexp(1)
  ), data=dat2 , chains=4 , cores=4 )
```



Innovation/loss model



Generalized linear model



Still have to deal with location as confound

Count GLMs

Distributions from constraints

Maximum entropy priors: Binomial, Poisson, and extensions

More event types: Multinomial and categorical

Robust regressions:
Beta-binomial, gamma-Poisson



